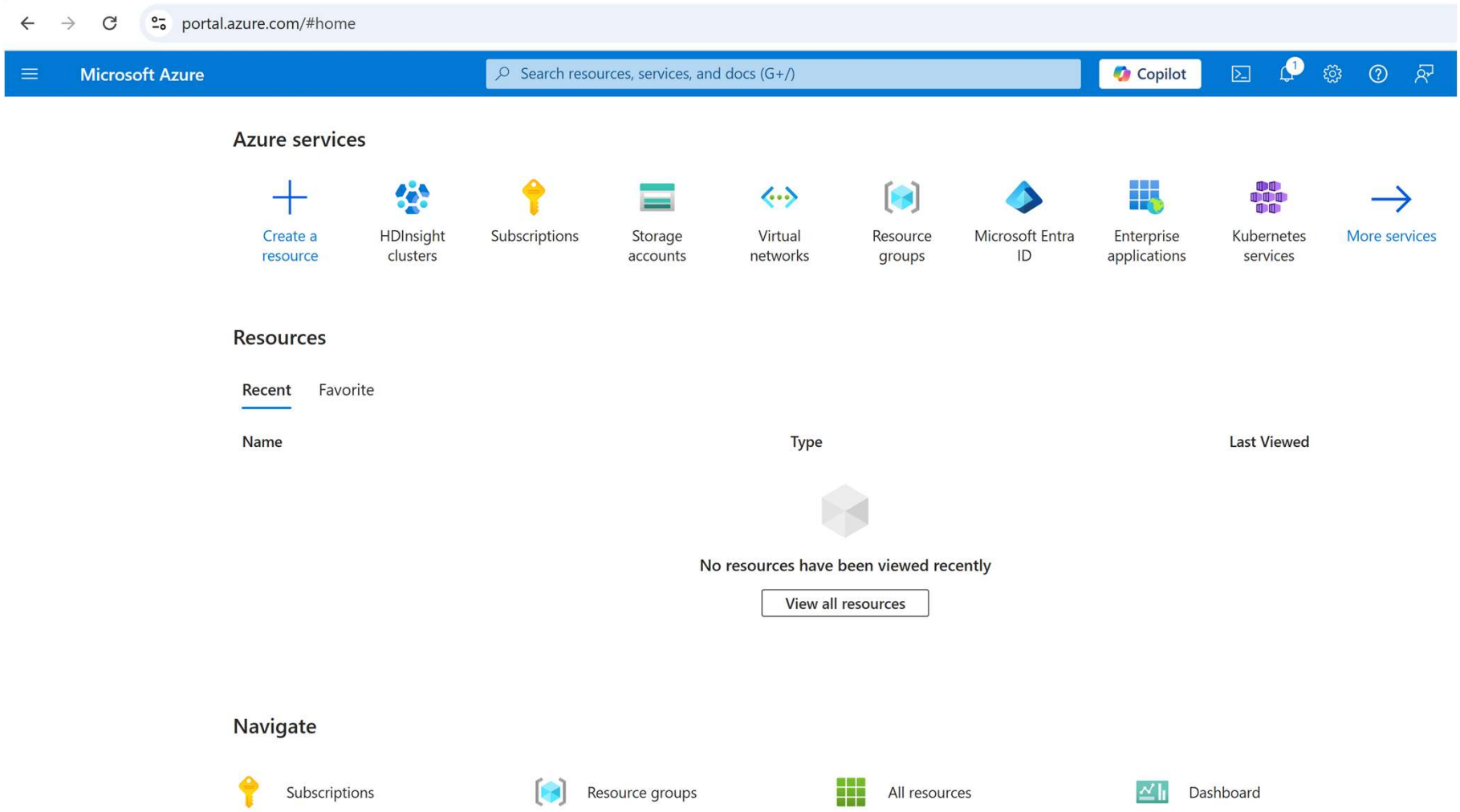
Azure HD-Insight
(Provisioning + Discovering Web UI)

arnaud.nauwynck@gmail.com

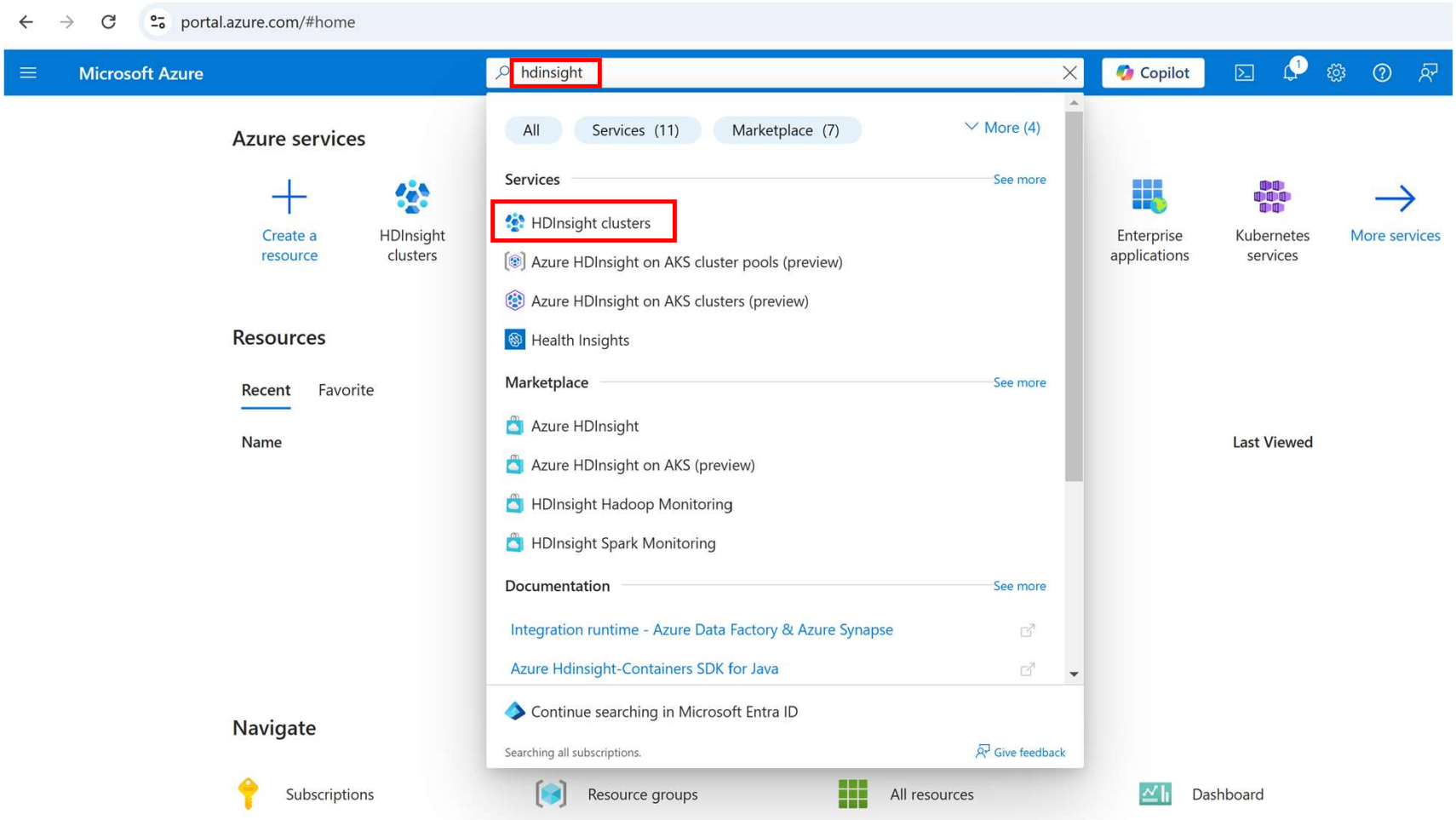
Esilv 2024

https://portal.azure.com
Home page => Azure Services

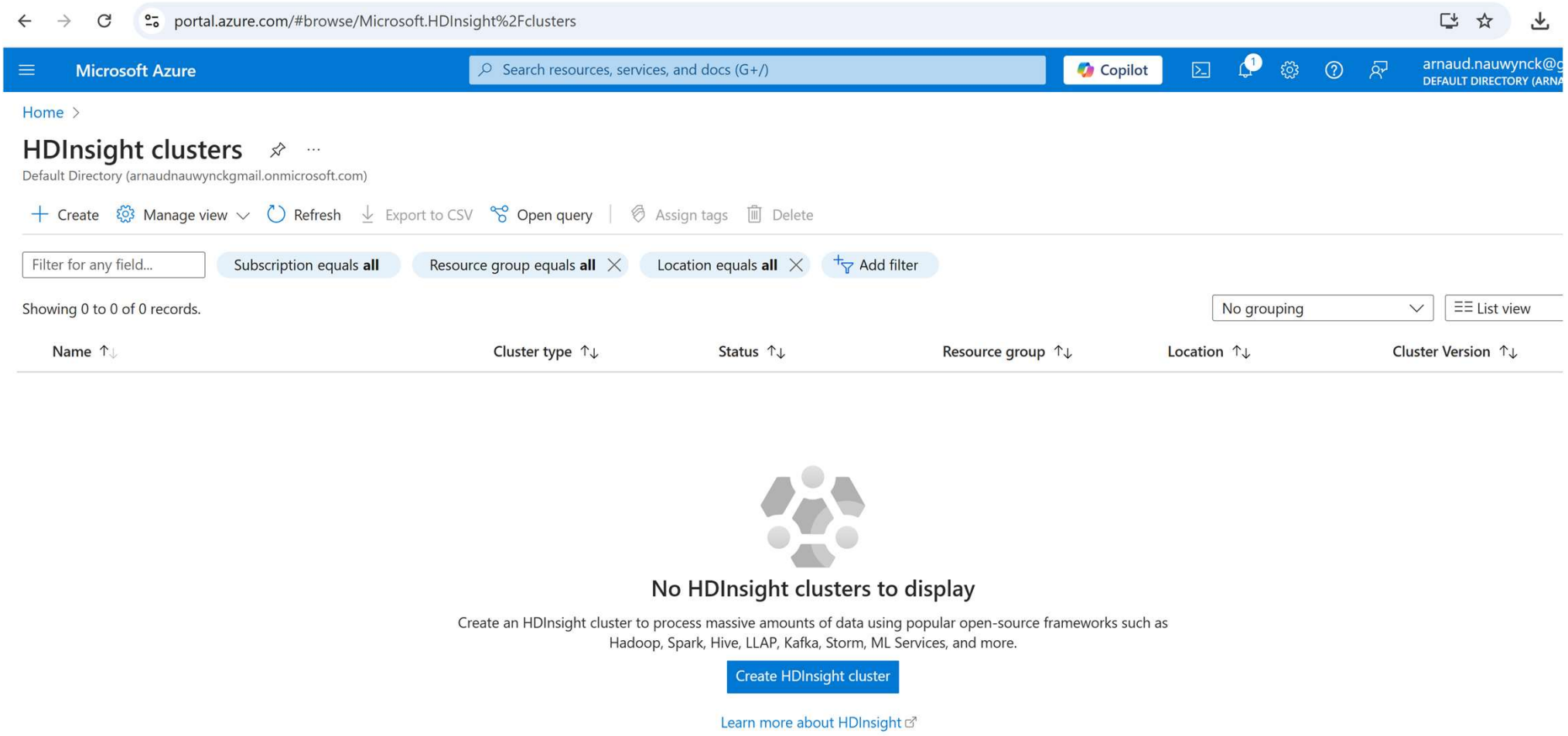


https://portal.azure.com

Search service "hdinsight"

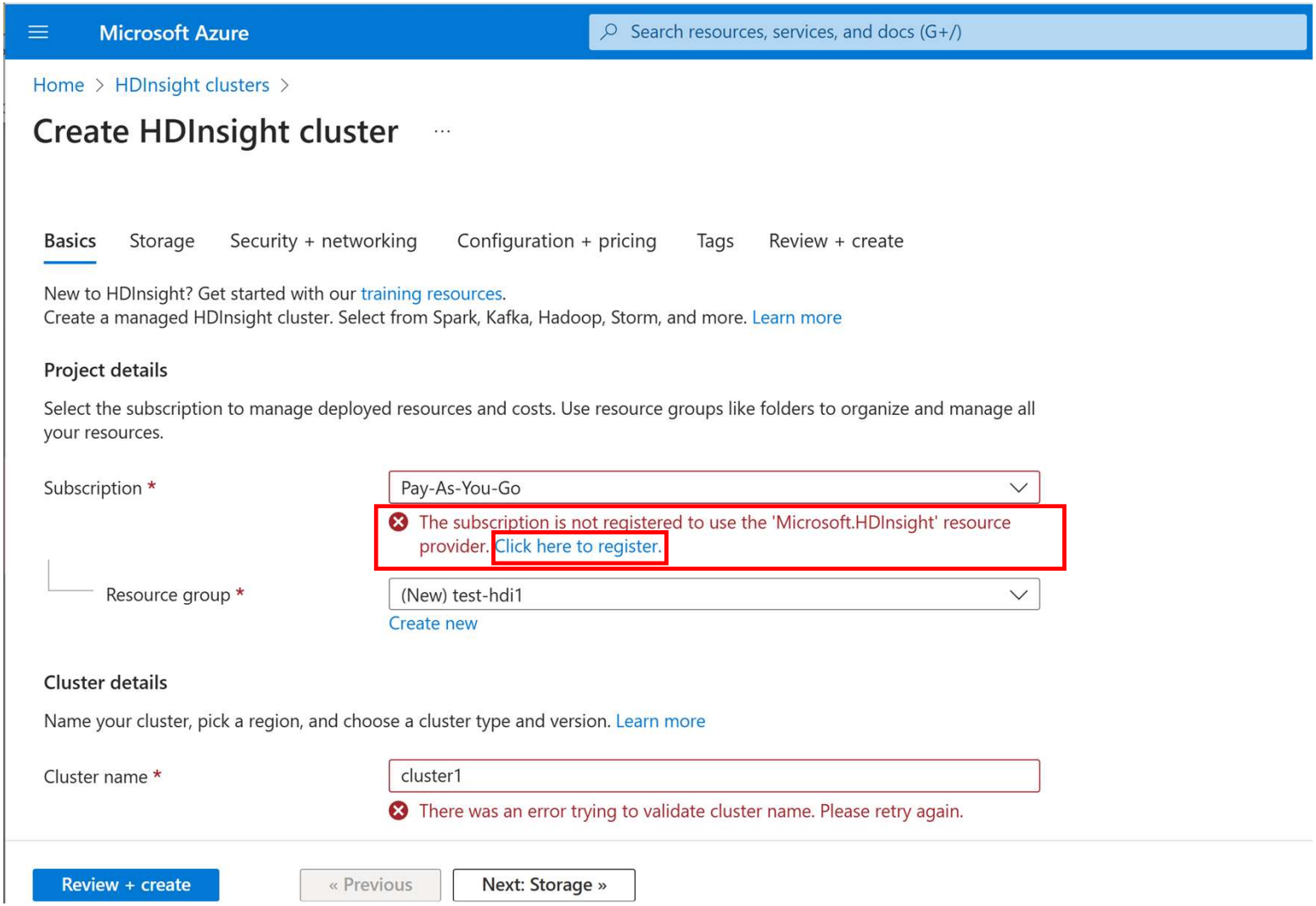


List of Provisioned HDInsight clusters

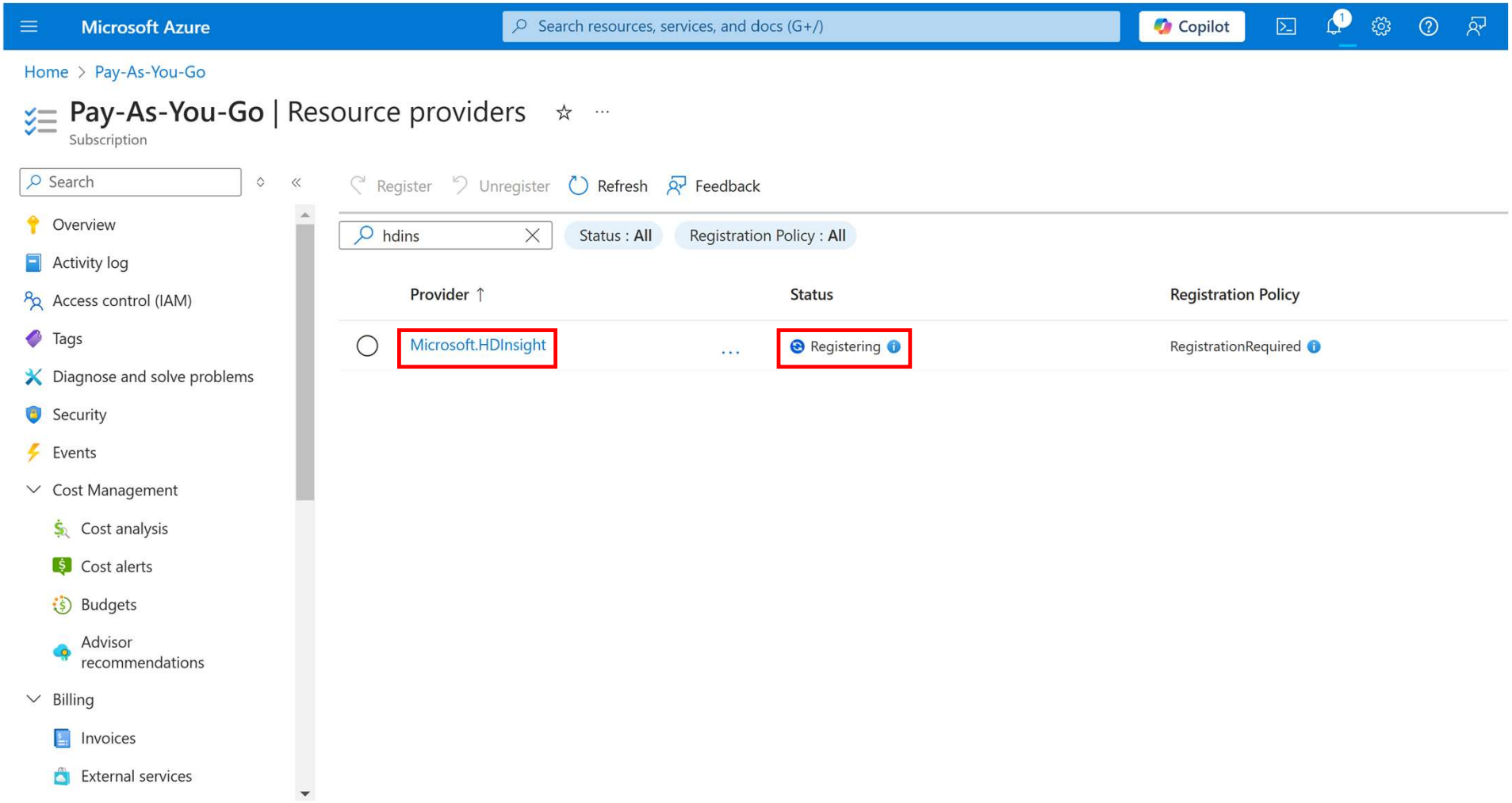


Create a Cluster

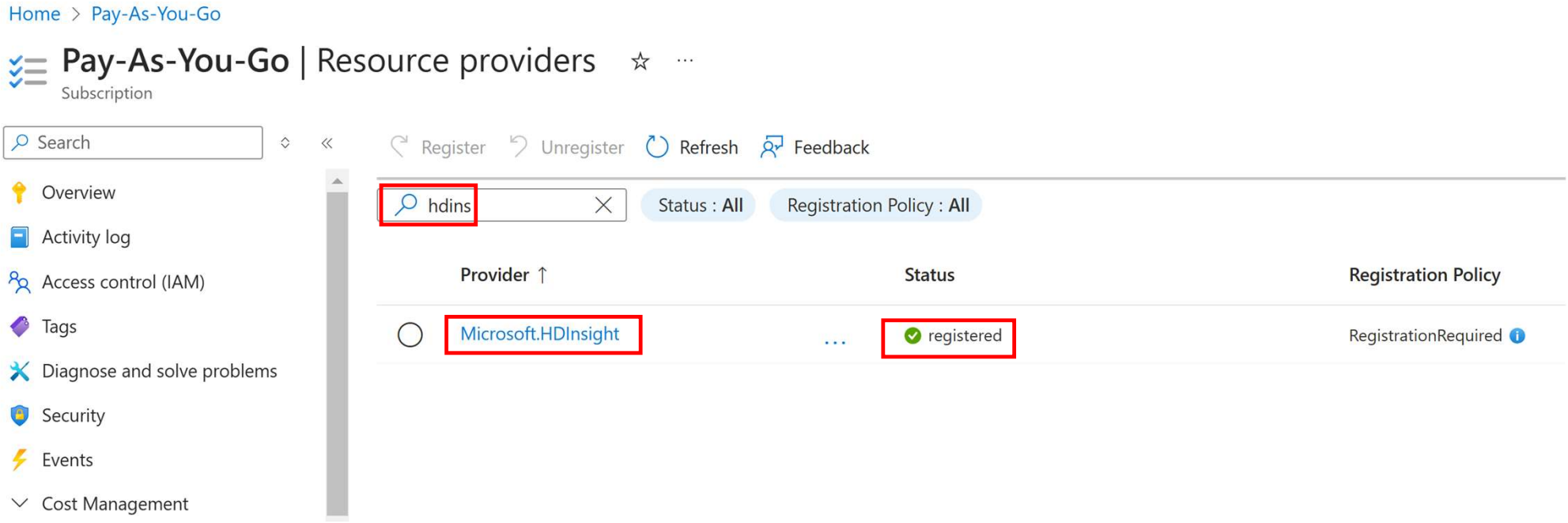
https://portal.azure.com/#create/Microsoft.HDInsightCluster



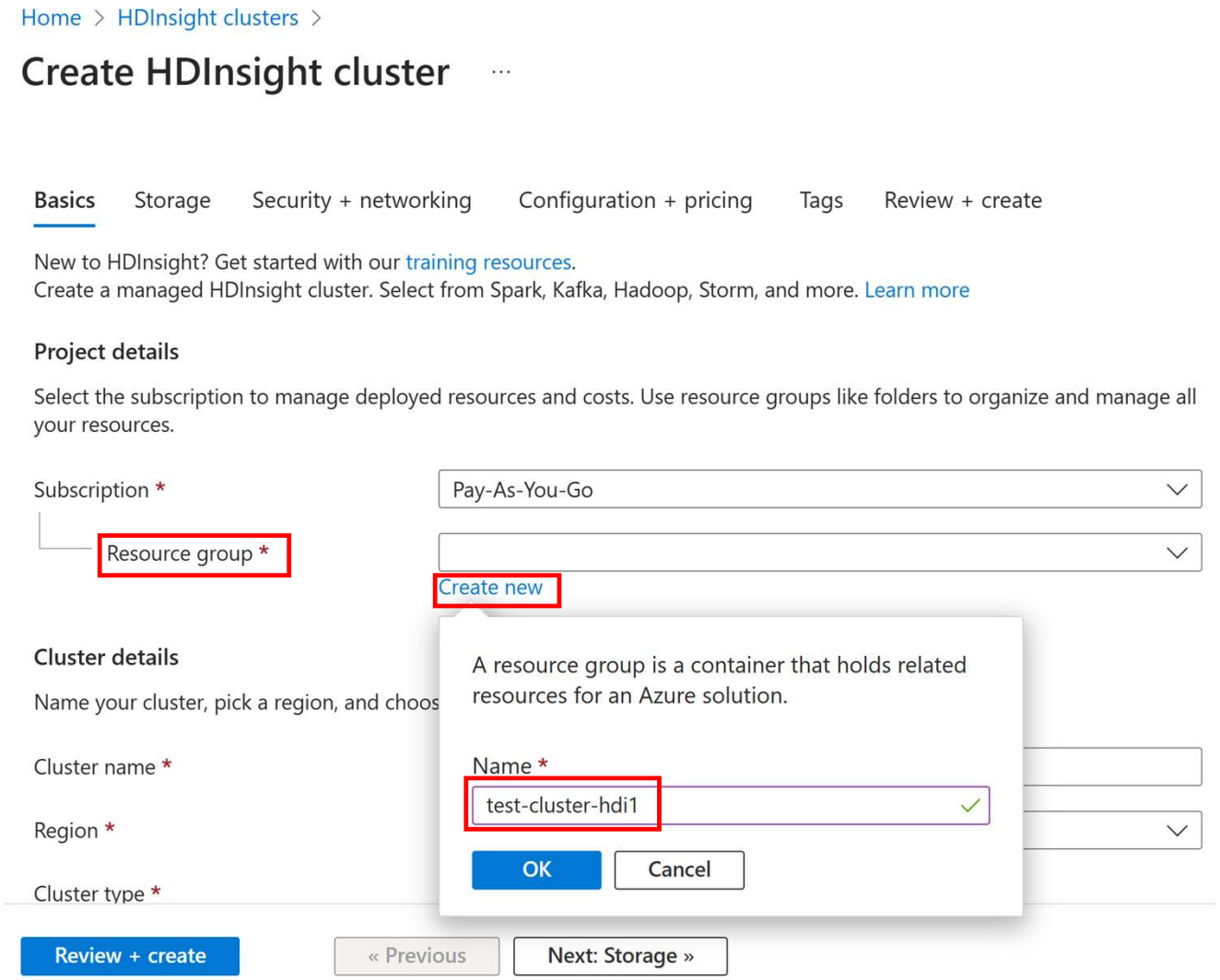
Need Register Feature



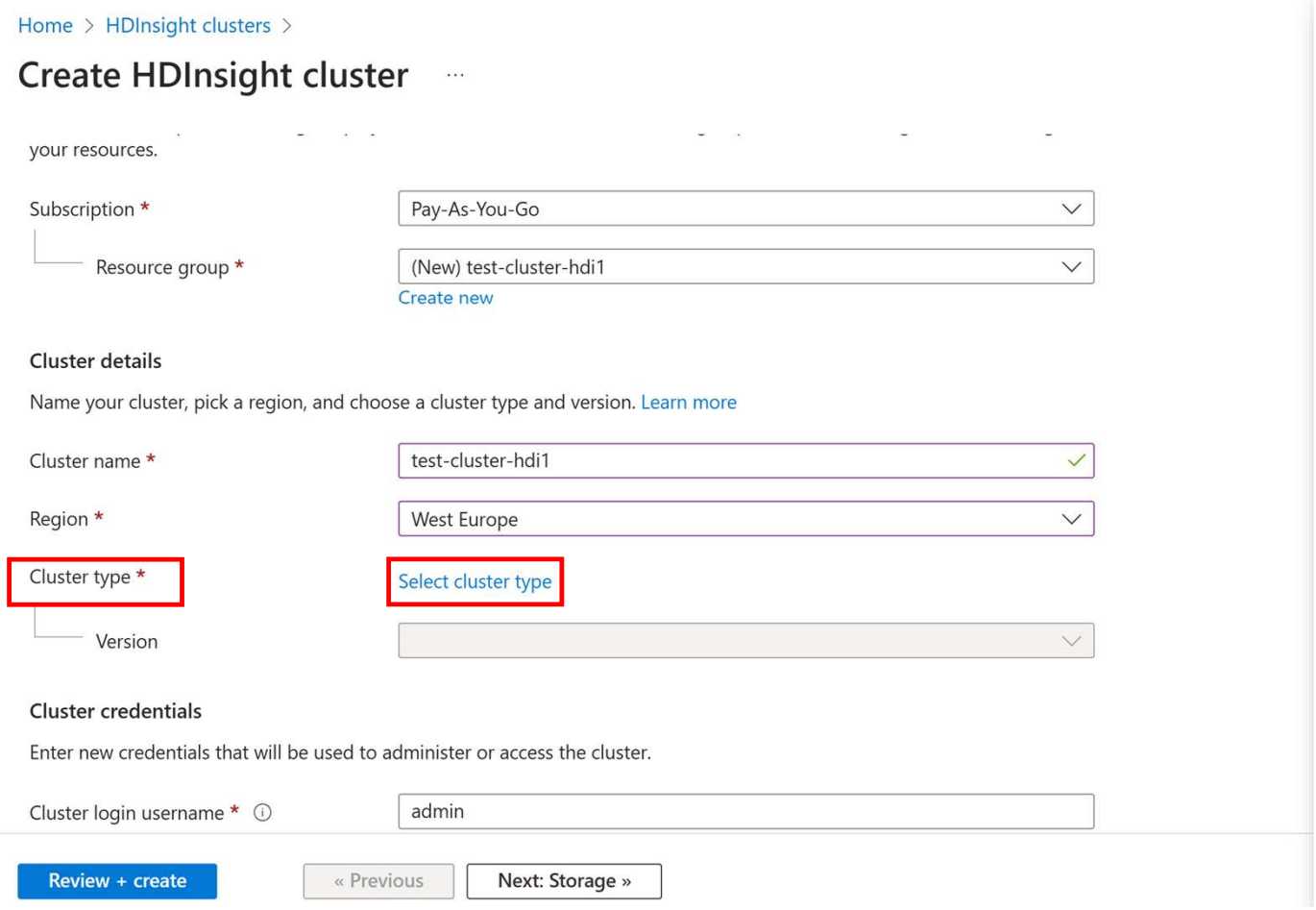
Select HDInsight, click "Register"
... wait few minutes

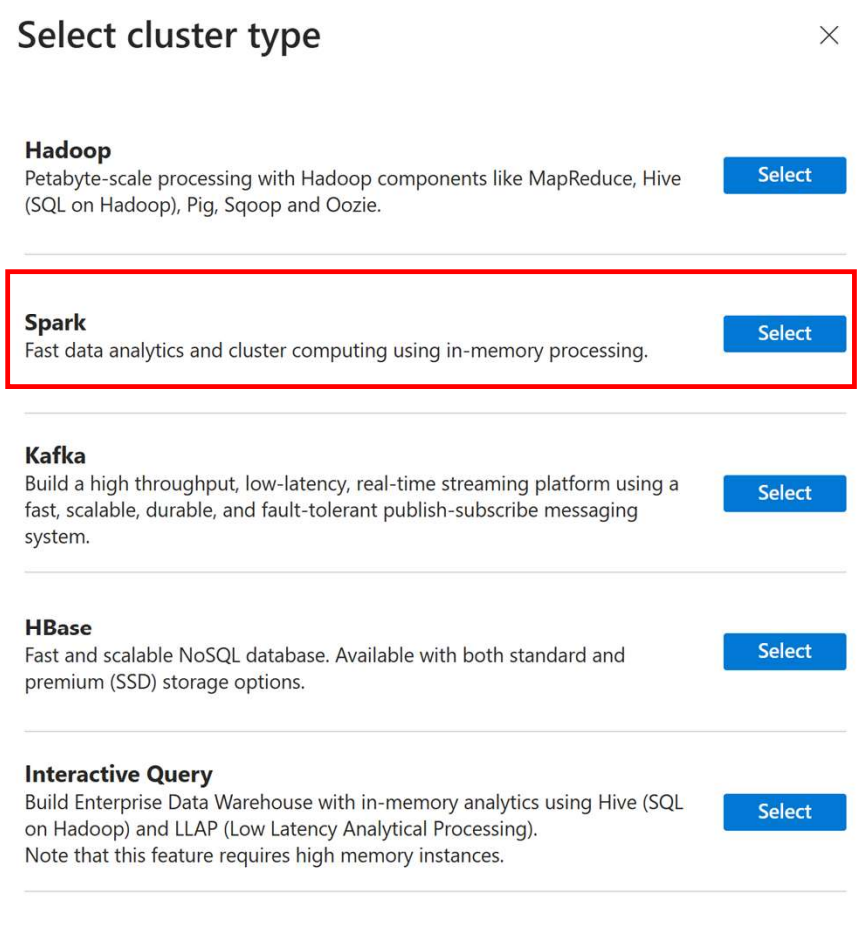


Create HDI - Step 1: Create Resource Group

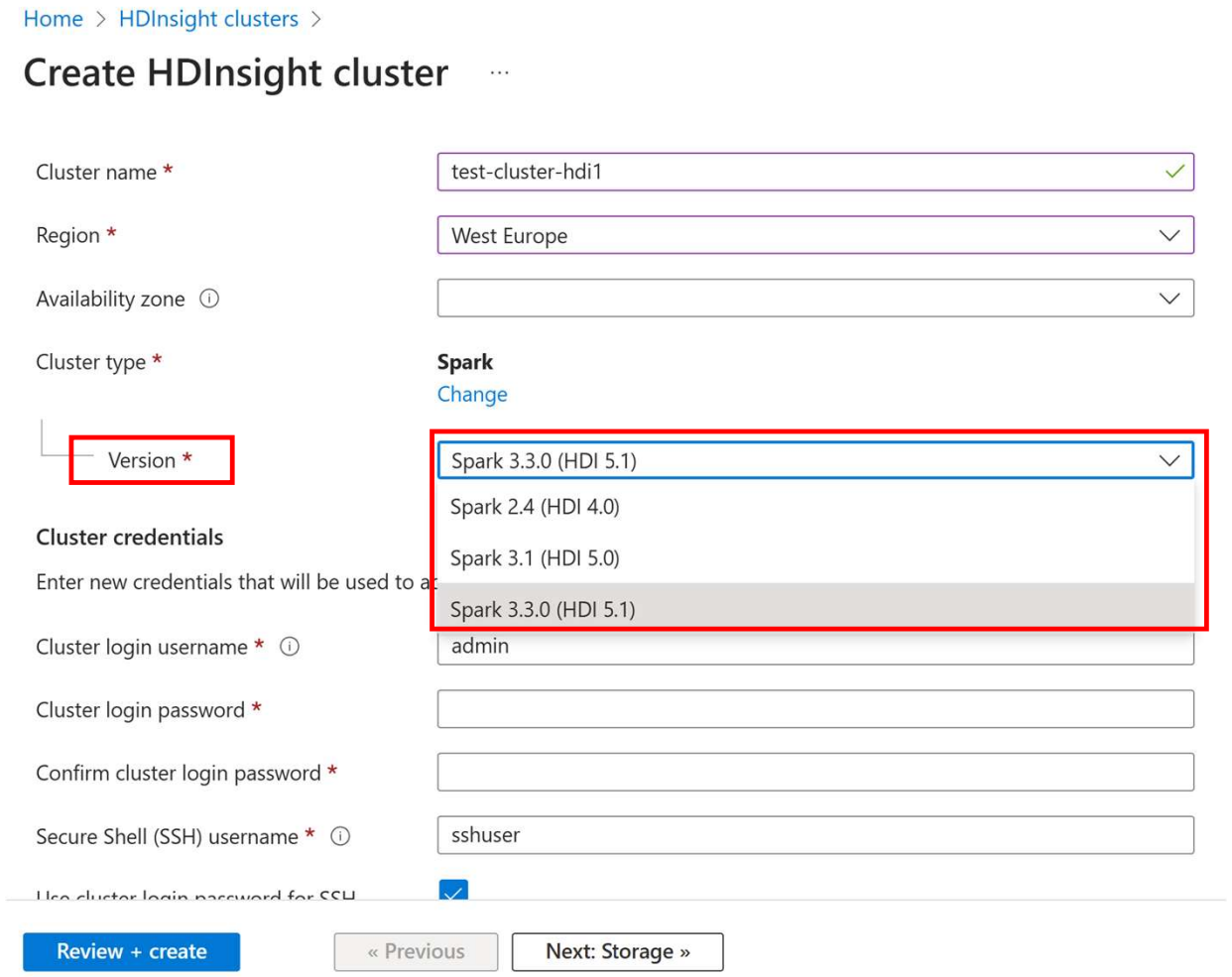


Create HDI - Choose Cluster Type (=> Spark)

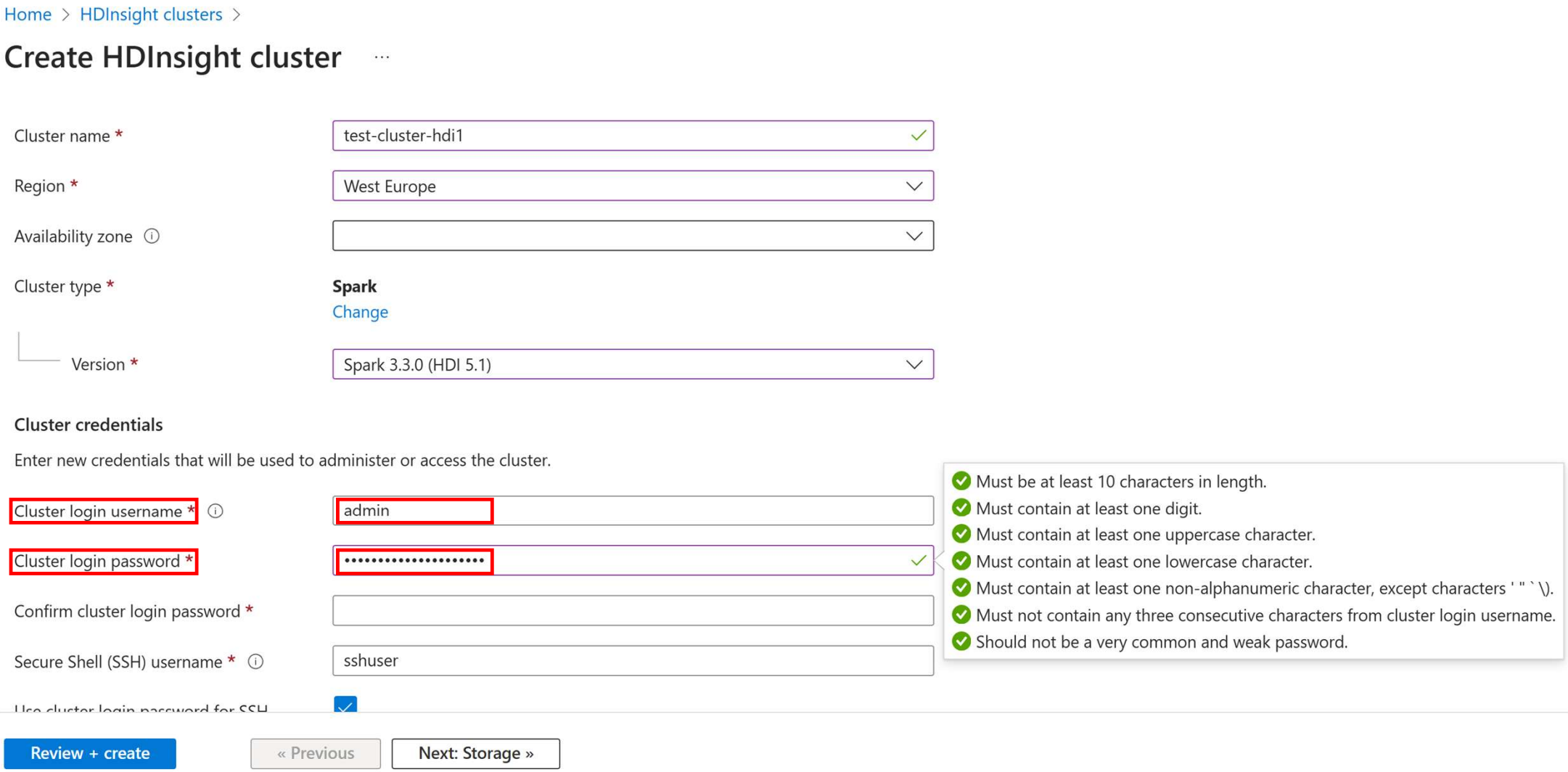




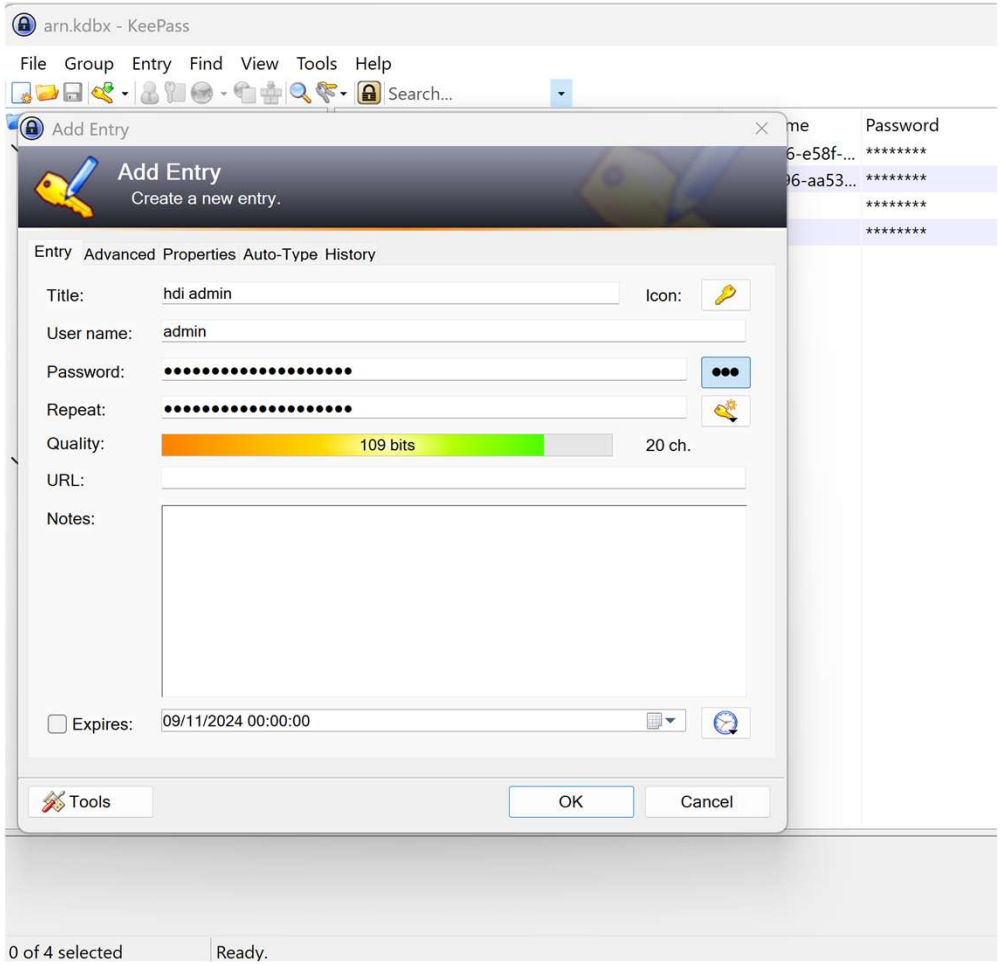
Create HDI - Choose Spark Version available 3.3.0 as of 2024 (but latest spark=3.5.3)



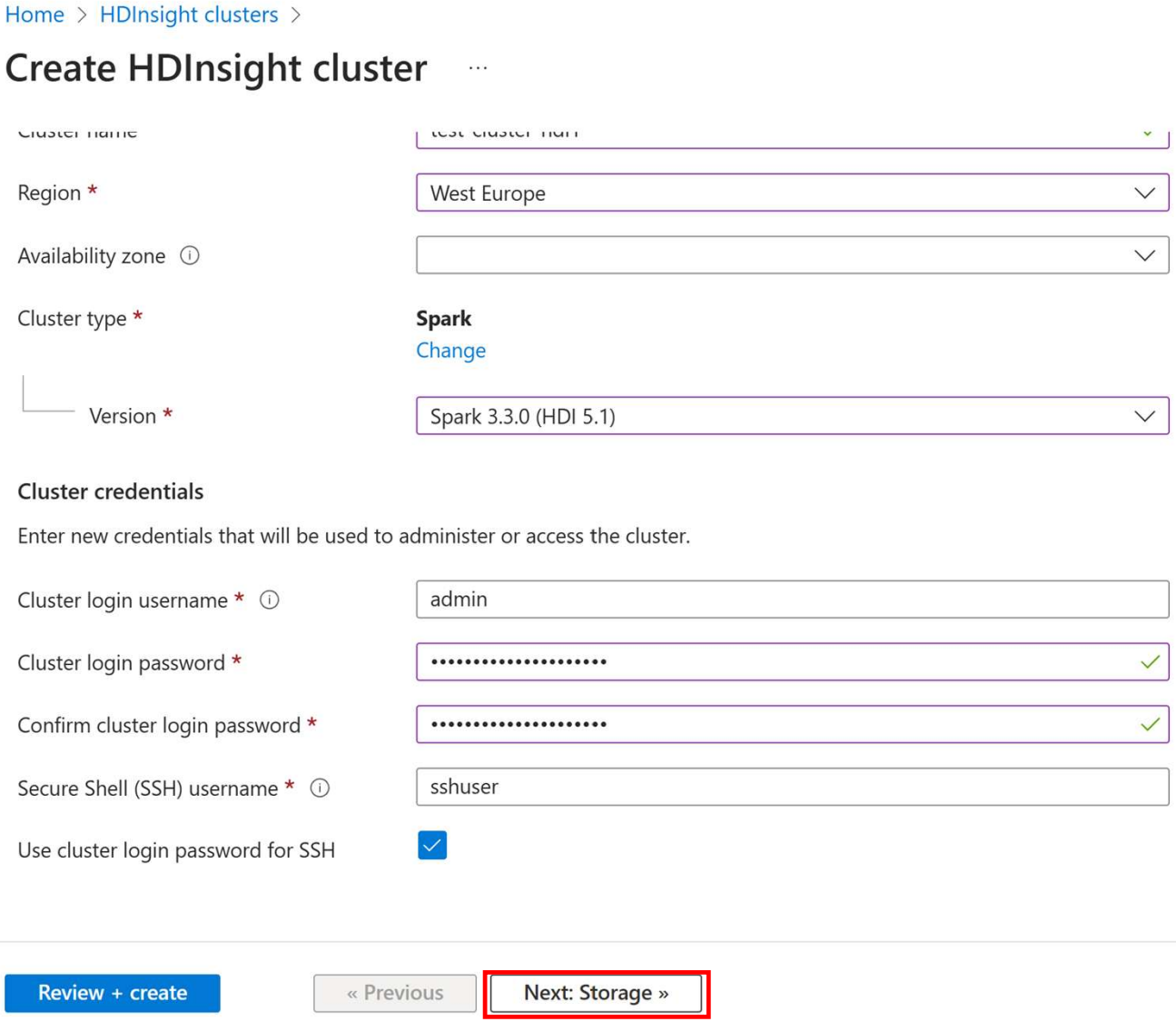
Create HDI - Admin Username + Password



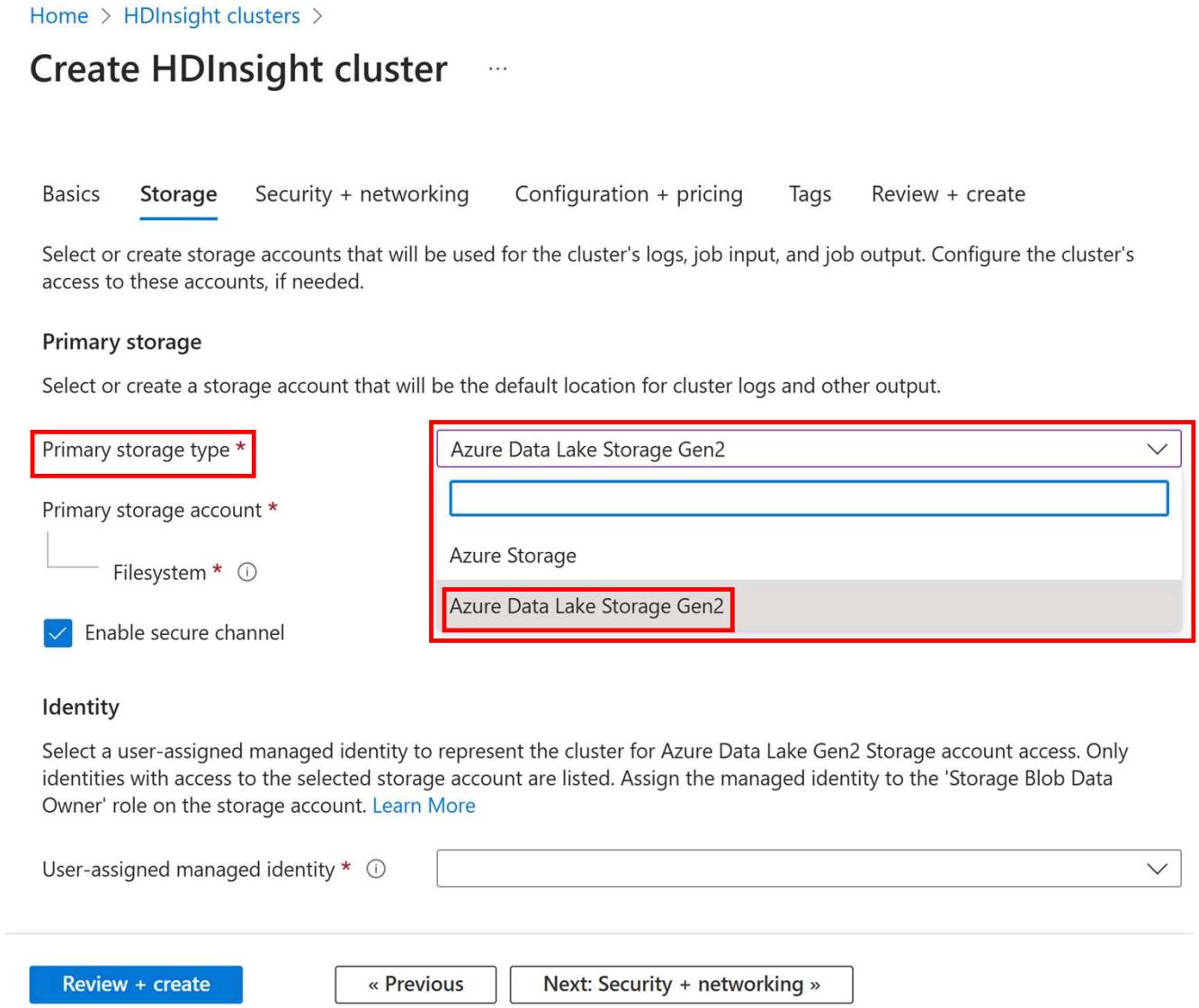
Always Generate your Random Passwords (and use KeePass)



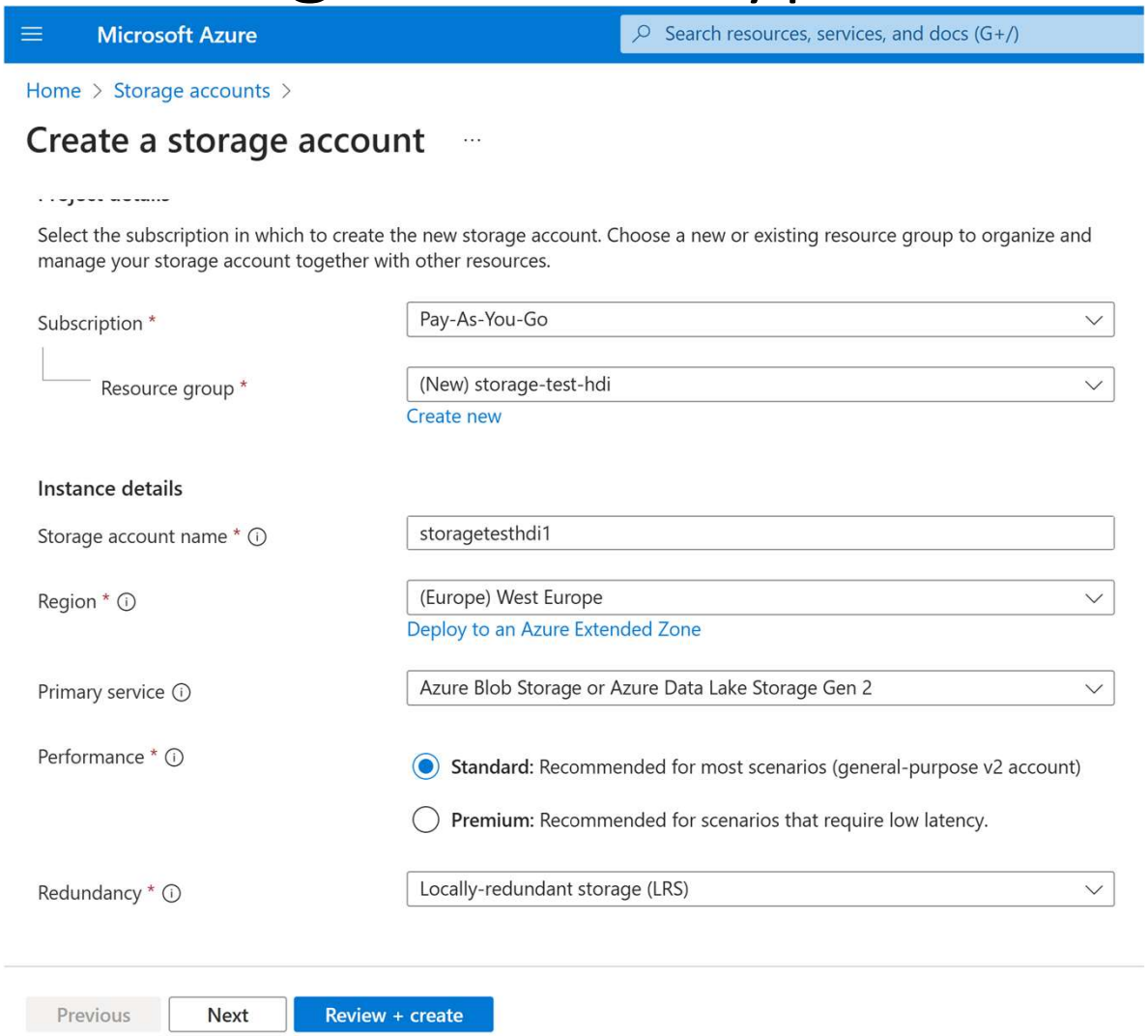
Create HDI - Next : Storage



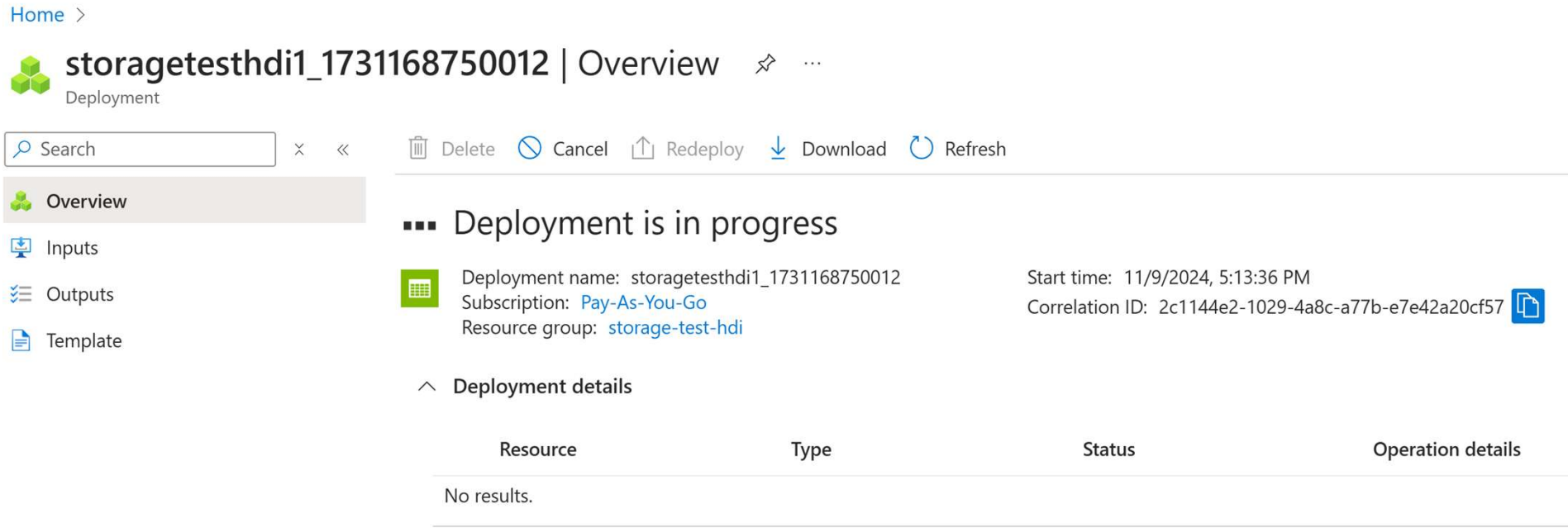
change default Storage Type "ADLS Gen 2"



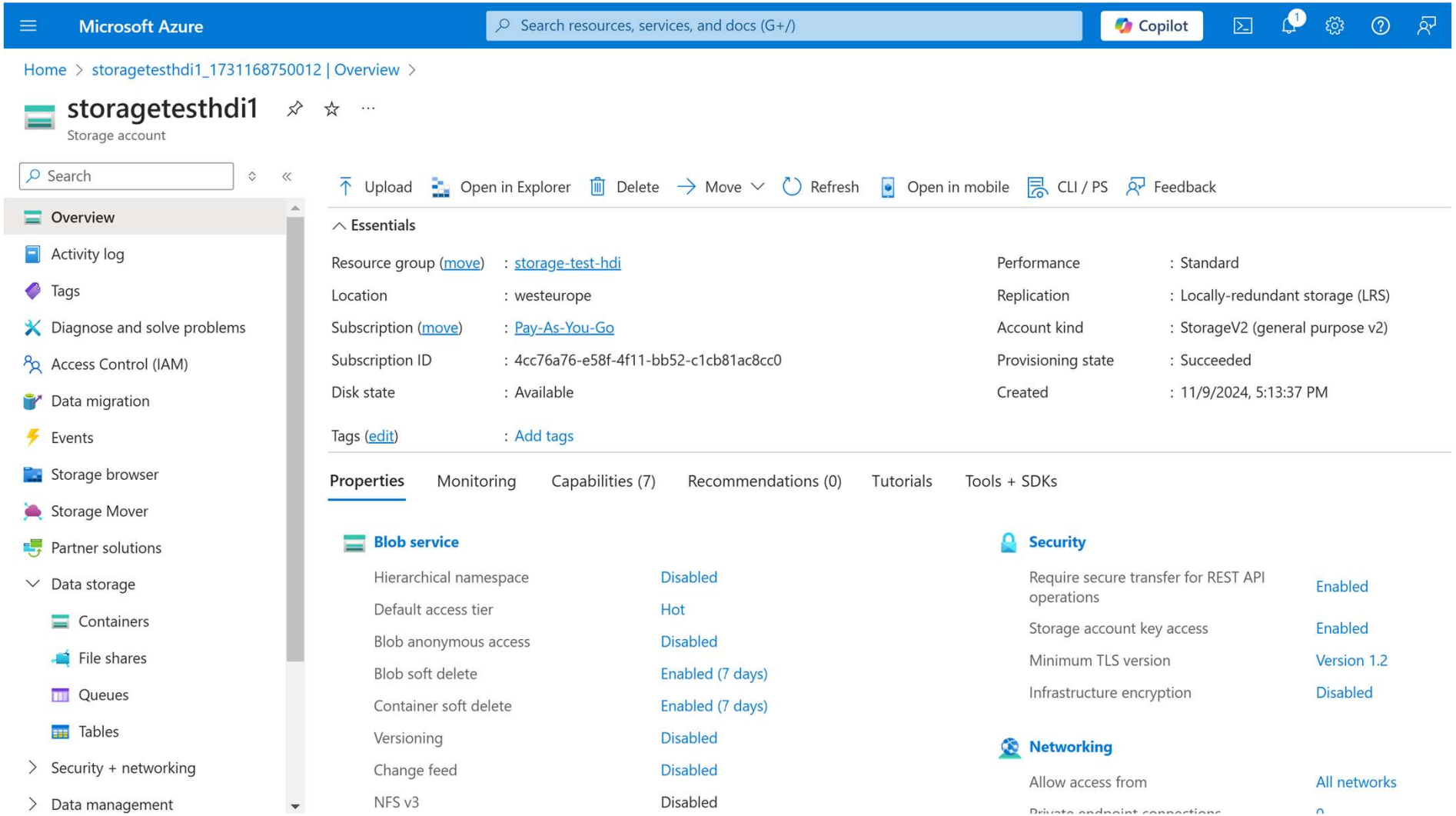
Create a Storage Account (in same Region, with type ADLSGen2)



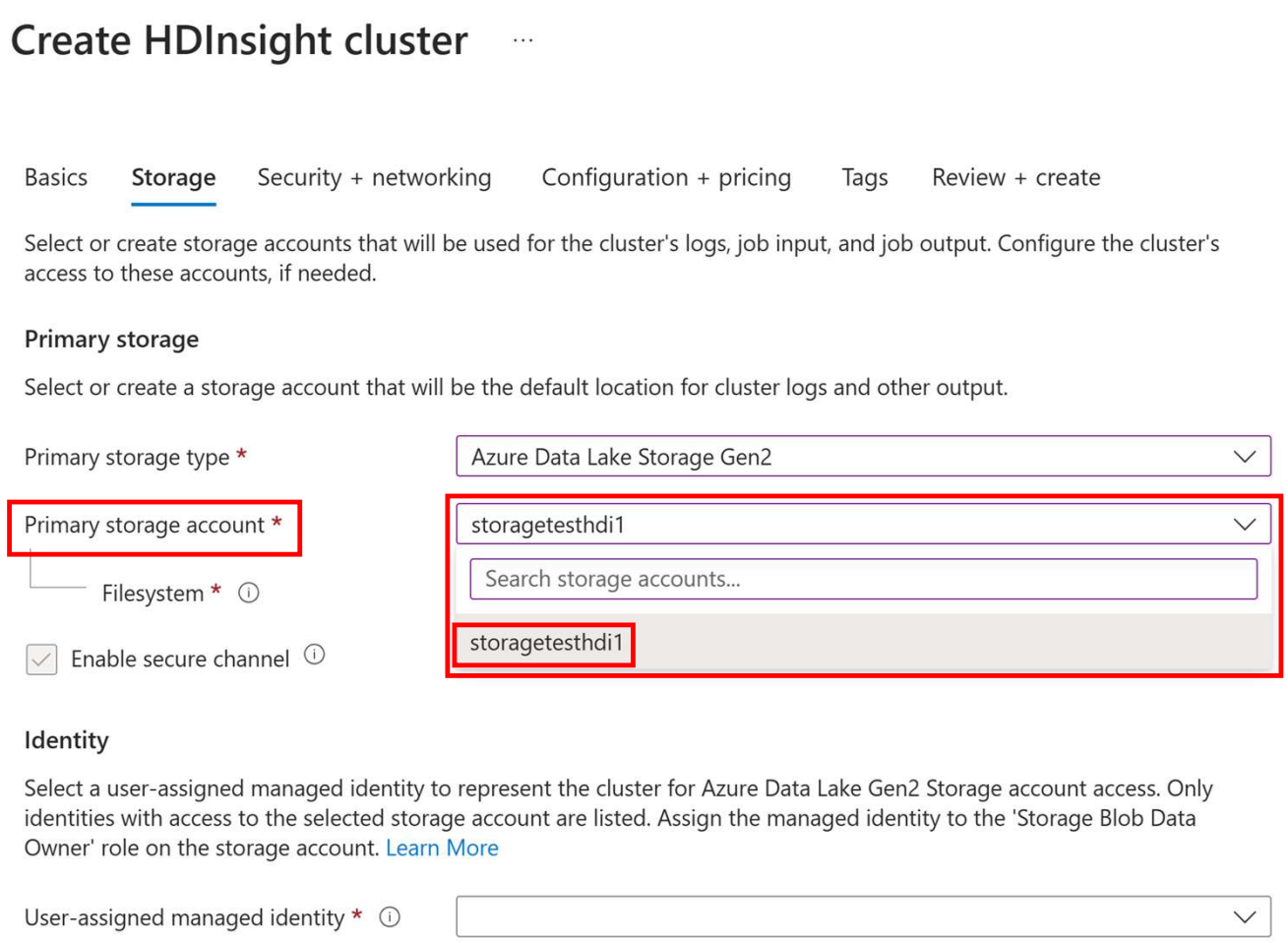
Creating Storage Account...



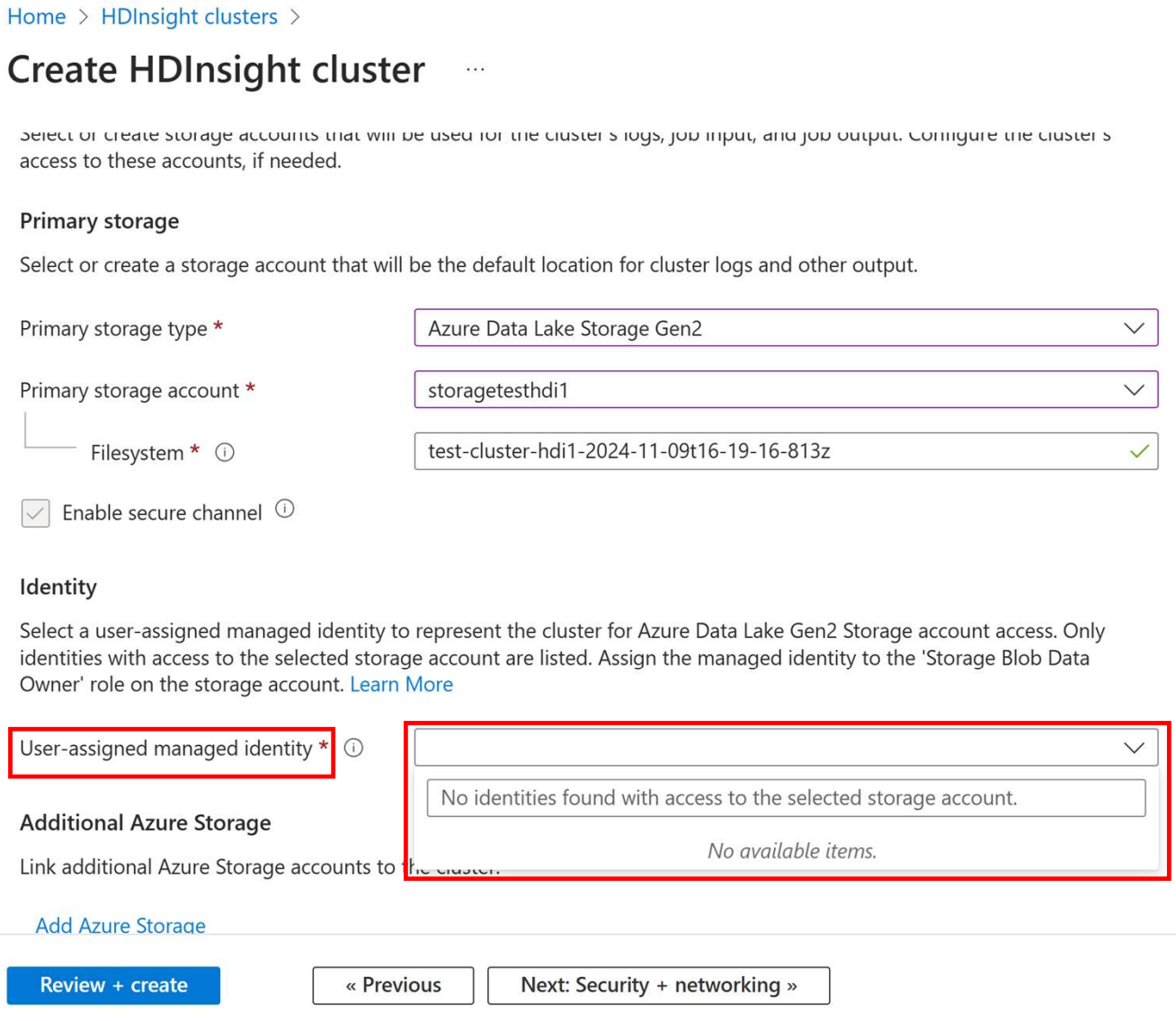
Created Storage Account



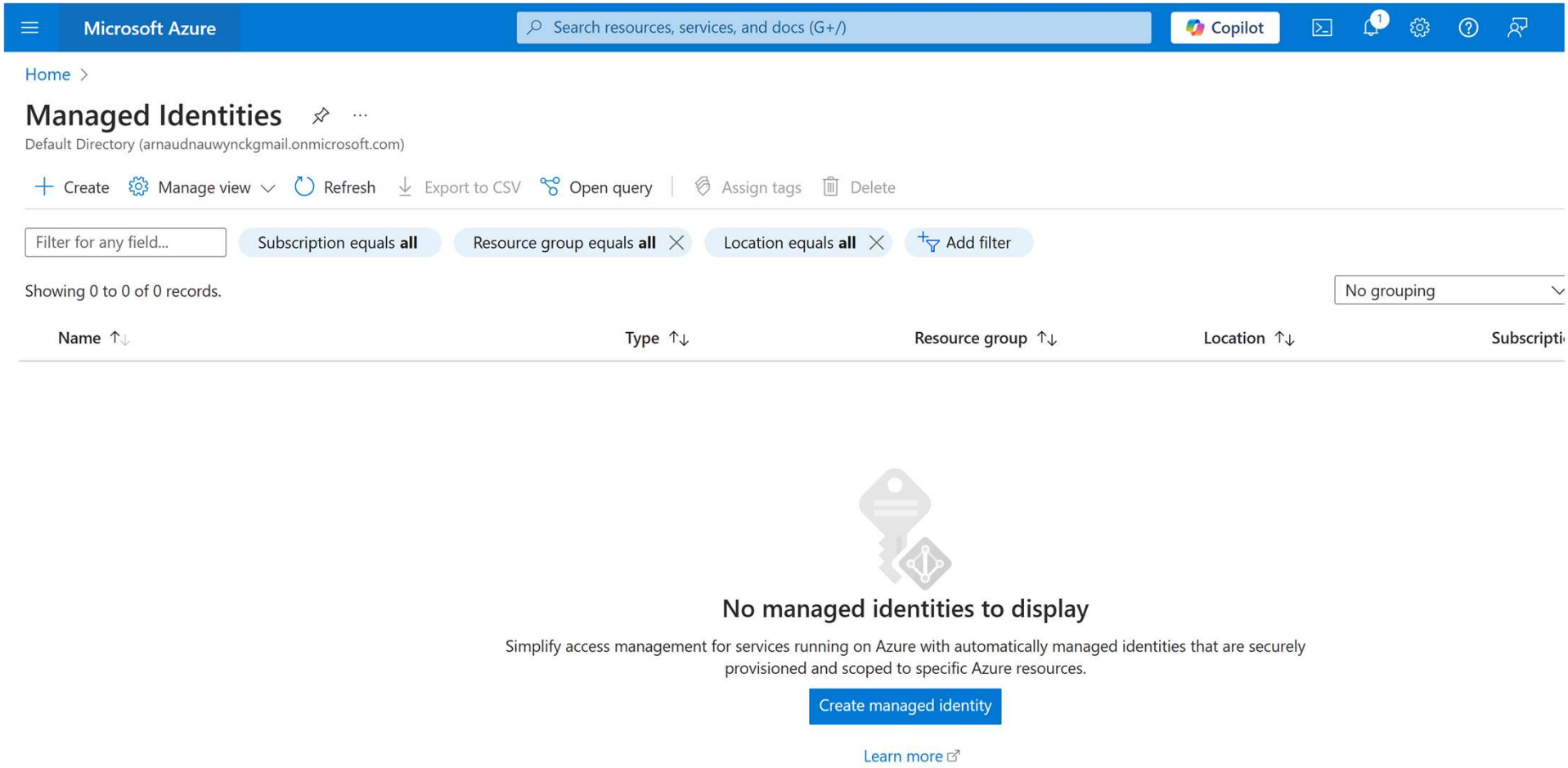
Create HDI - Select your Storage Account (bug Azure ... need close+reenter HDI Form)



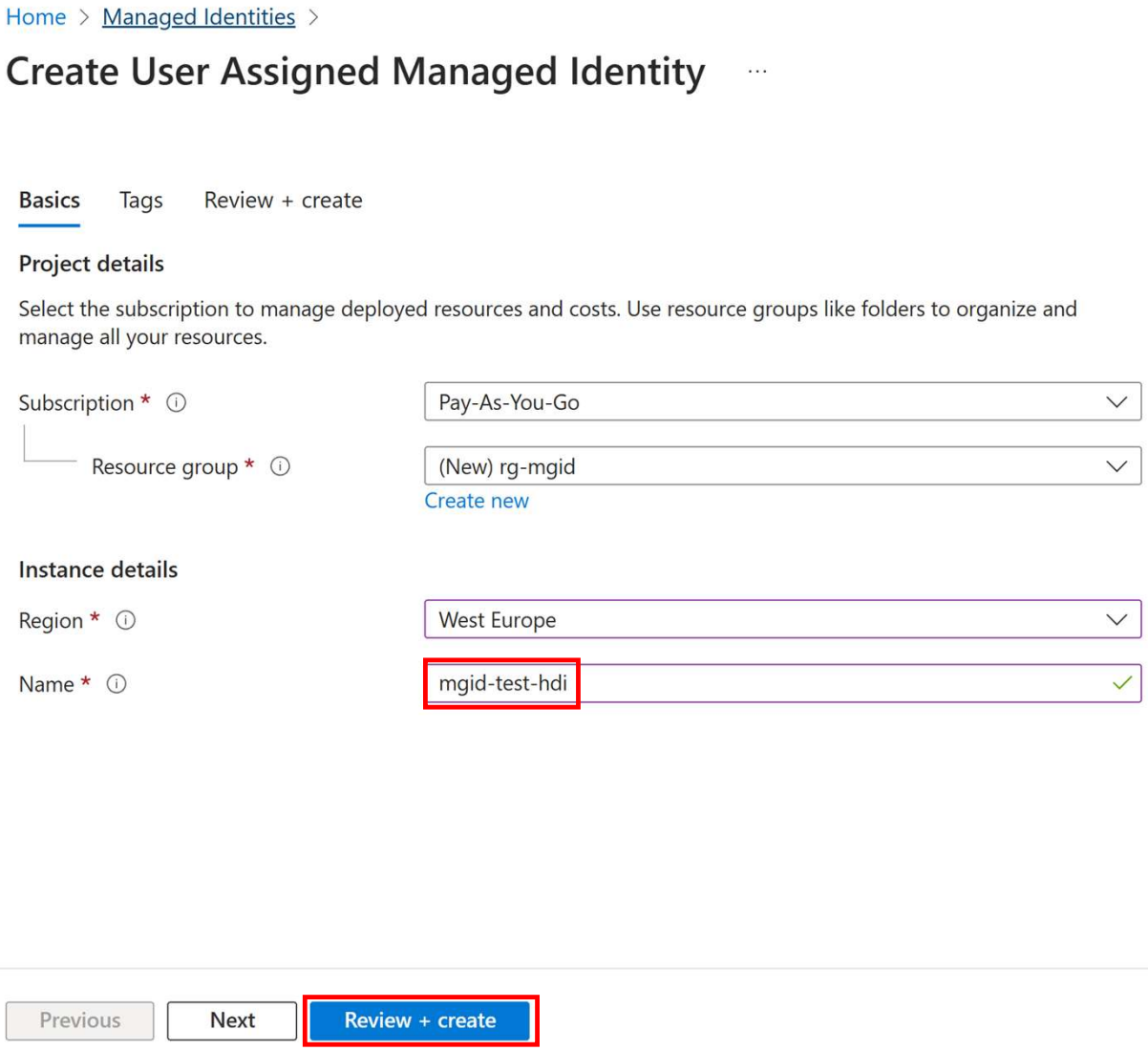
… Need a Managed Identity



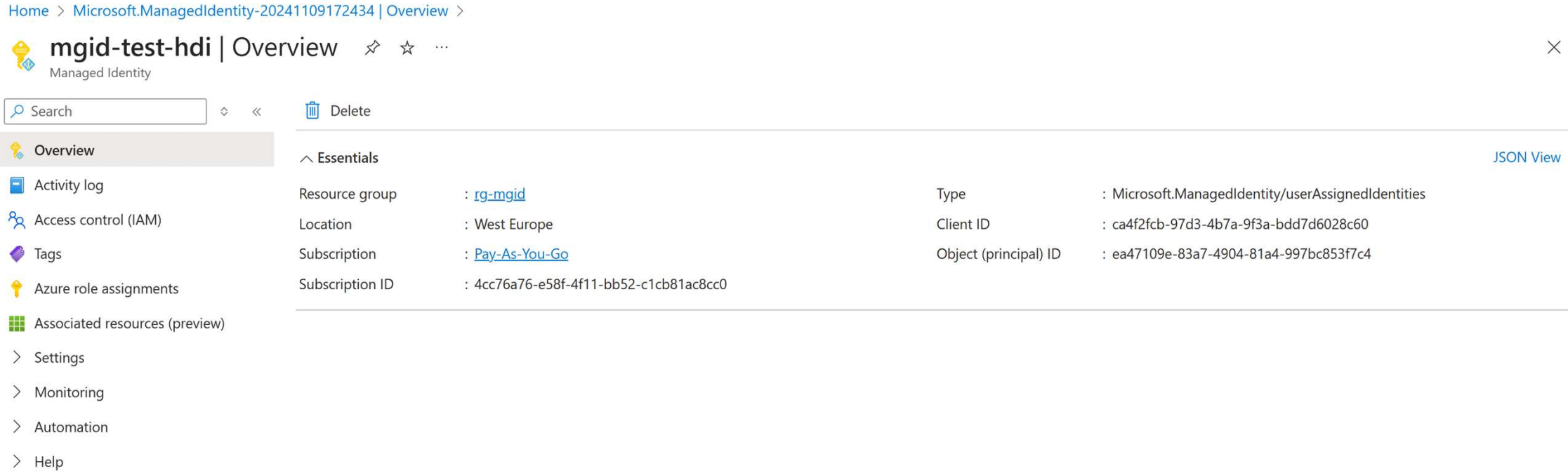
Managed Identity



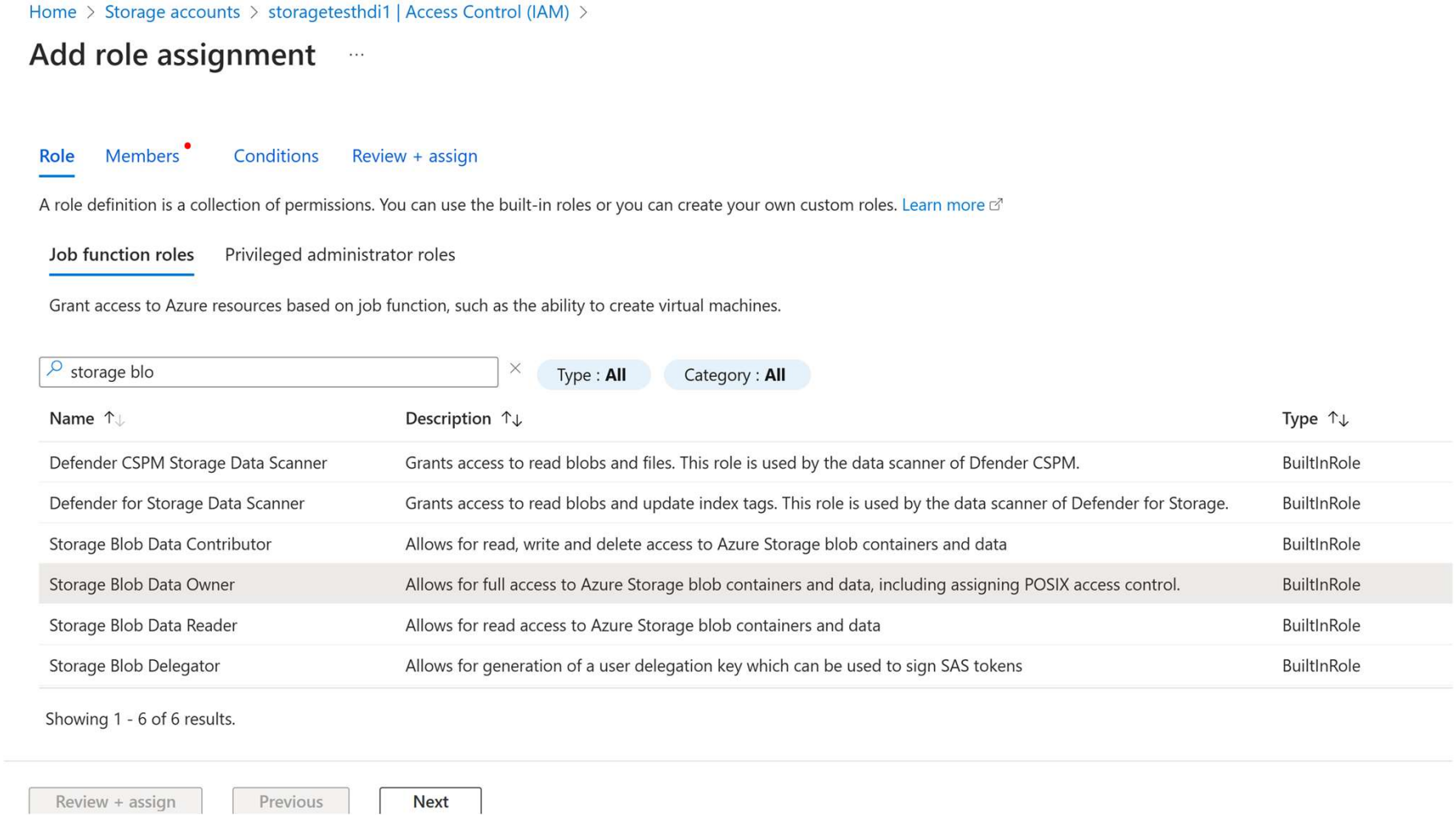
Create Managed Identity



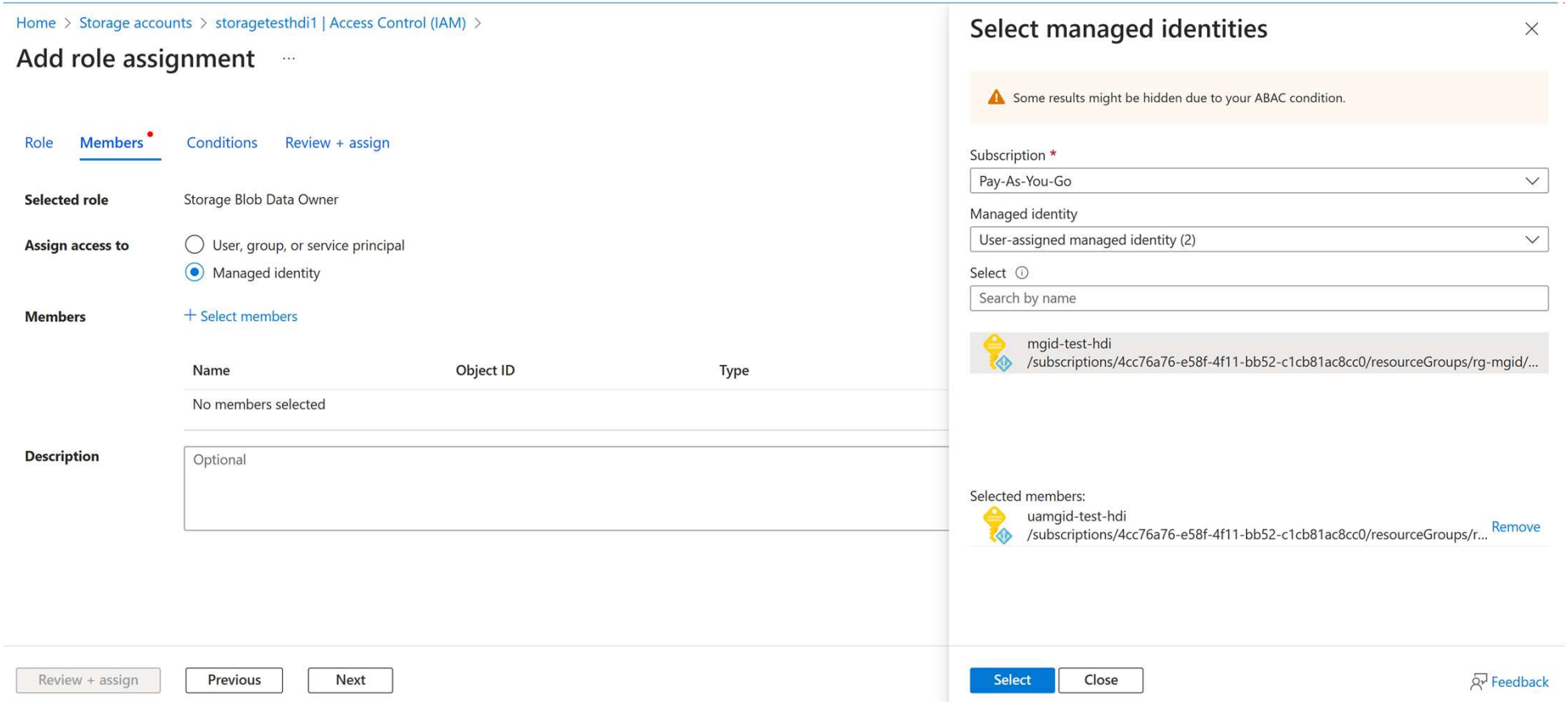
Managed Identity



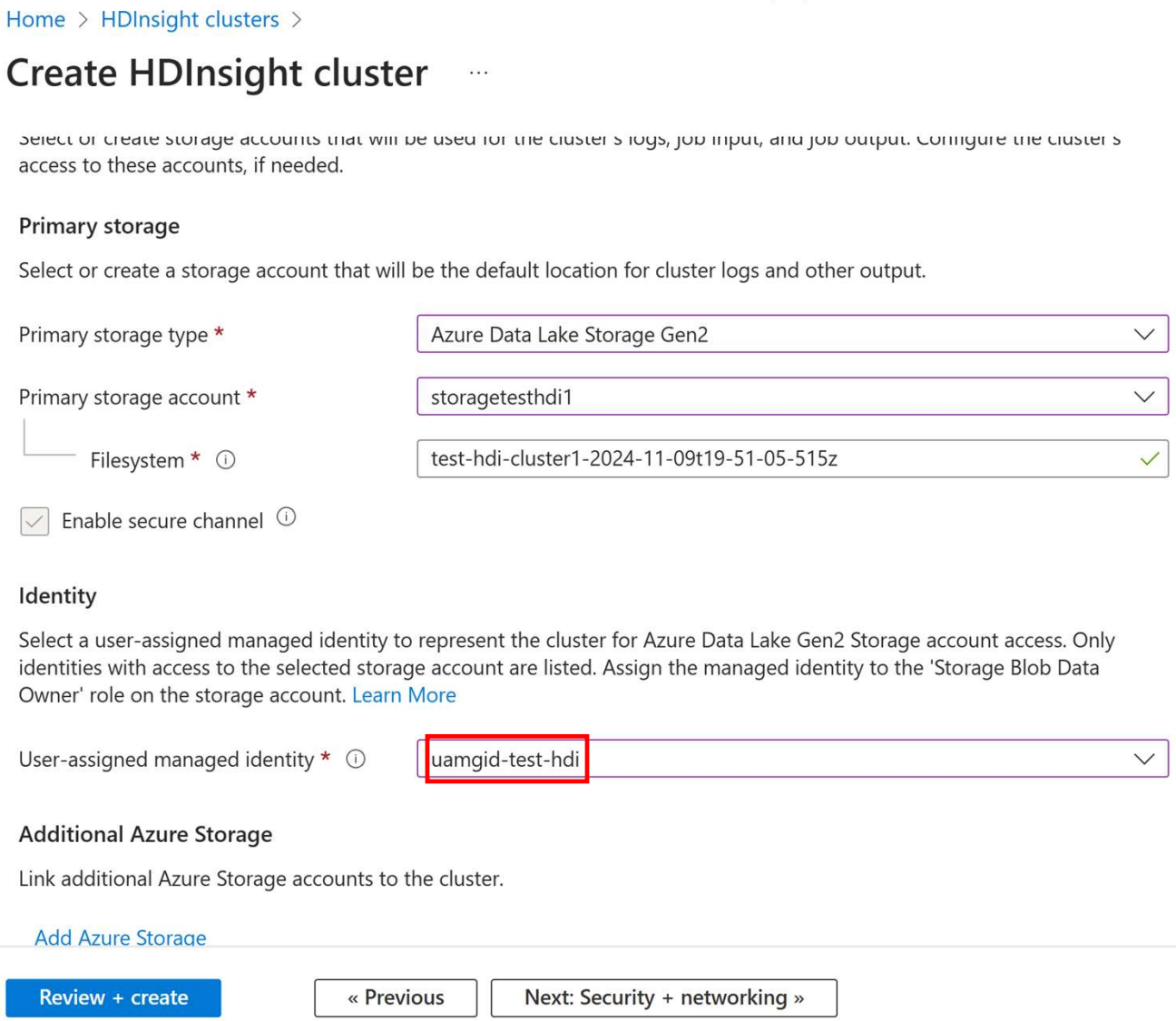
Adding "Storage Blob Data Owner" Role to Managed Identity



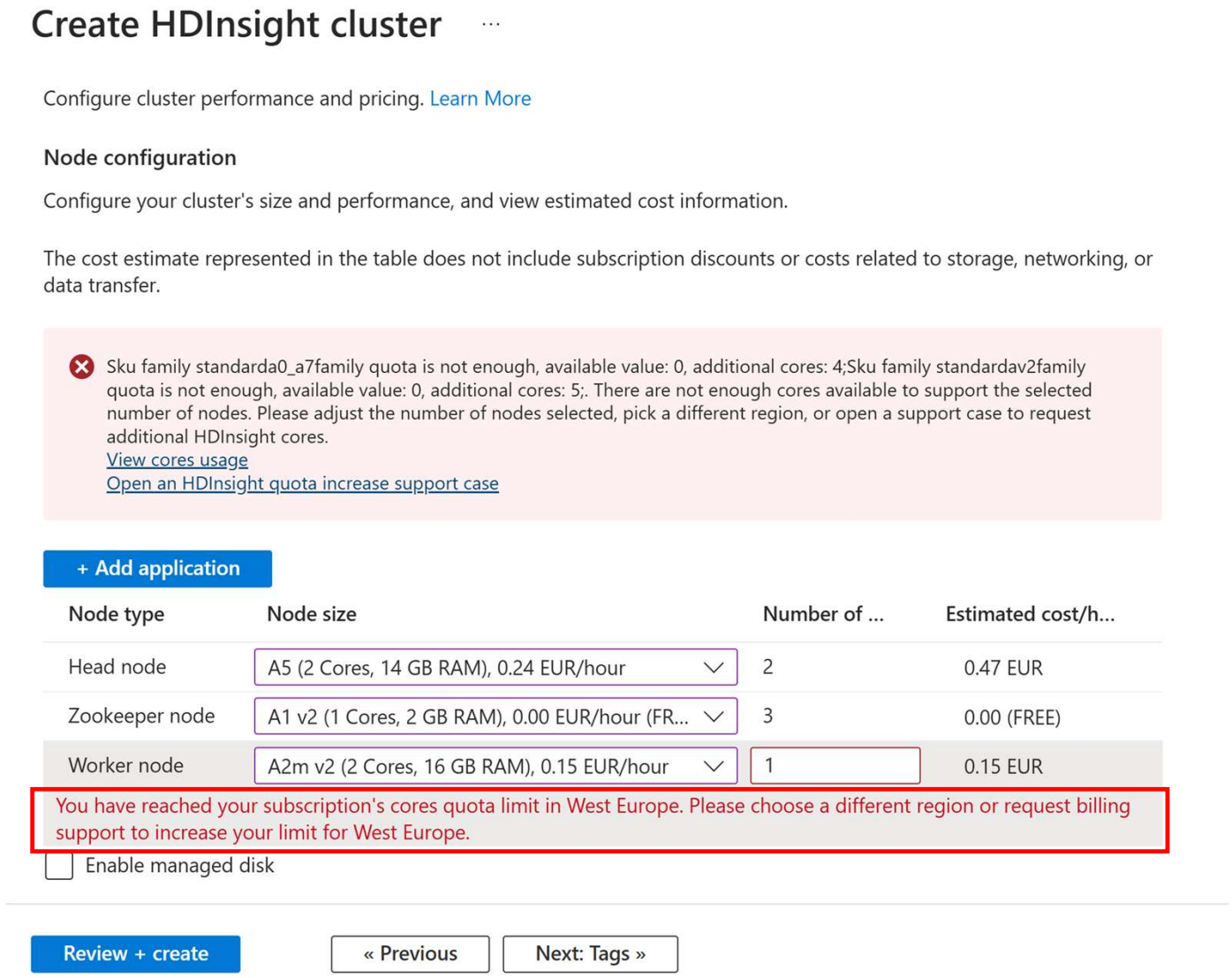
Adding Role..


Home > Storage accounts > storagetesthdi1 | Access Control (IAM) >
Add role assignment
Role
Members
Conditions
Review + assign
Selected role
Storage Blob Data Owner
Assign access to
User, group, or service principal
Managed identity
Members
+ Select members
Name
Object ID
Type
No members selected
Description
Optional
Review + assign
Previous
Next
Select managed identities
Some results might be hidden due to your ABAC condition.
Subscription *
Pay-As-You-Go
Managed identity
User-assigned managed identity (2)
Select
Search by name
mgid-test-hdi
/subscriptions/4cc76a76-e58f-4f11-bb52-c1cb81ac8cc0/resourceGroups/rg-mgid/...
Selected members:
uamgid-test-hdi
/subscriptions/4cc76a76-e58f-4f11-bb52-c1cb81ac8cc0/resourceGroups/r...
Remove
Select
Close
Feedback

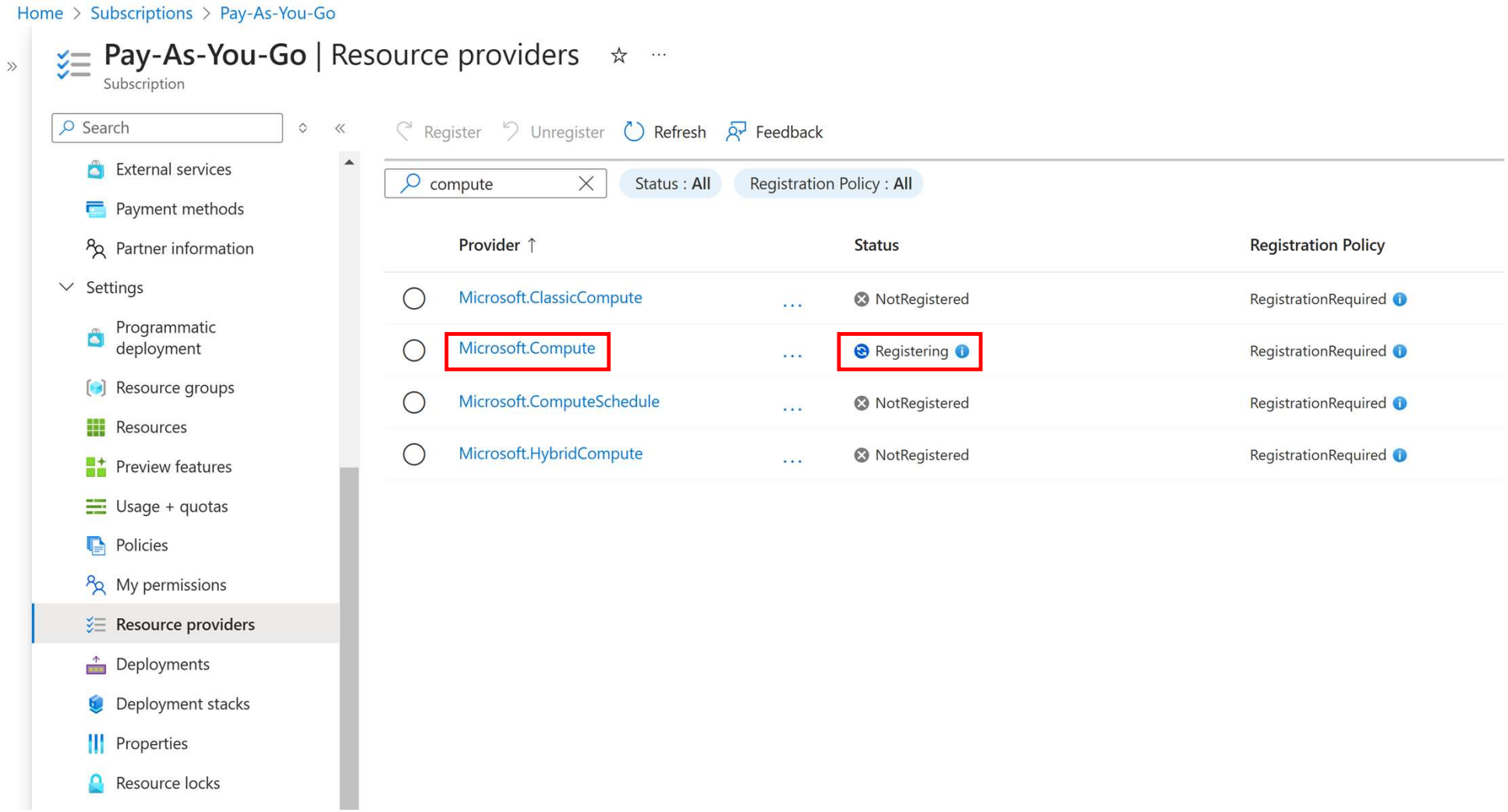
Create HDI - Managed Identity



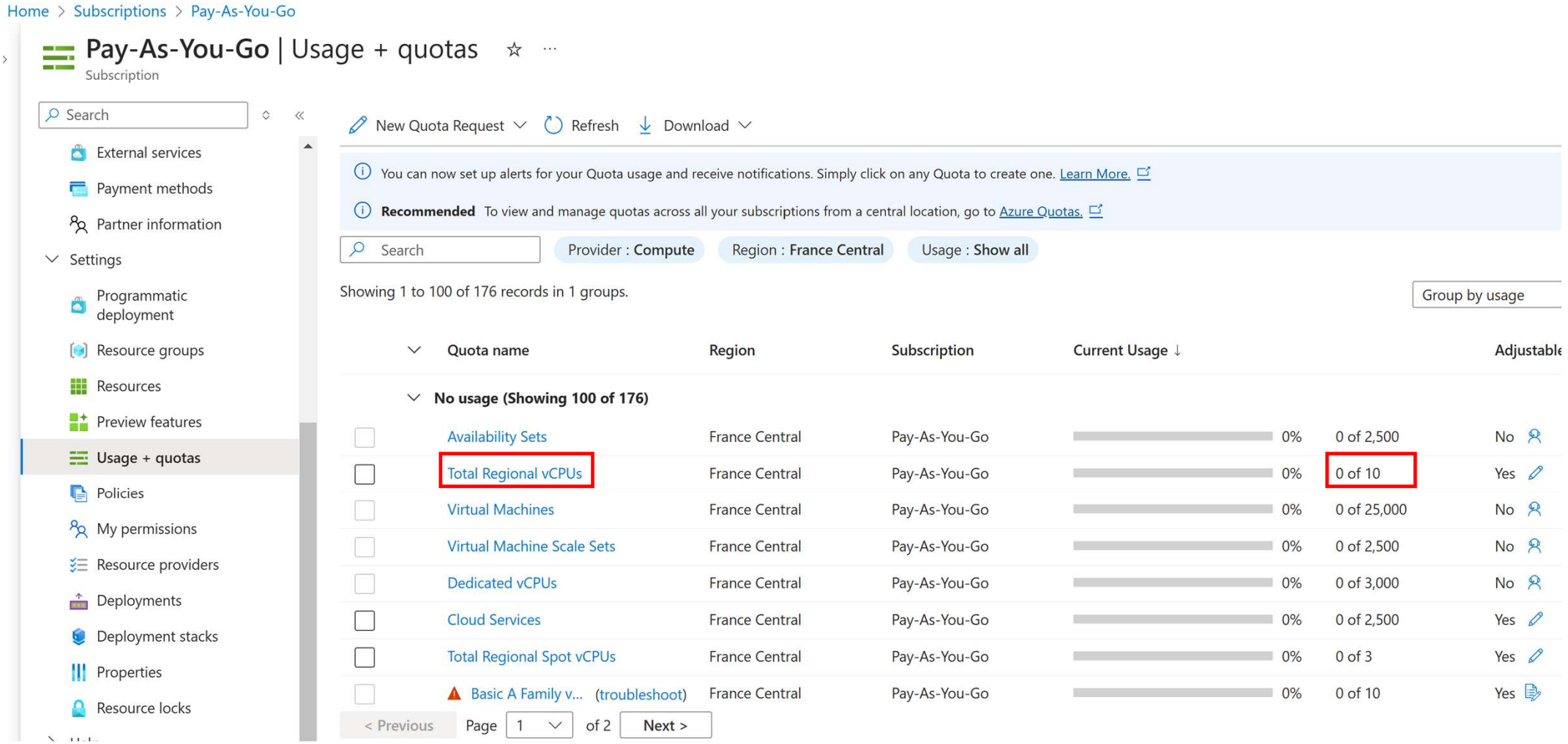
HDI.. Need Increase Subscription Cores Quota Limit



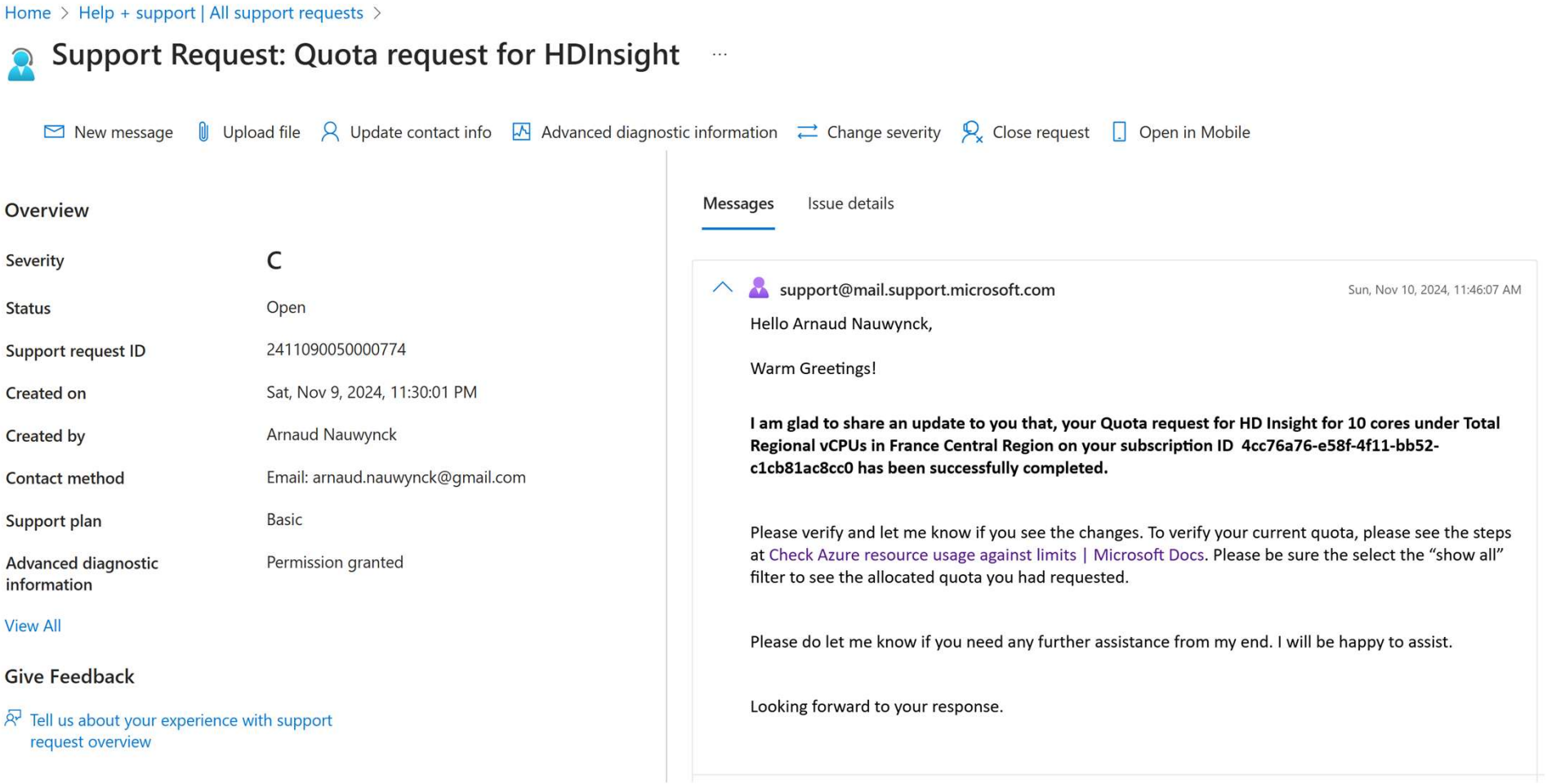
For Quota .. need Resource provider "Microsoft.Compute"



Quota vCPUs


Home > Subscriptions > Pay-As-You-Go
Pay-As-You-Go | Usage + quotas
Subscription
Search
External services
Payment methods
Partner information
Settings
Programmatic deployment
Resource groups
Resources
Preview features
Usage + quotas
Policies
My permissions
Resource providers
Deployments
Deployment stacks
Properties
Resource locks
New Quota Request
Refresh
Download
You can now set up alerts for your Quota usage and receive notifications. Simply click on any Quota to create one. Learn More.
Recommended To view and manage quotas across all your subscriptions from a central location, go to Azure Quotas.
Search
Provider : Compute
Region : France Central
Usage : Show all
Showing 1 to 100 of 176 records in 1 groups.
Group by usage
Quota name
Region
Subscription
Current Usage
Adjustable
No usage (Showing 100 of 176)
Availability Sets France Central Pay-As-You-Go 0% 0 of 2,500 No
Total Regional vCPUs France Central Pay-As-You-Go 0% 0 of 10 Yes
Virtual Machines France Central Pay-As-You-Go 0% 0 of 25,000 No
Virtual Machine Scale Sets France Central Pay-As-You-Go 0% 0 of 2,500 No
Dedicated vCPUs France Central Pay-As-You-Go 0% 0 of 3,000 No
Cloud Services France Central Pay-As-You-Go 0% 0 of 2,500 Yes
Total Regional Spot vCPUs France Central Pay-As-You-Go 0% 0 of 3 Yes
Basic A Family v... (troubleshoot) France Central Pay-As-You-Go 0% 0 of 10 Yes
< Previous
Page 1 of 2
Next >

Support Request Answer: Quota Increased



Home > Help + support | All support requests >

Support Request: Quota request for HDInsight ...

New message | Upload file | Update contact info | Advanced diagnostic information | Change severity | Close request | Open in Mobile

Overview

Severity	C
Status	Open
Support request ID	2411090050000774
Created on	Sat, Nov 9, 2024, 11:30:01 PM
Created by	Arnaud Nauwynck
Contact method	Email: arnaud.nauwynck@gmail.com
Support plan	Basic
Advanced diagnostic information	Permission granted

View All

Give Feedback

Tell us about your experience with support request overview

Messages | Issue details

support@mail.support.microsoft.com Sun, Nov 10, 2024, 11:46:07 AM

Hello Arnaud Nauwynck,

Warm Greetings!

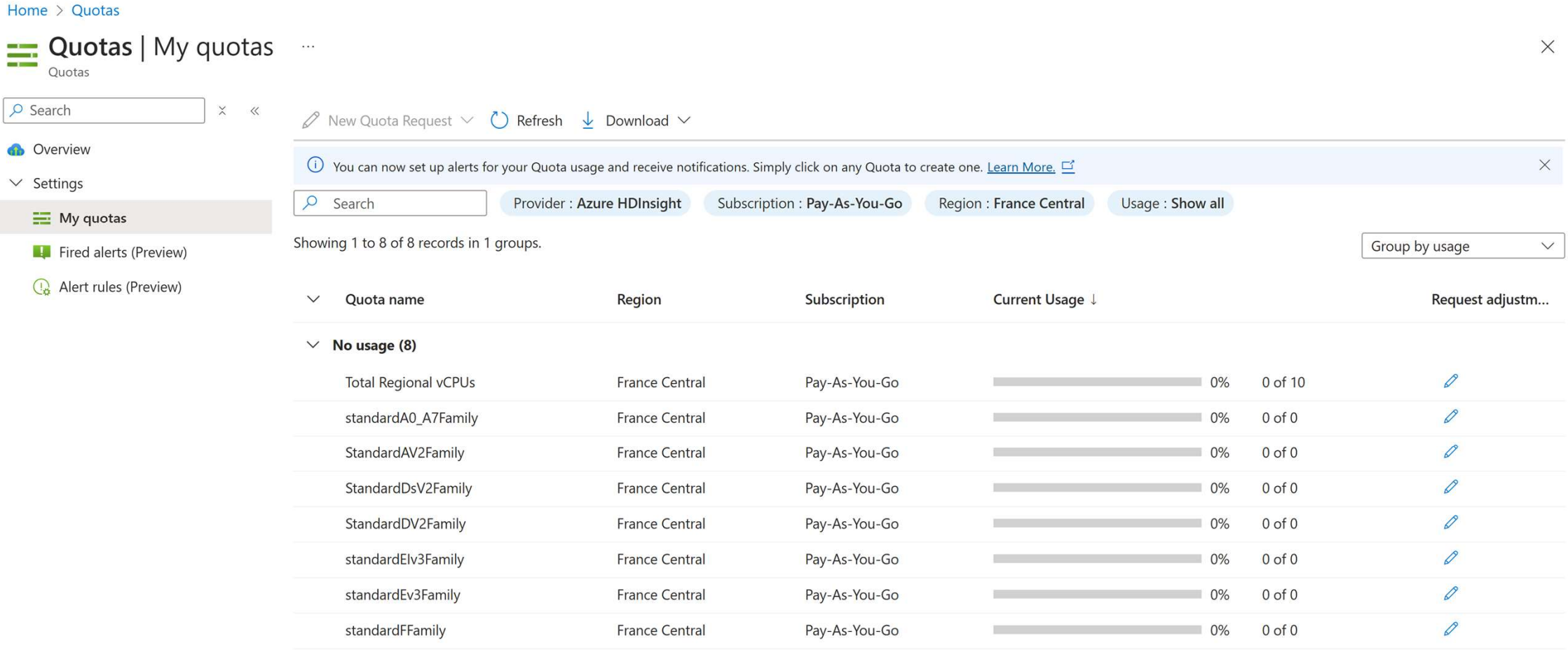
I am glad to share an update to you that, your Quota request for HD Insight for 10 cores under Total Regional vCPUs in France Central Region on your subscription ID 4cc76a76-e58f-4f11-bb52-c1cb81ac8cc0 has been successfully completed.

Please verify and let me know if you see the changes. To verify your current quota, please see the steps at Check Azure resource usage against limits | Microsoft Docs. Please be sure the select the "show all" filter to see the allocated quota you had requested.

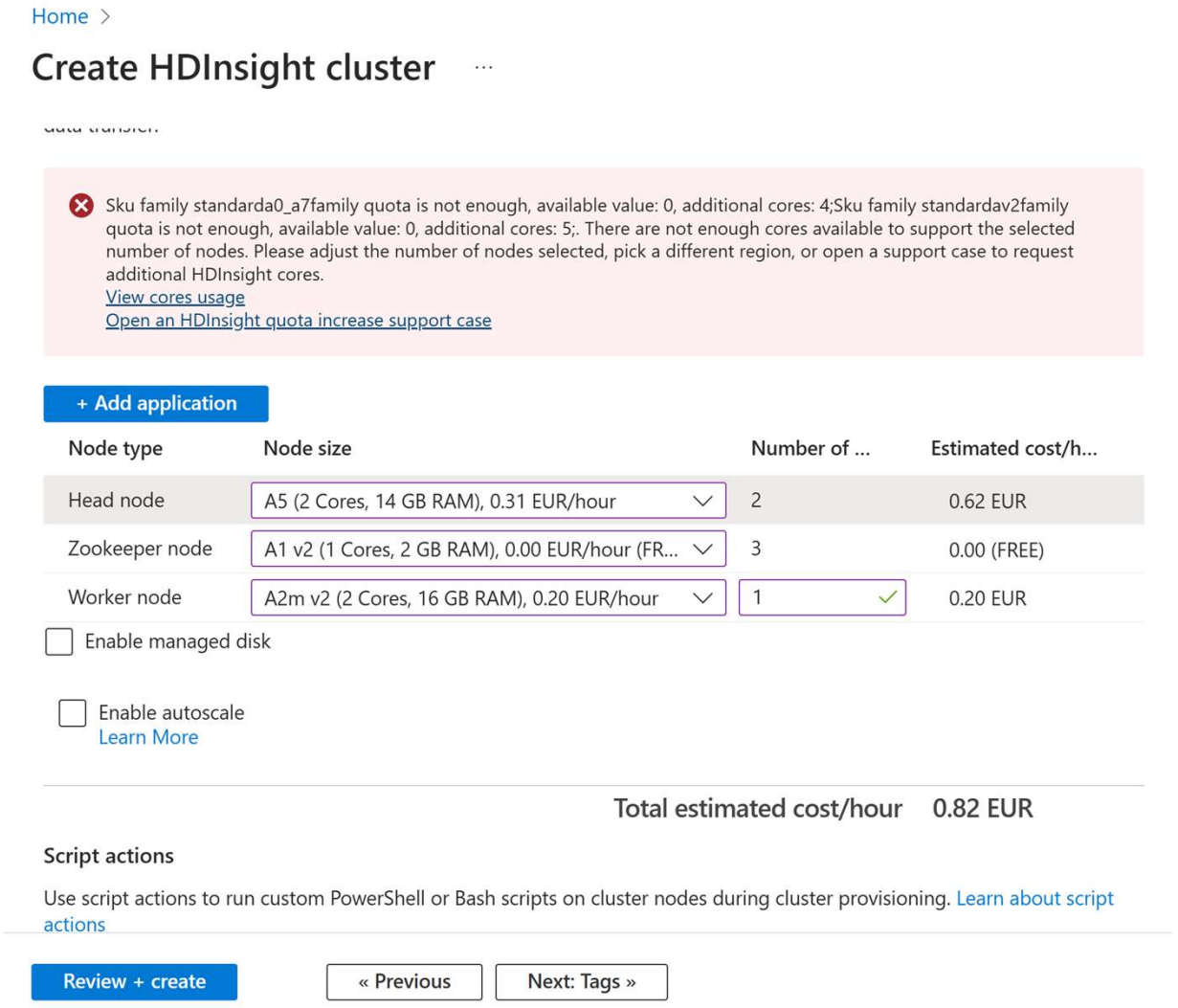
Please do let me know if you need any further assistance from my end. I will be happy to assist.

Looking forward to your response.

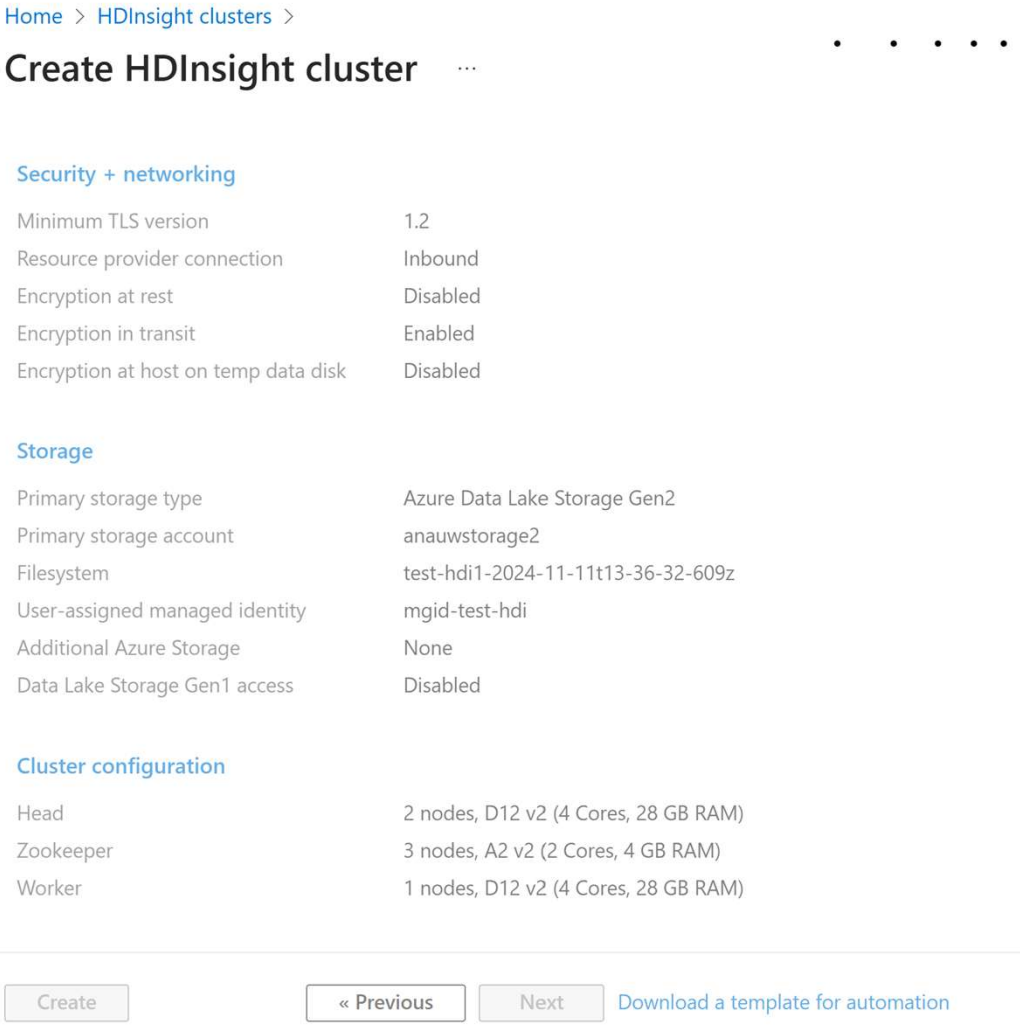
Quota ...


Home > Quotas
Quotas | My quotas
Quotas
Search
Overview
Settings
My quotas
Fired alerts (Preview)
Alert rules (Preview)
New Quota Request
Refresh
Download
You can now set up alerts for your Quota usage and receive notifications. Simply click on any Quota to create one. Learn More.
Search
Provider : Azure HDInsight
Subscription : Pay-As-You-Go
Region : France Central
Usage : Show all
Showing 1 to 8 of 8 records in 1 groups.
Group by usage
Quota name
Region
Subscription
Current Usage
Request adjustm...
No usage (8)
Total Regional vCPUs France Central Pay-As-You-Go 0% 0 of 10
standardA0_A7Family France Central Pay-As-You-Go 0% 0 of 0
StandardAV2Family France Central Pay-As-You-Go 0% 0 of 0
StandardDsV2Family France Central Pay-As-You-Go 0% 0 of 0
StandardDV2Family France Central Pay-As-You-Go 0% 0 of 0
standardEIv3Family France Central Pay-As-You-Go 0% 0 of 0
standardEv3Family France Central Pay-As-You-Go 0% 0 of 0
standardFFamily France Central Pay-As-You-Go 0% 0 of 0

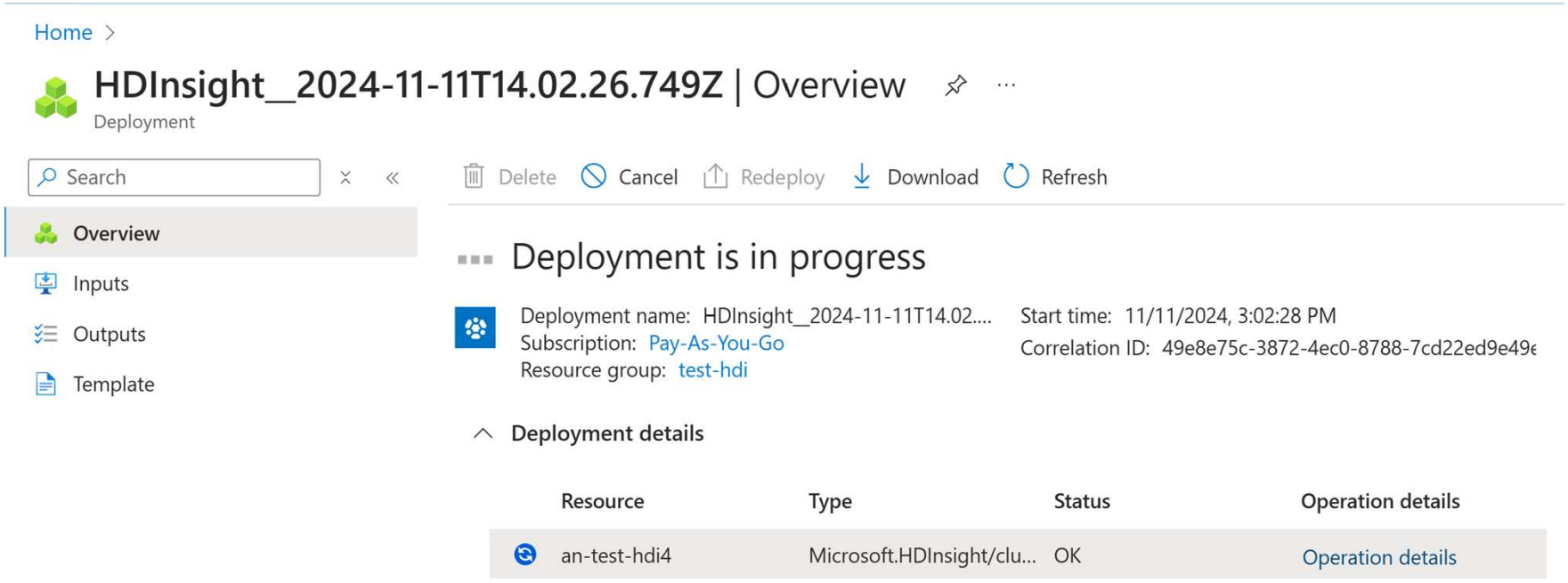
Create Cluster: 2xA5 + 3xA1+ 1xA2



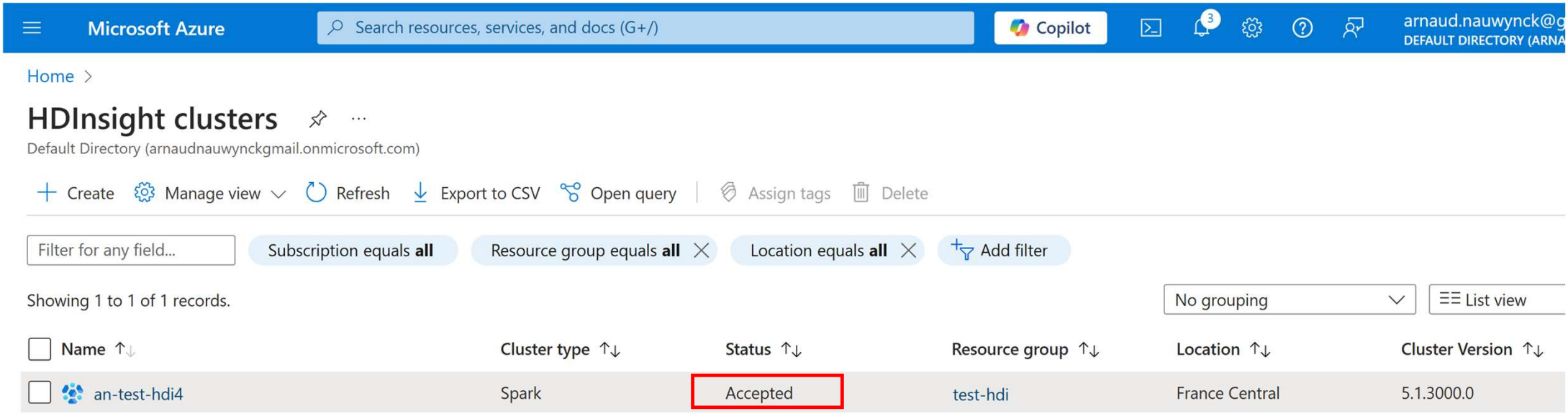
Create HDI ...


Home > HDInsight clusters >
Create HDInsight cluster ...
Security + networking
Minimum TLS version 1.2
Resource provider connection Inbound
Encryption at rest Disabled
Encryption in transit Enabled
Encryption at host on temp data disk Disabled
Storage
Primary storage type Azure Data Lake Storage Gen2
Primary storage account anauwstorage2
Filesystem test-hdi1-2024-11-11t13-36-32-609z
User-assigned managed identity mgid-test-hdi
Additional Azure Storage None
Data Lake Storage Gen1 access Disabled
Cluster configuration
Head 2 nodes, D12 v2 (4 Cores, 28 GB RAM)
Zookeeper 3 nodes, A2 v2 (2 Cores, 4 GB RAM)
Worker 1 nodes, D12 v2 (4 Cores, 28 GB RAM)
Create
« Previous
Next
Download a template for automation

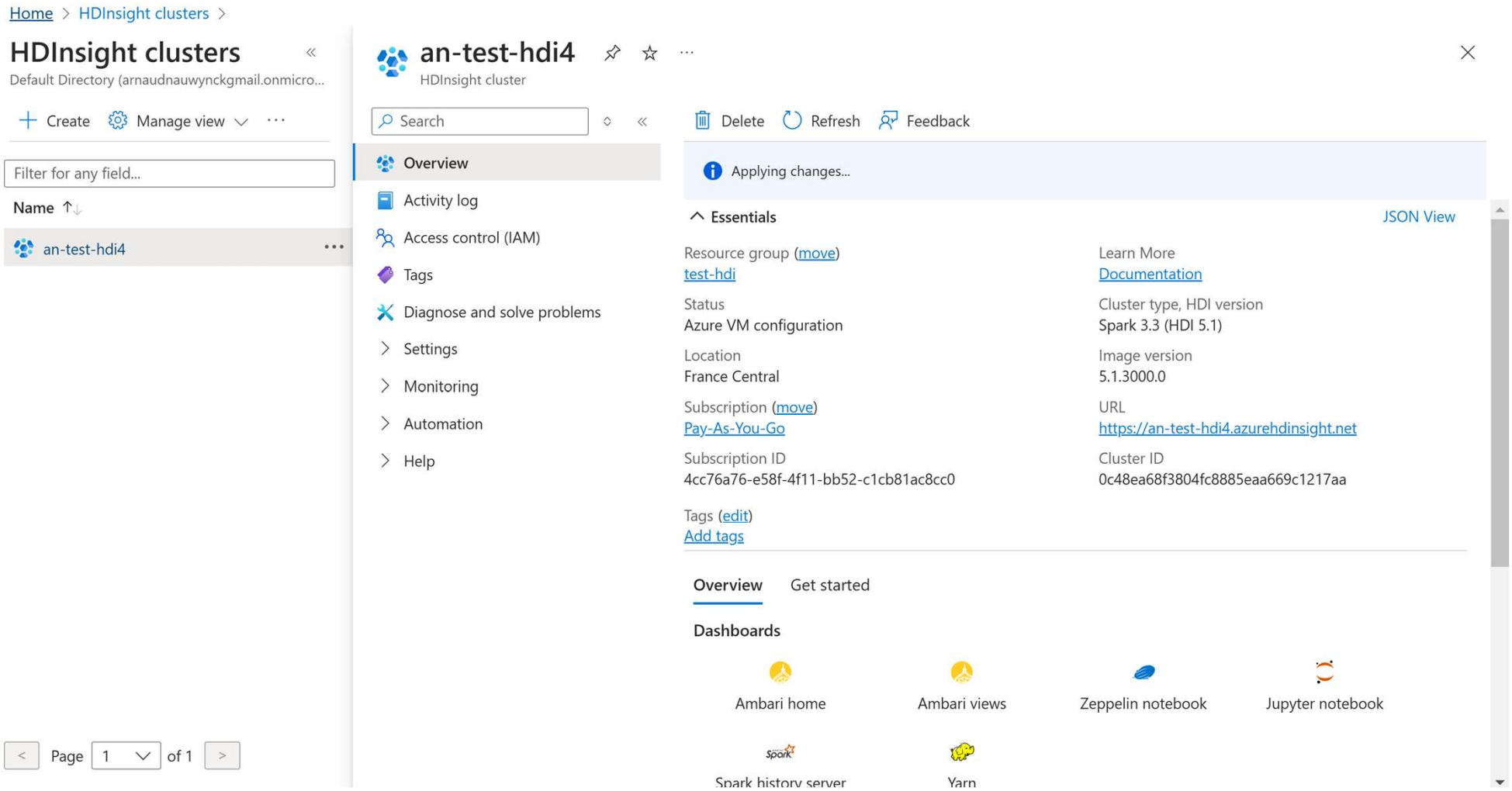
"Deployment" object, in progress



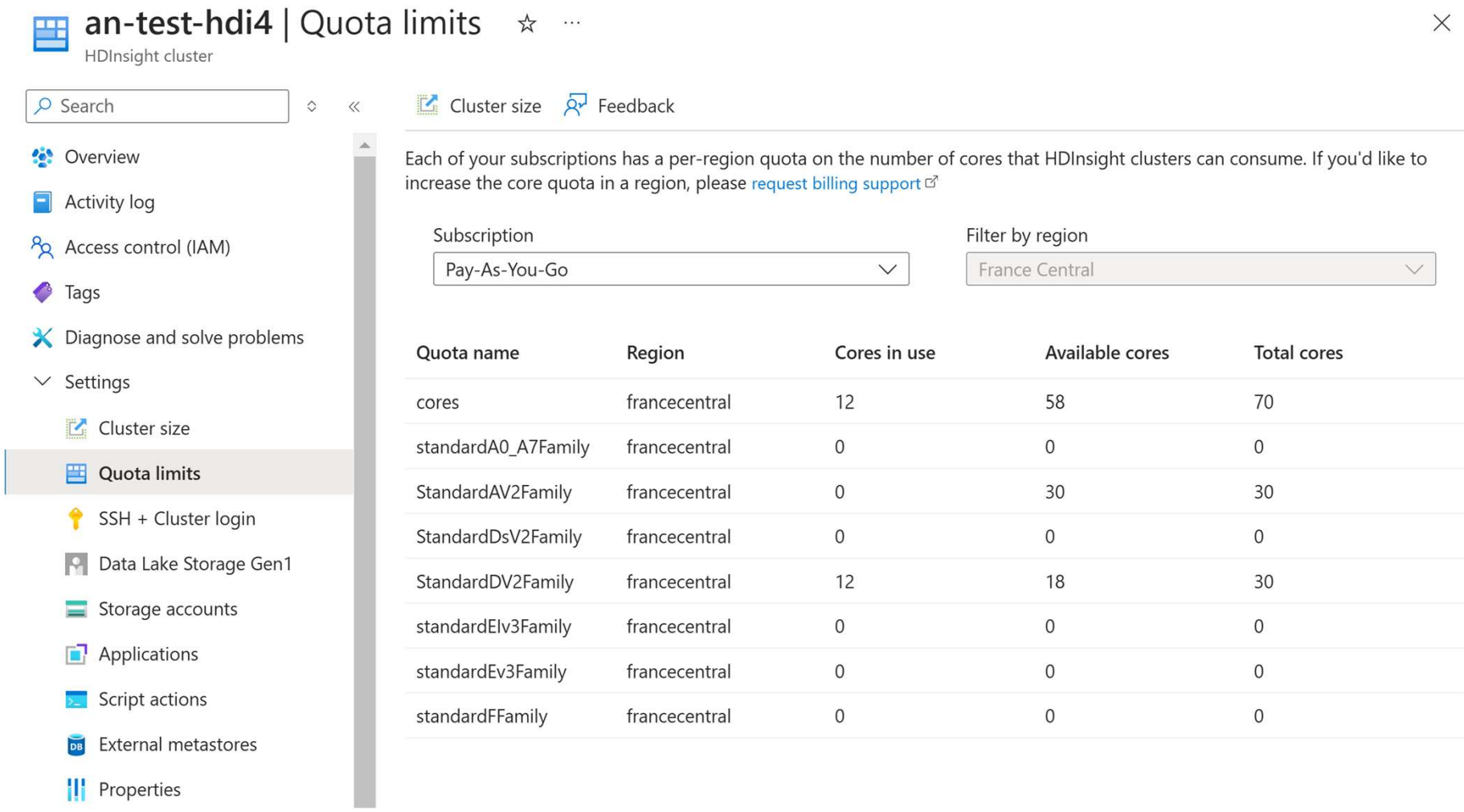
HDInsight Resource, Status=Accepted (!= Running)



HDInsight - Resource Overview

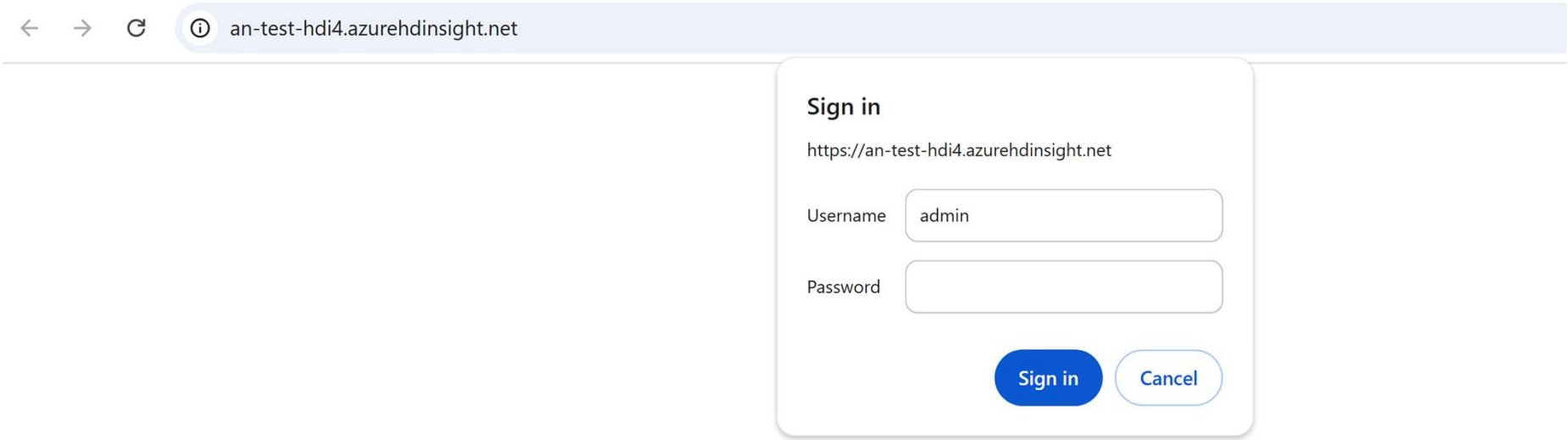


HDInsight - Quota "Core in use"

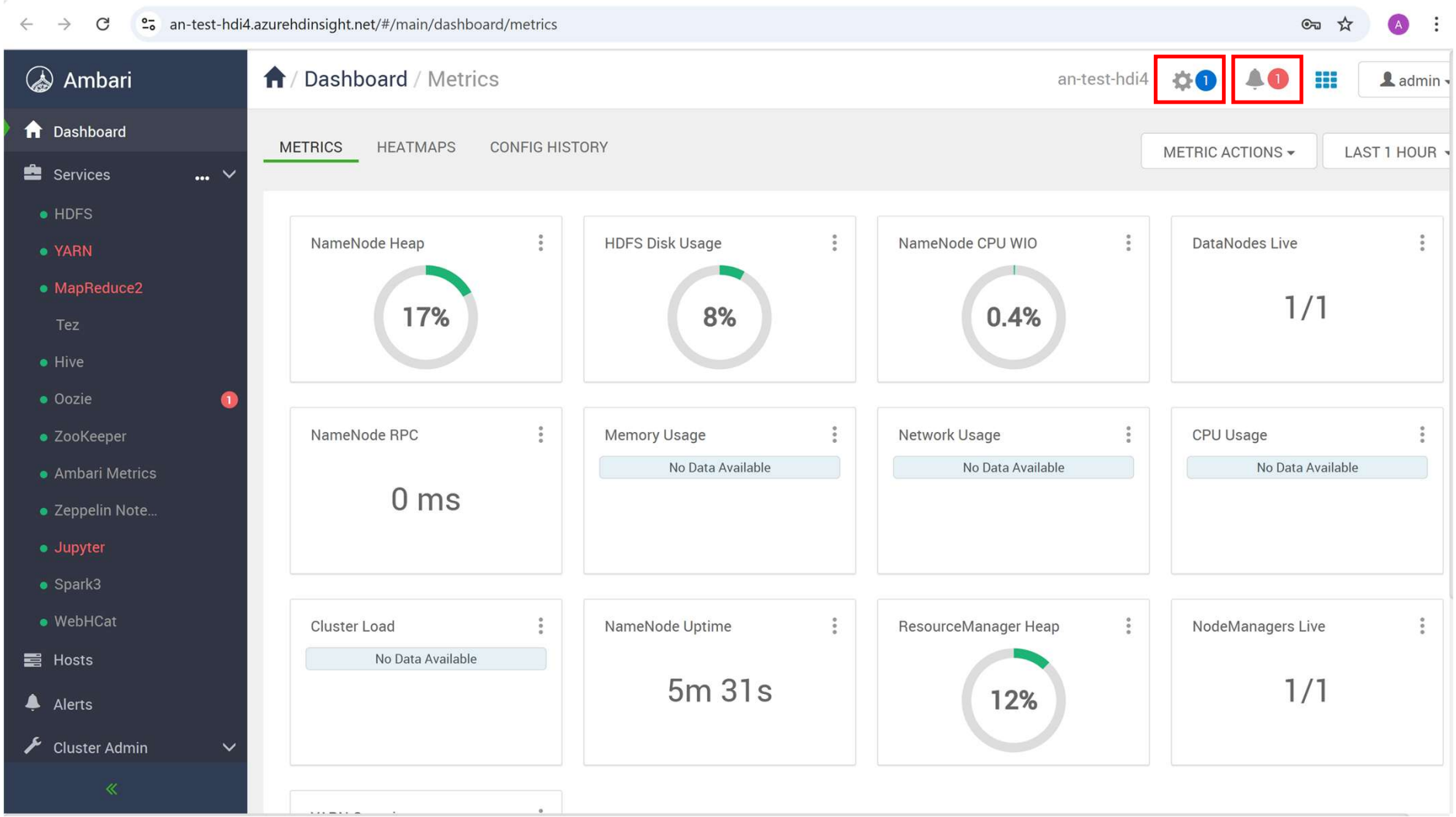


Using HDI .. Discovering the UI

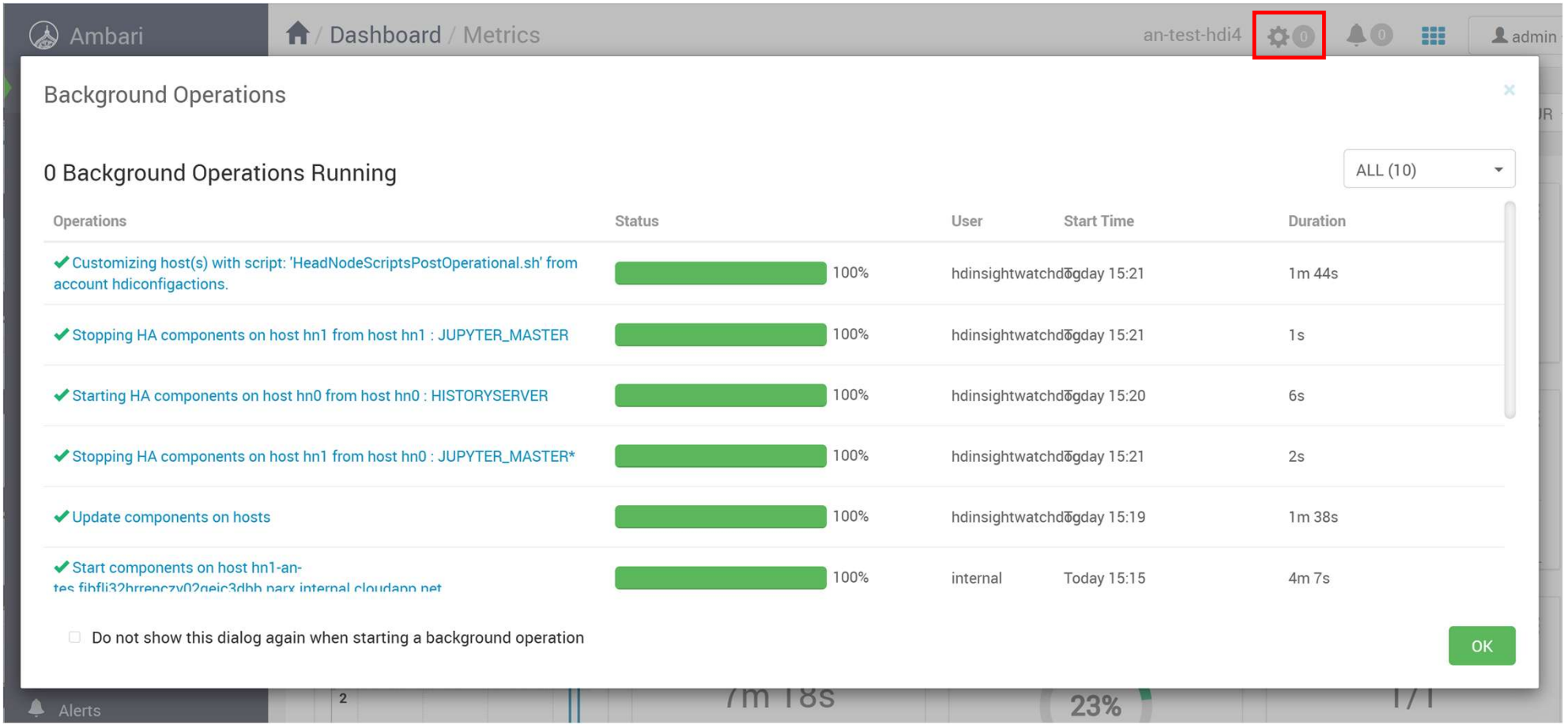
login Http Gateway (=Ambari)



Still in progress => Ambari Alerts



Ambari Scripts



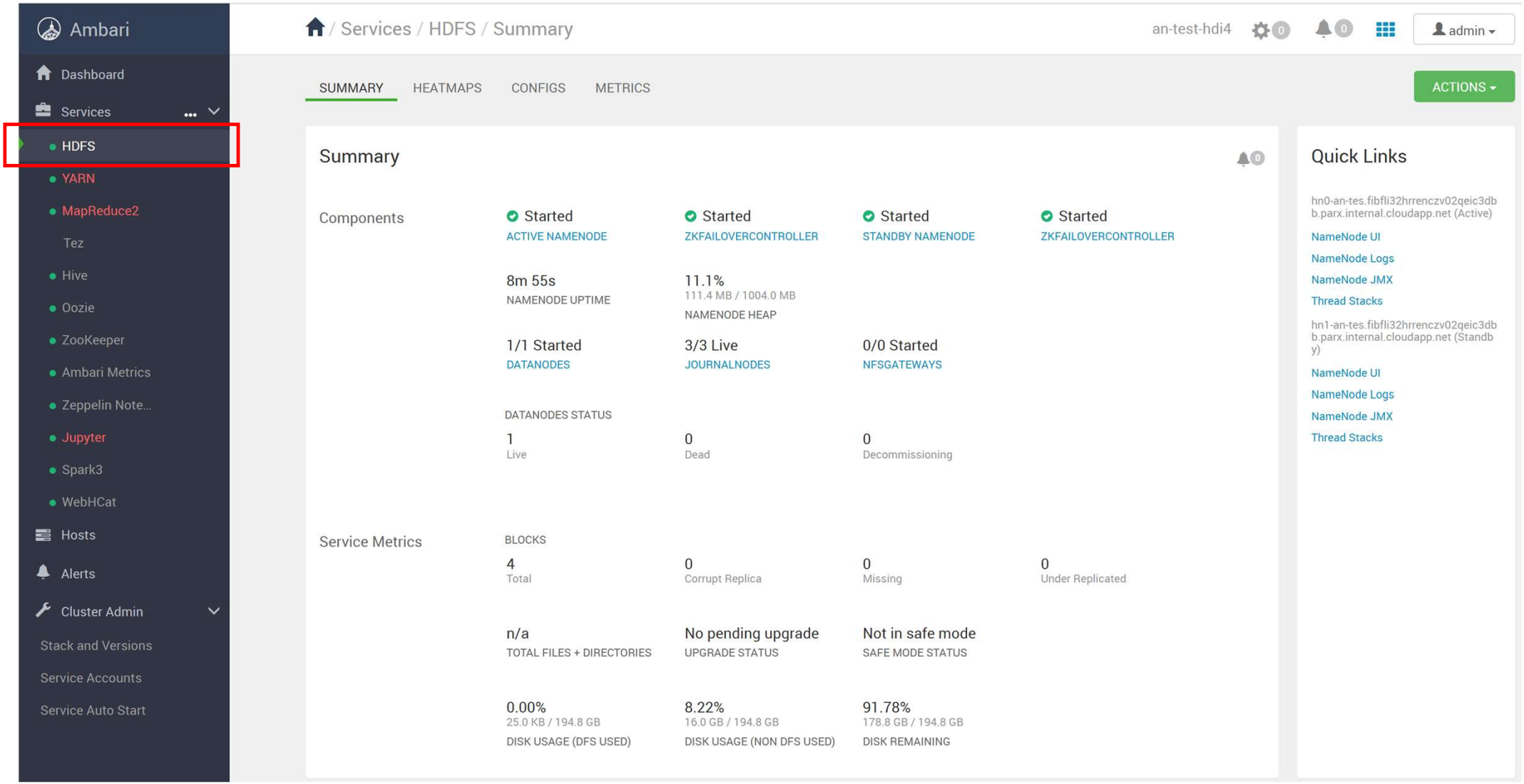
Exploring Ambari Services ...

Ambari > HDFS

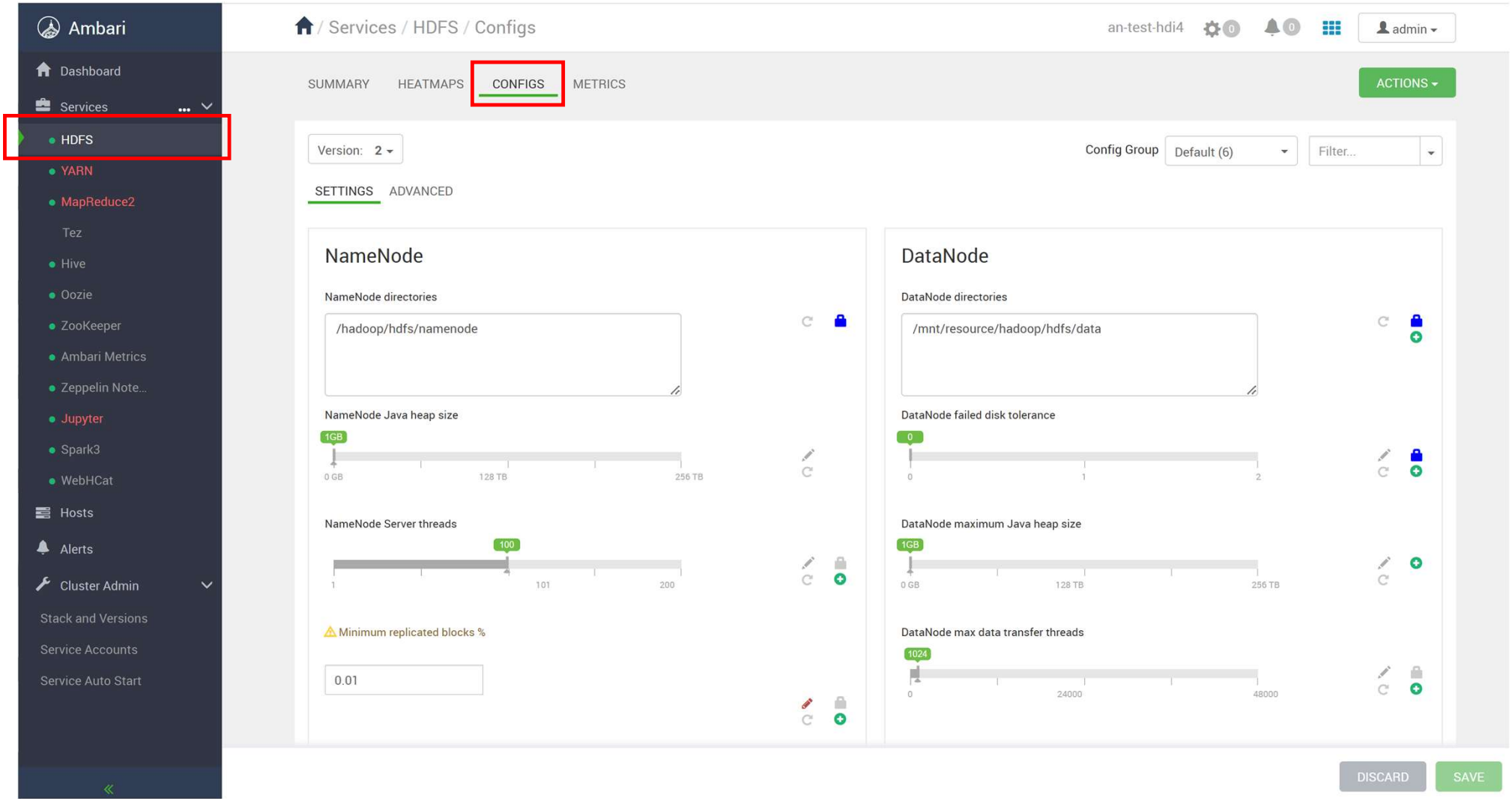
Reminder ... No classical "HDFS"
on HD-Insight !

=> using Azure Storage (AdlsGen2)
+ Adapter class for Hadoop
"VirtualFileSystem"

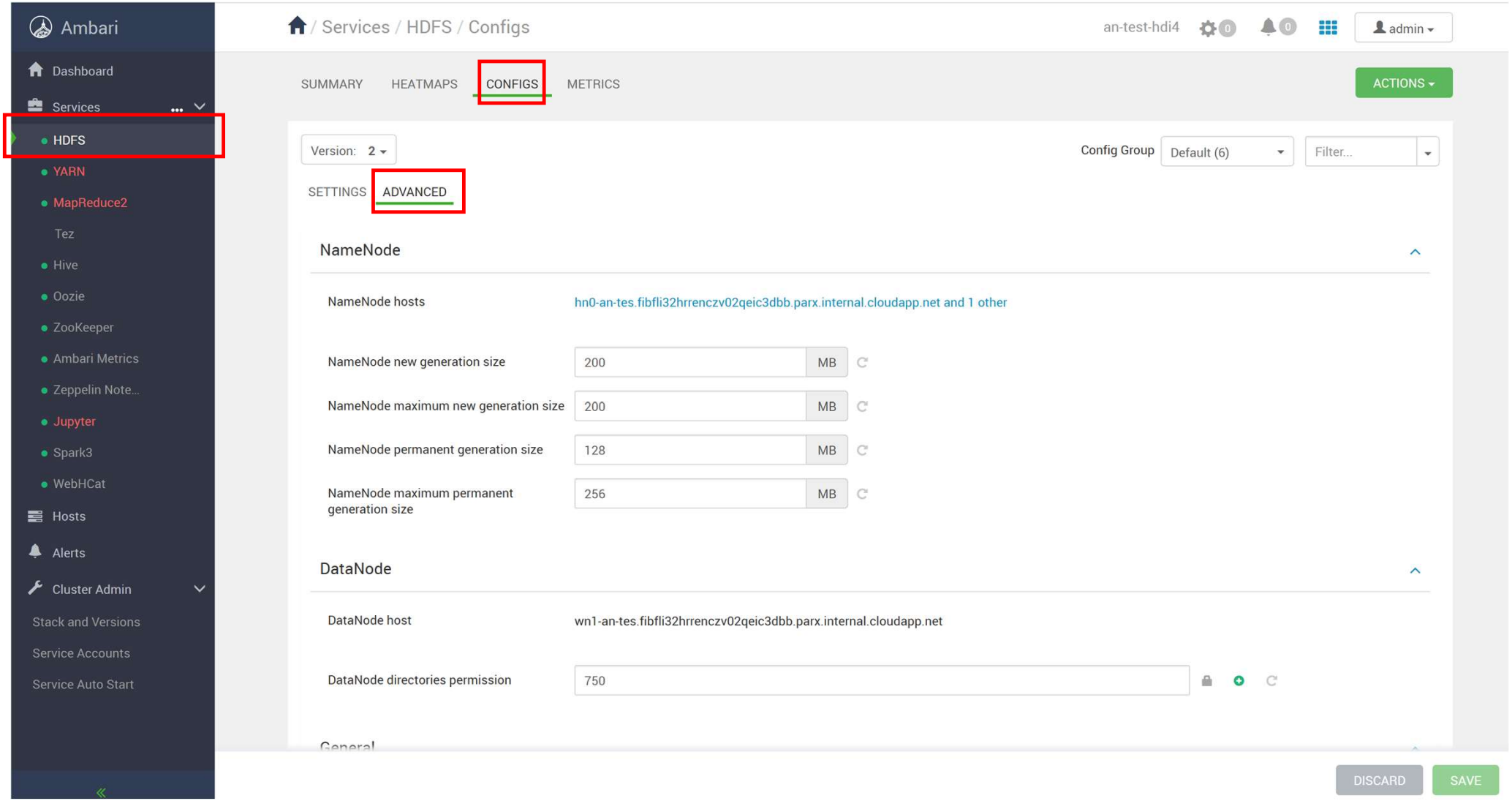
Ambari > Services > HDFS



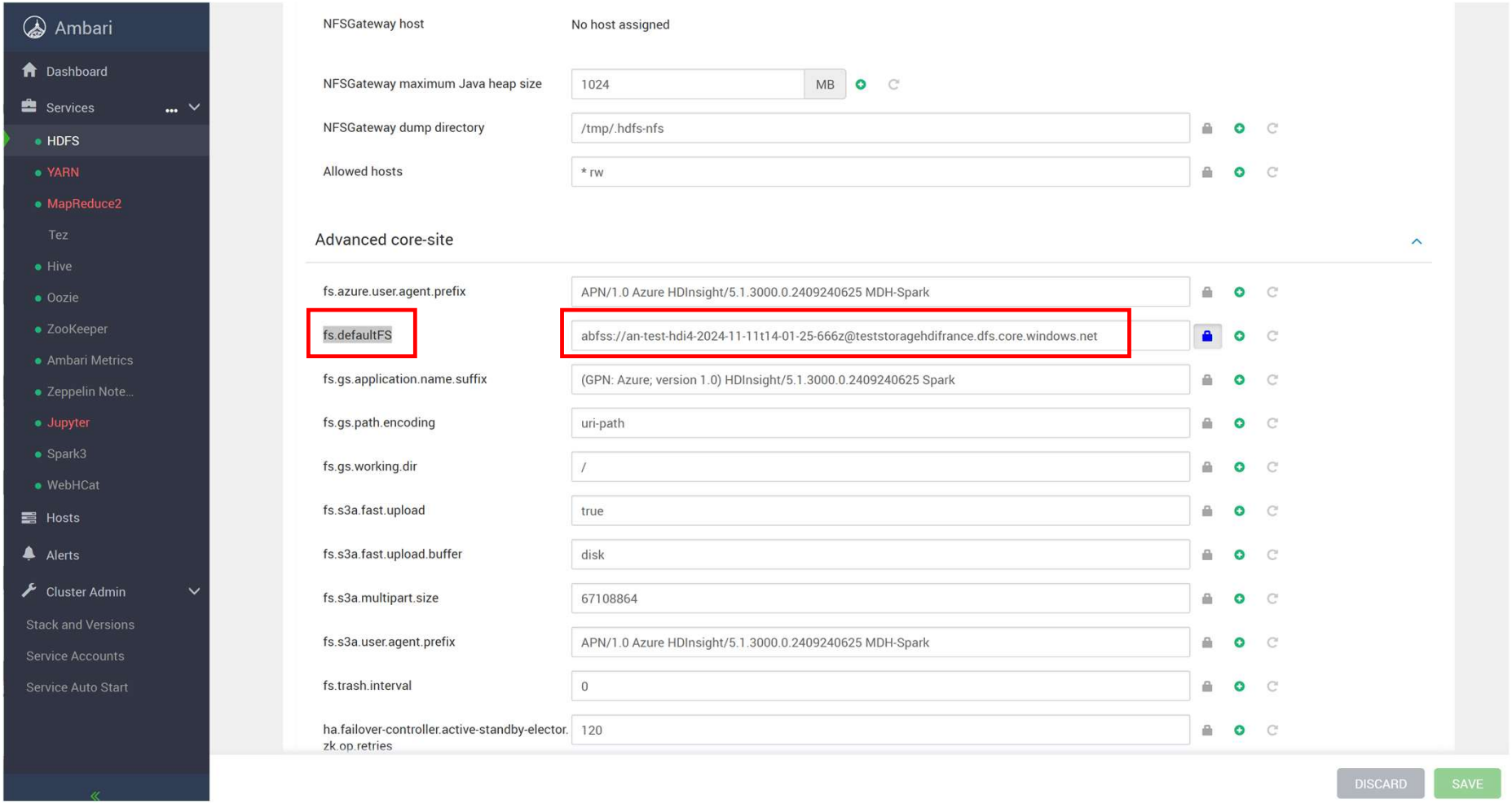
Ambari > HDFS > Configs



Ambari > HDFS > Configs > Advanced

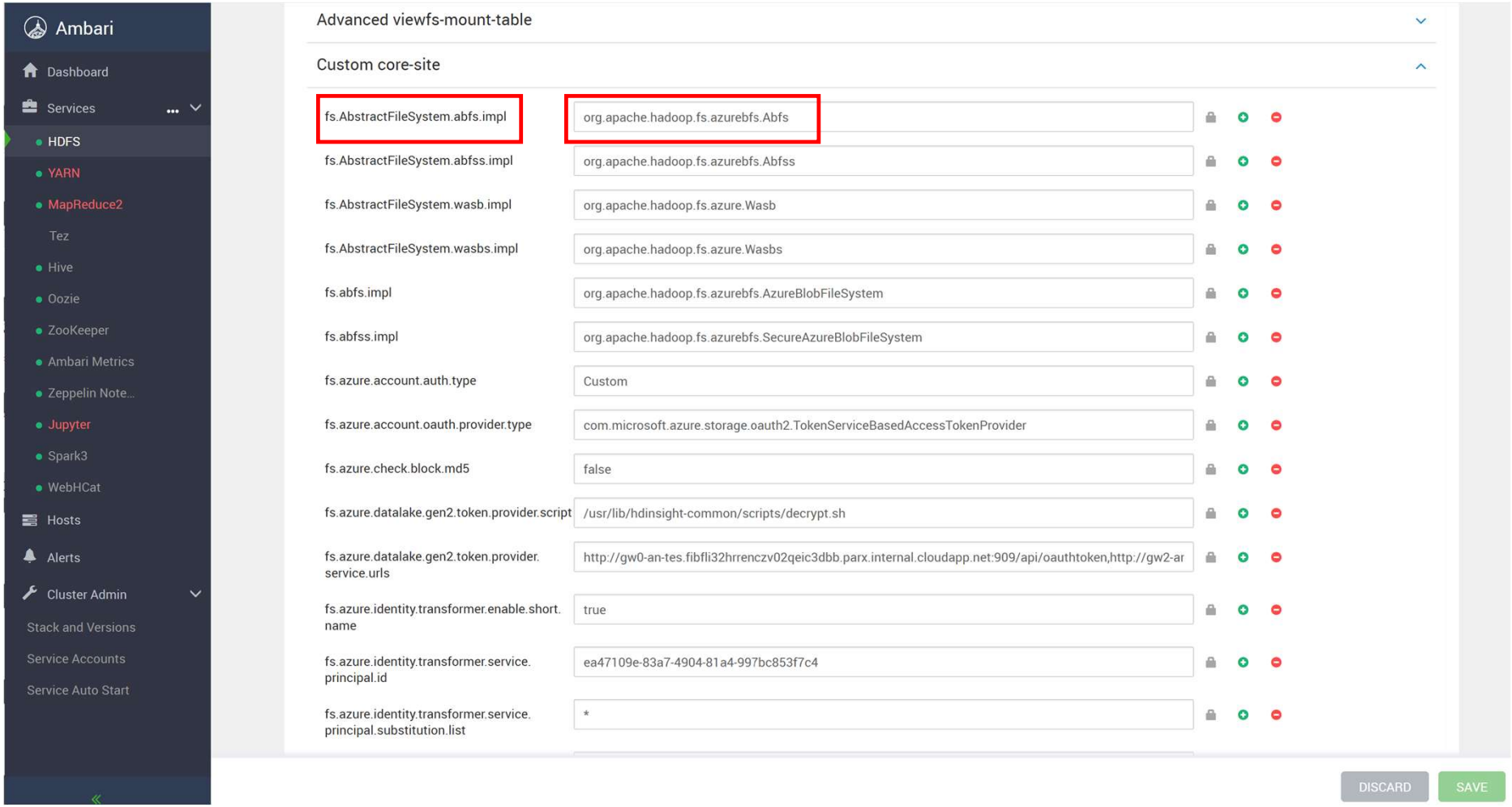


Ambari > HDFS > Configs > Advanced fs.defaultFS in core-site.xml

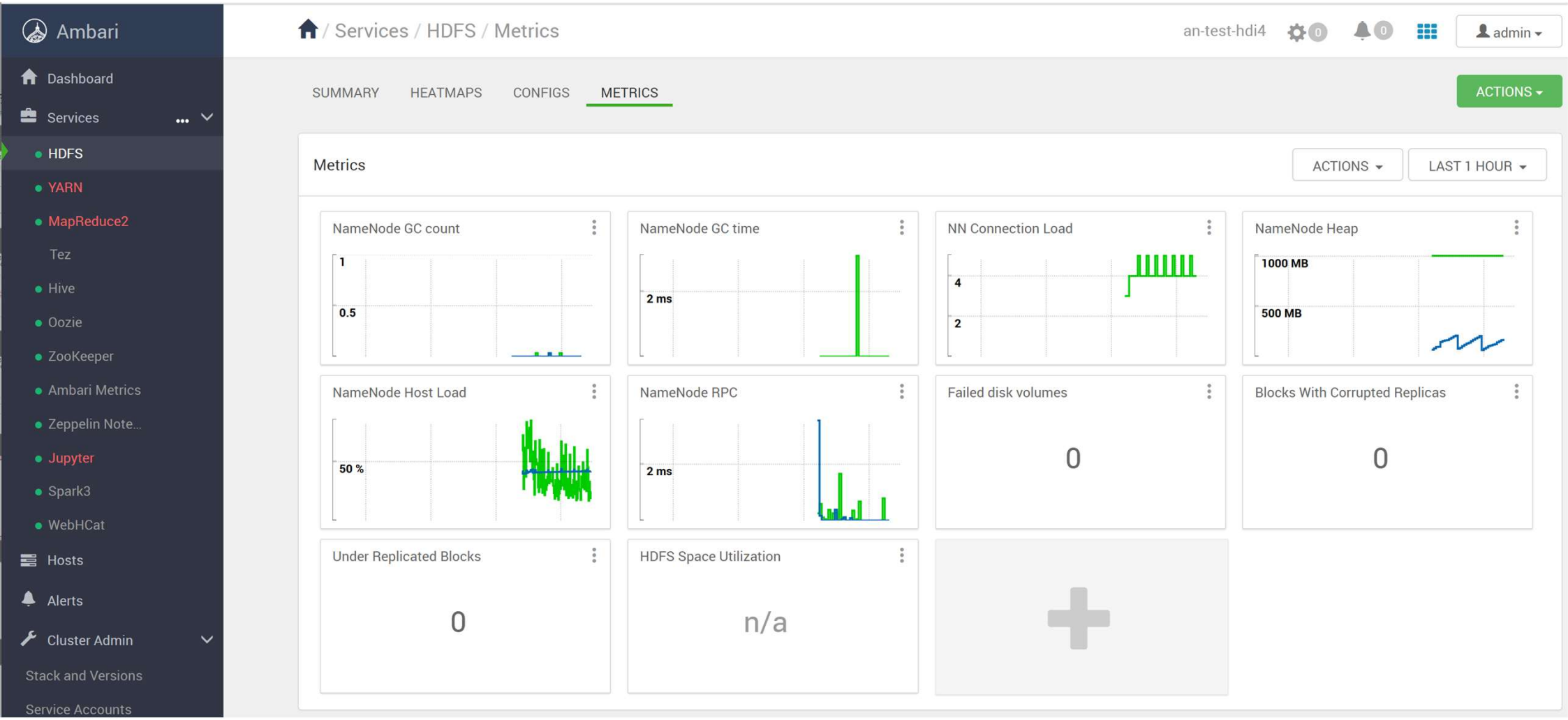


Ambari > HDFS > Configs > Advanced custom hdfs-site.xml

Hook for support Azure Storage AdlsGen2 (abfs://...) in Hadoop

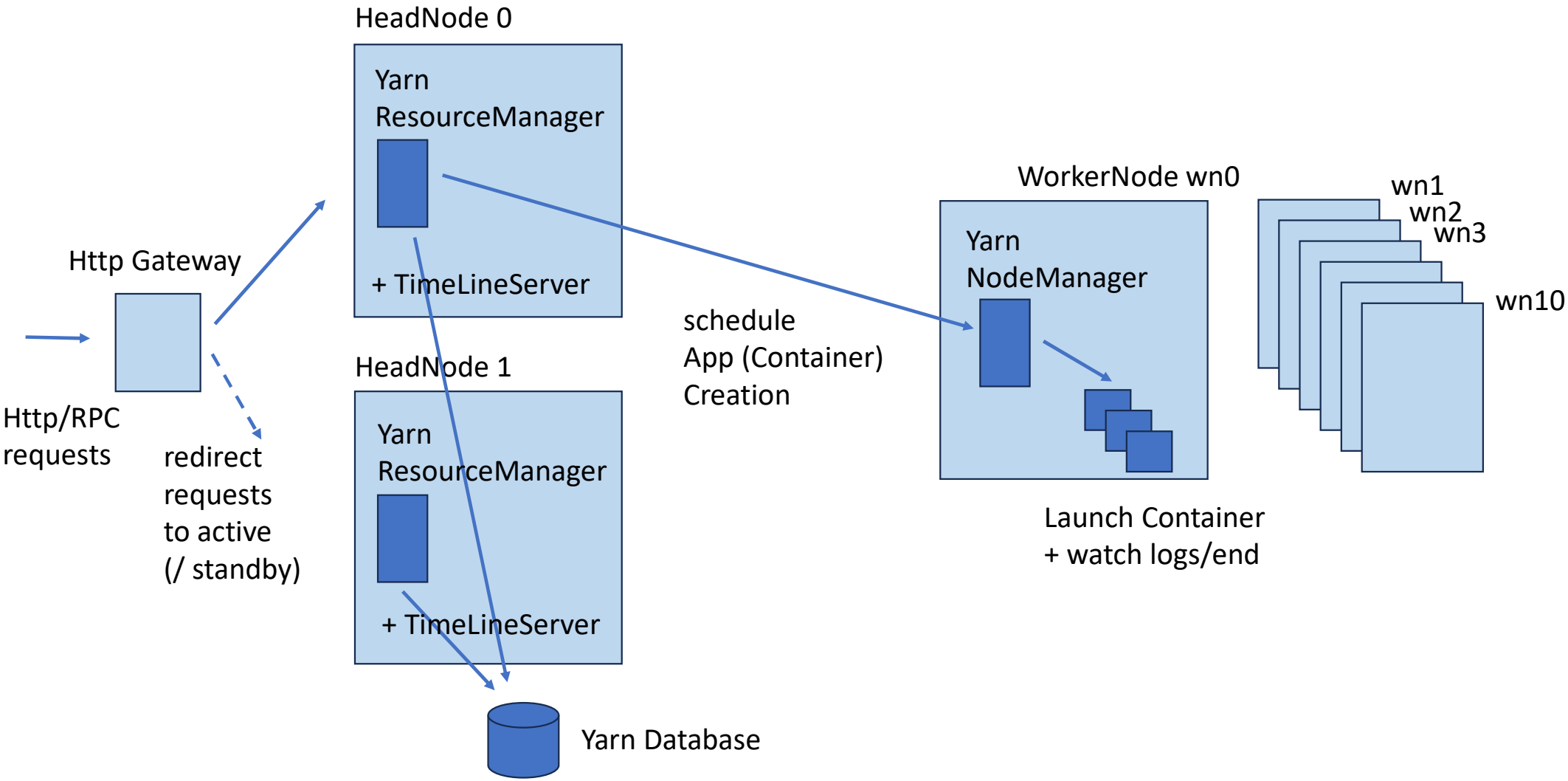


Ambari > HDFS > Metrics

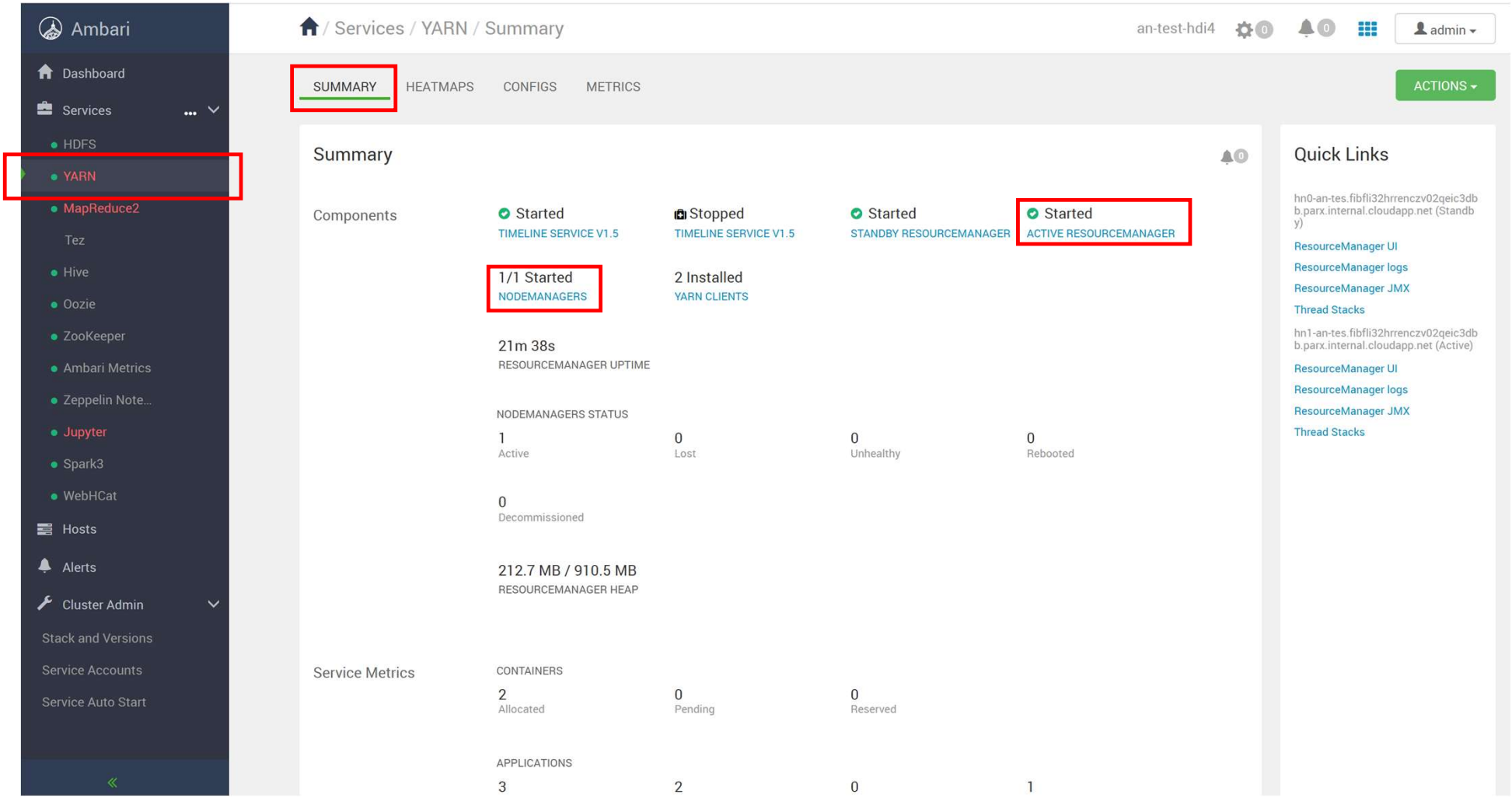


Ambari > Yarn

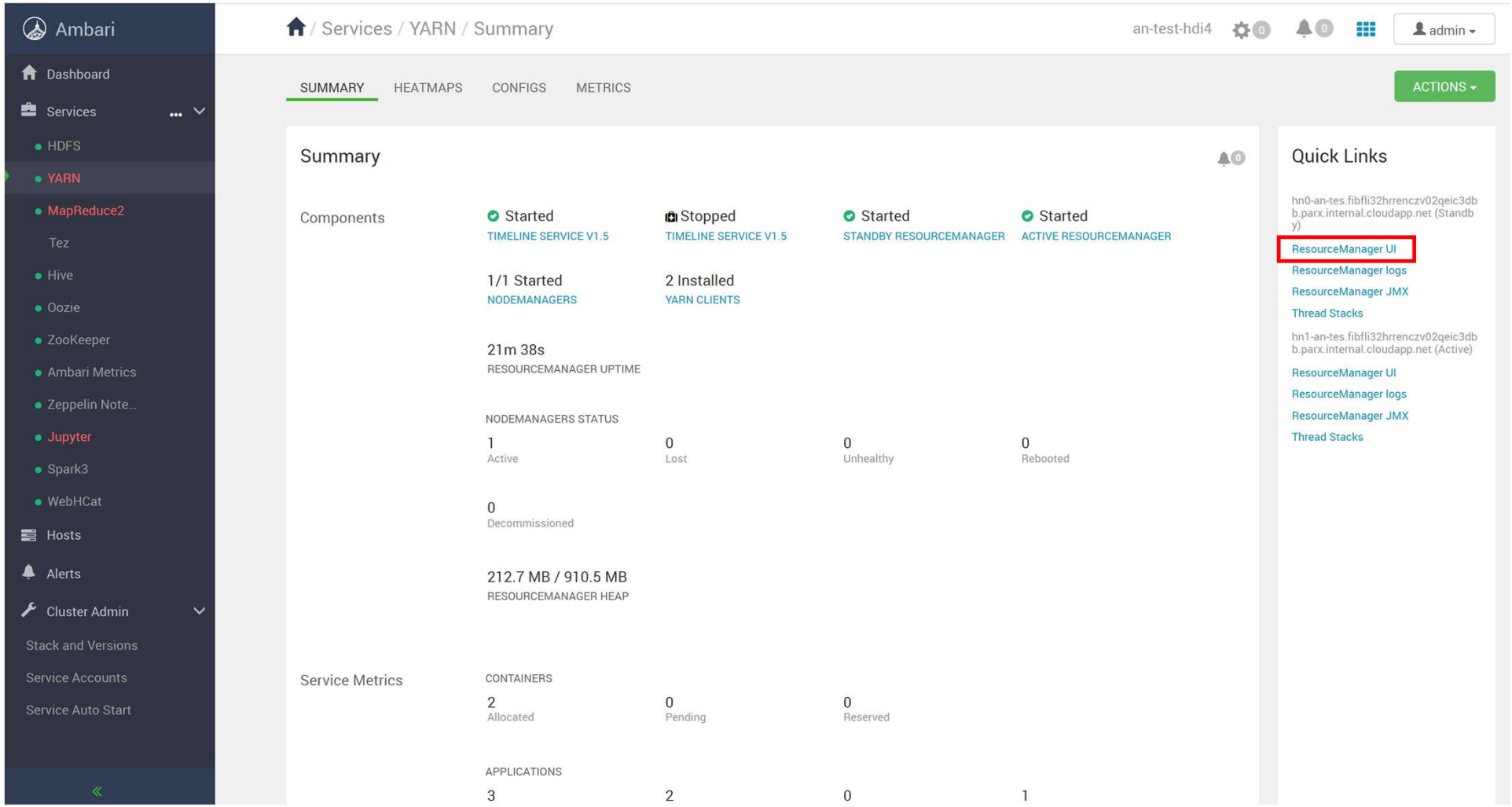
Reminder Yarn Architecture



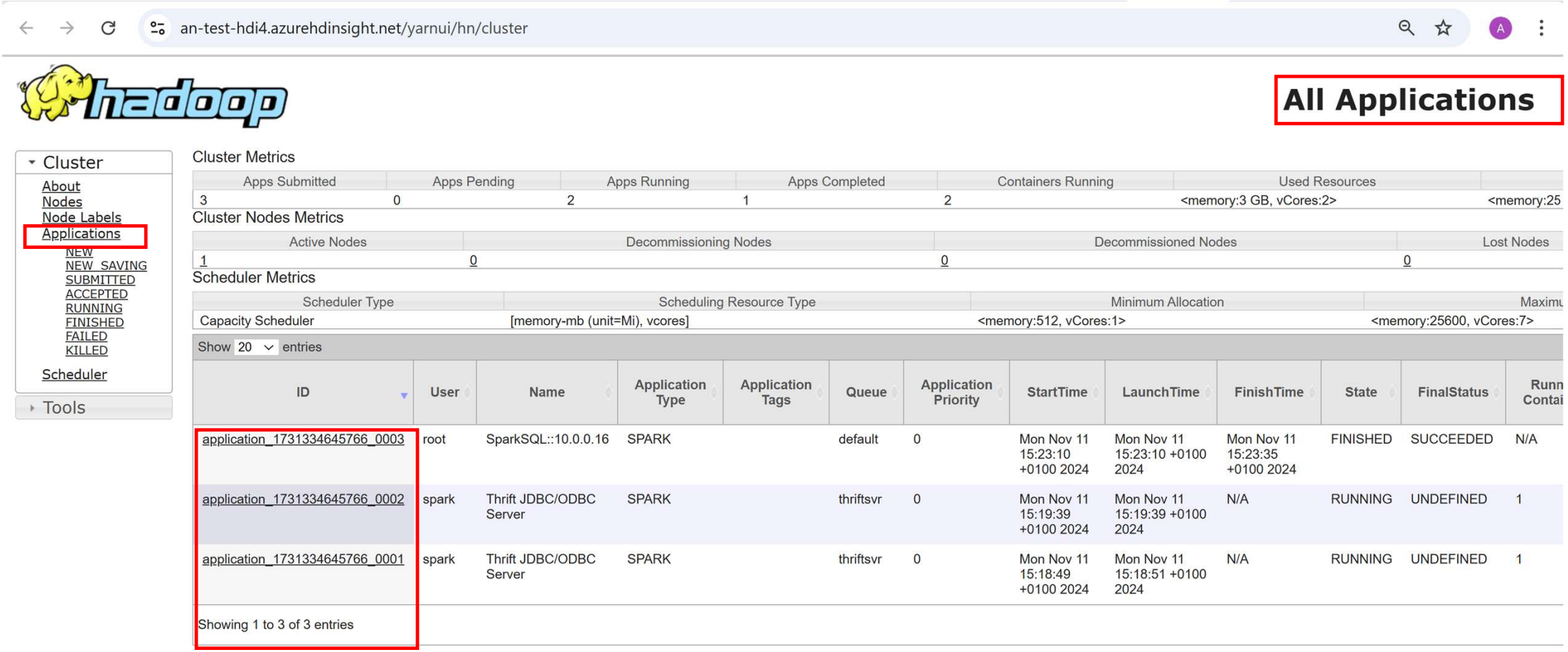
Ambari > Yarn > Summary



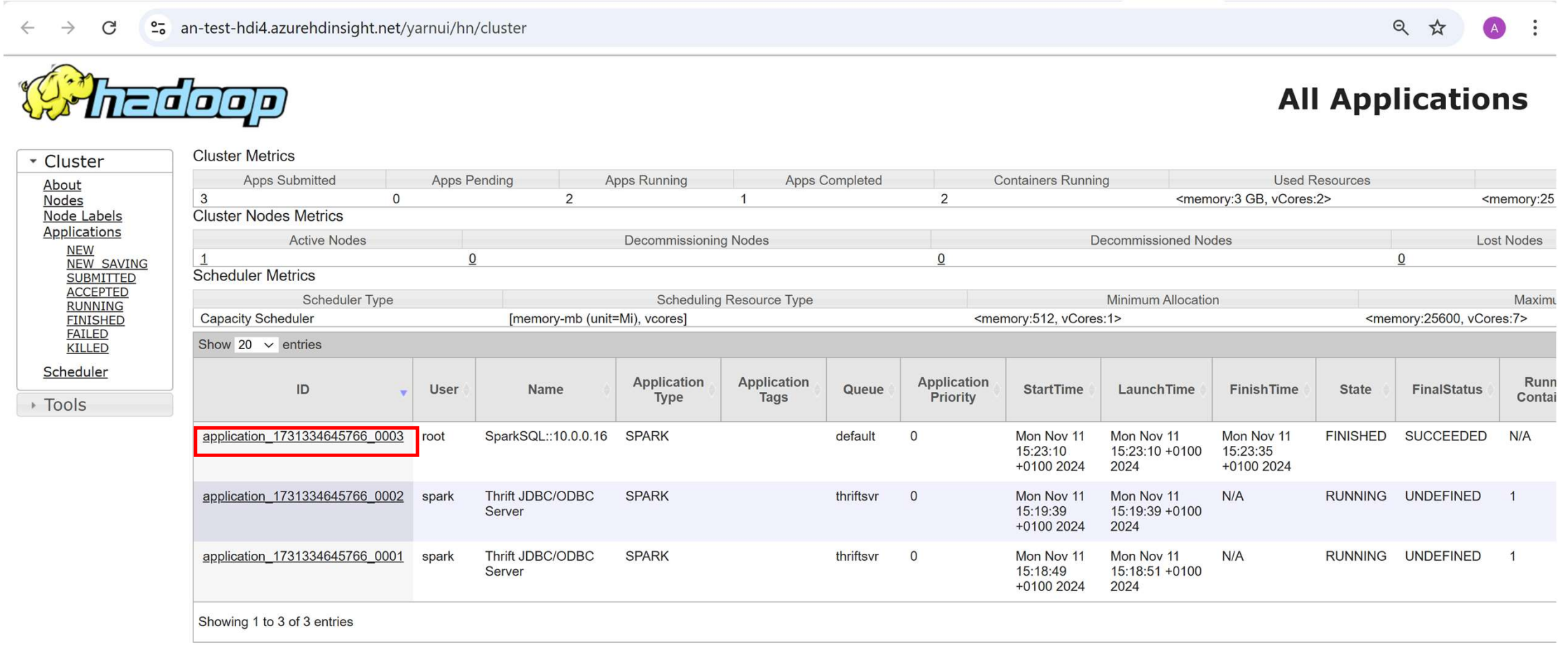
Click on Quick Links "Resource Manager UI"



<<baseurl>> /yarnui
(=Webapp for Yarn ResourceManager)

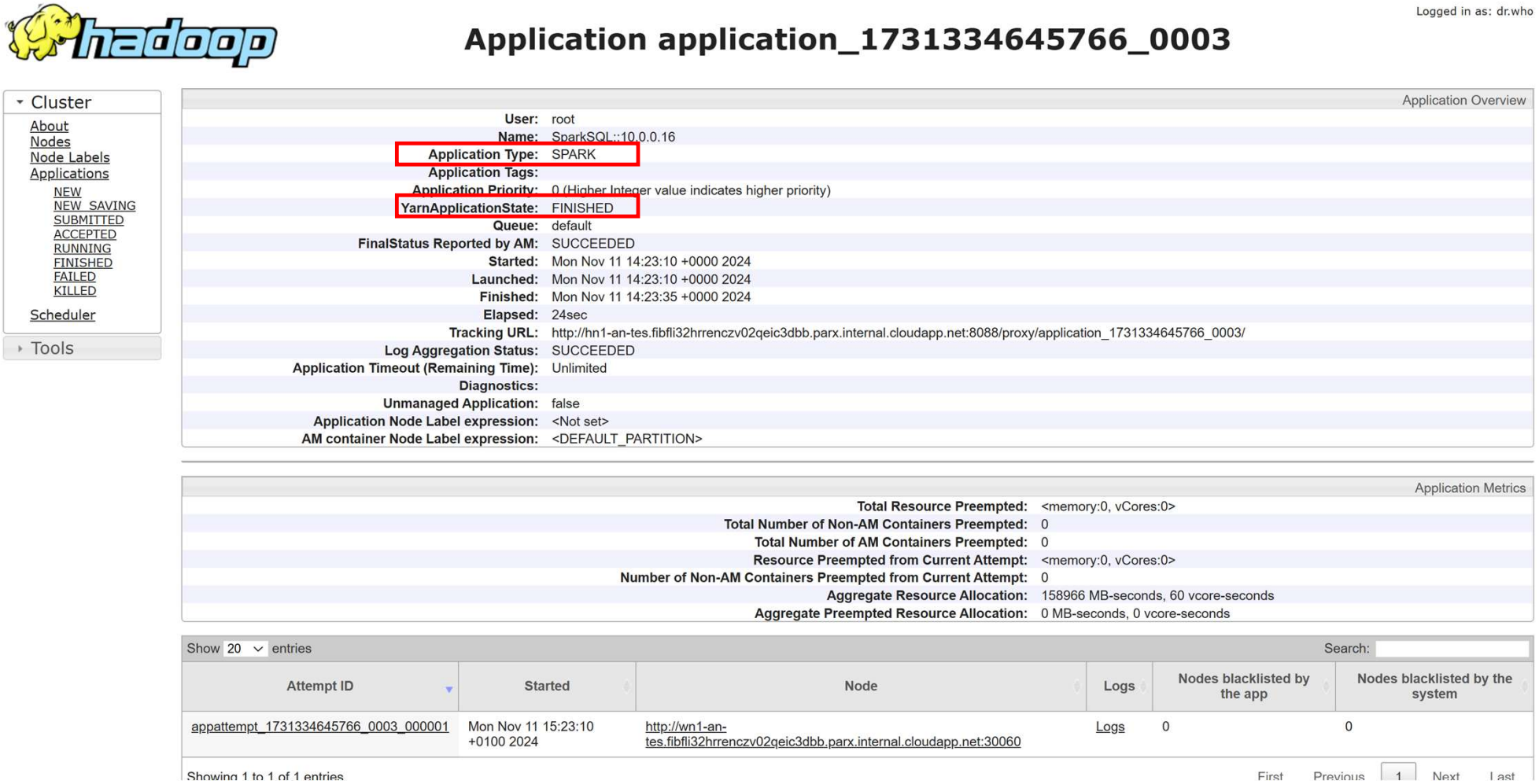


Click on 1 Application
(Stil Running OR Finished)



/yarnui/hn/cluster/app/application_<<id>>

example: a FINISHED Spark Application



Yarn Application Internals : Attempt -> (AM) Master -> Containers

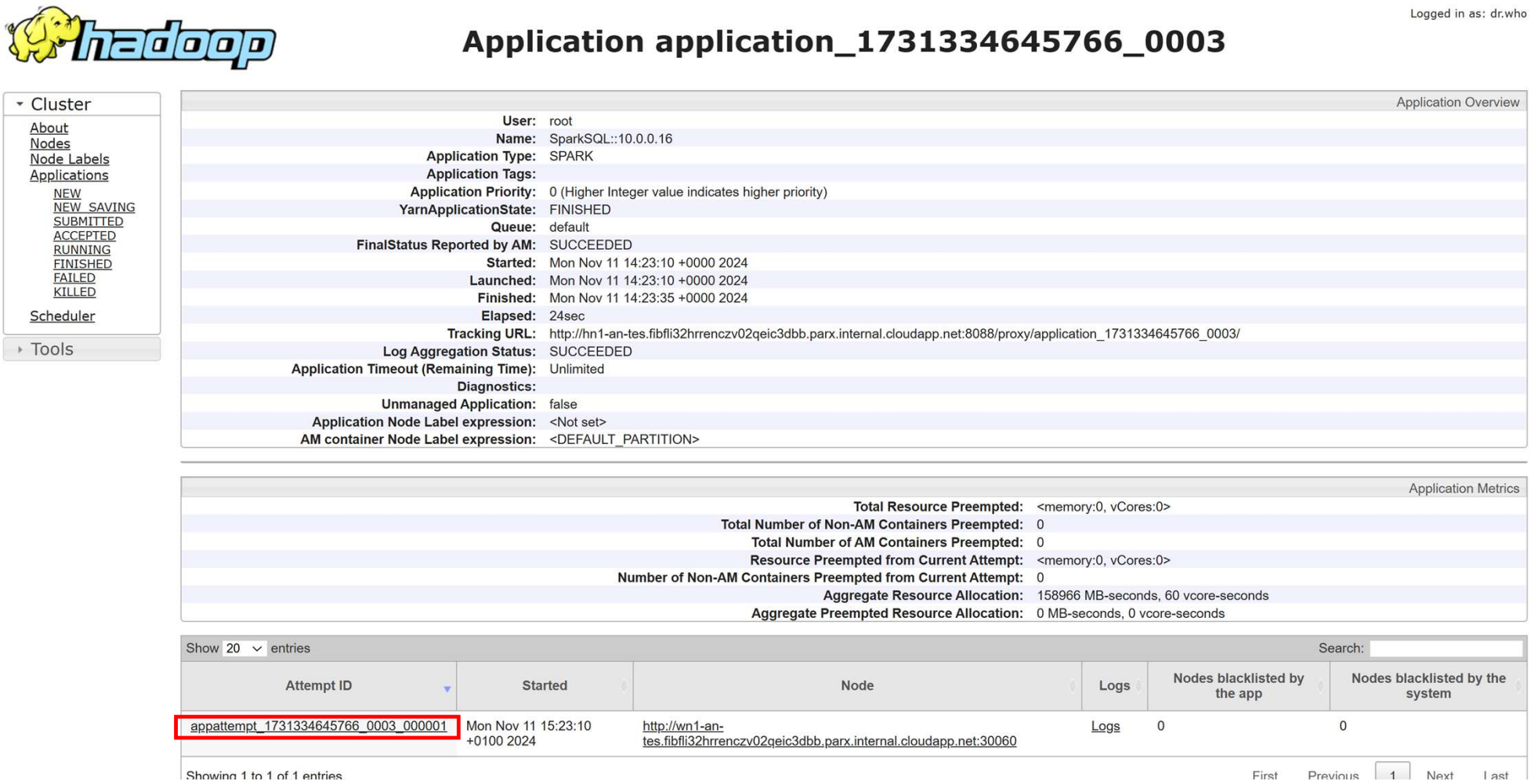
Admitedly : Yarn is over complex ... Think analogies with Kubernetes

Attempt => first try to launch, then maybe retries on failure

AM Master => idem "K8s Operator", but restricted to 1 Application

Container => mostly java apps (Docker images also supported after)

Click on Application > Attempt "appattempt_<<id>>_000001"



Application Attempt



Logged in as: dr.who

Application Attempt appattempt_1731334645766_0003_000001

- Cluster
  - About
  - Nodes
  - Node Labels
  - Applications
    - NEW
    - NEW_SAVING
    - SUBMITTED
    - ACCEPTED
    - RUNNING
    - FINISHED
    - FAILED
    - KILLED
  - Scheduler
- Tools

Application Attempt Overview

Application Attempt State:	FINISHED
Started:	Mon Nov 11 14:23:10 +0000 2024
Elapsed:	24sec
AM Container:	container_1731334645766_0003_01_000001
Node:	N/A
Tracking URL:	History
Diagnostics Info:	
Nodes blacklisted by the application:	-
Nodes blacklisted by the system:	-

Total Allocated Containers: 3

Each table cell represents the number of NodeLocal/RackLocal/OffSwitch containers satisfied by NodeLocal/RackLocal/OffSwitch resource requests.

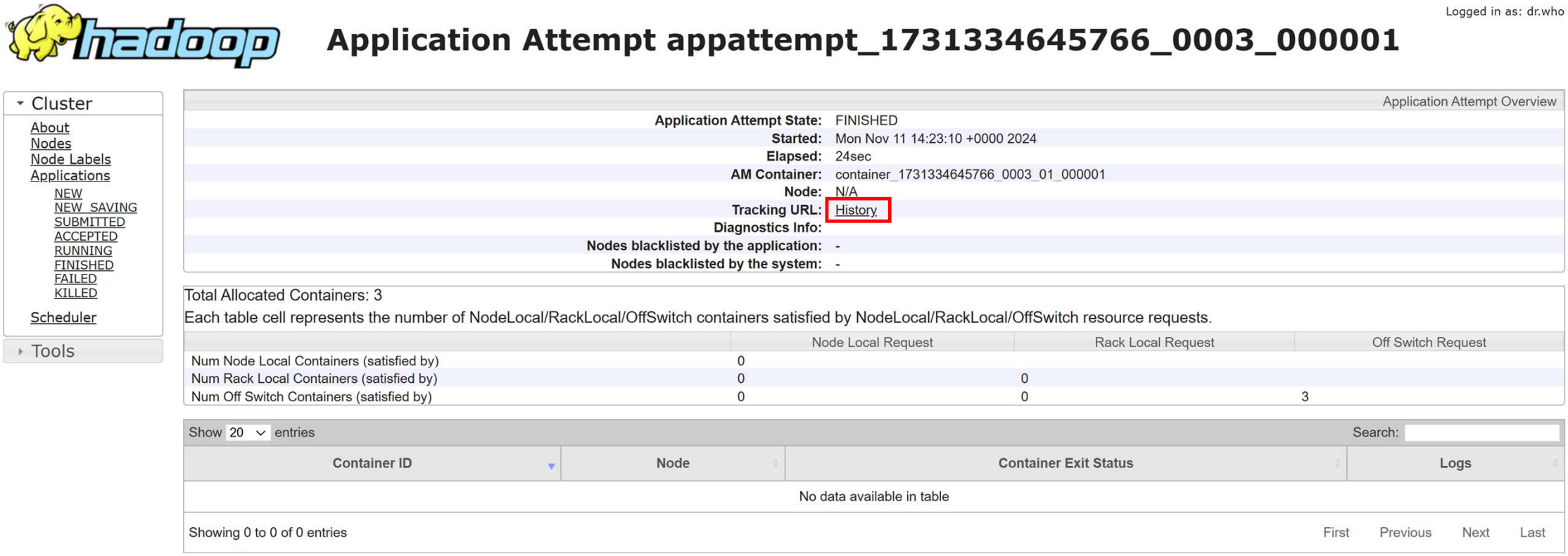
	Node Local Request	Rack Local Request	Off Switch Request
Num Node Local Containers (satisfied by)	0		
Num Rack Local Containers (satisfied by)	0	0	
Num Off Switch Containers (satisfied by)	0	0	3

Show 20 entries Search:

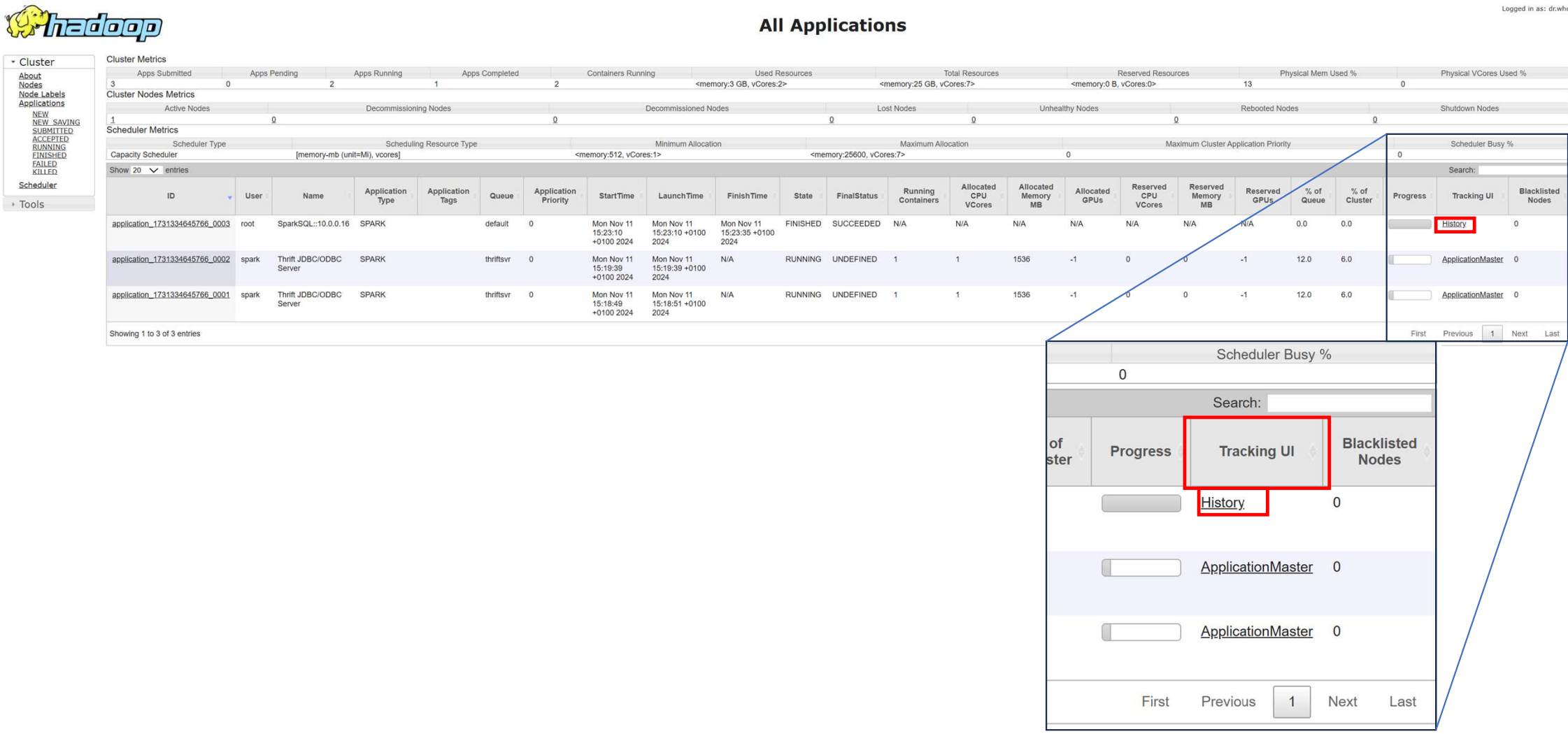
Container ID	Node	Container Exit Status	Logs
No data available in table			

Showing 0 to 0 of 0 entries First Previous Next Last

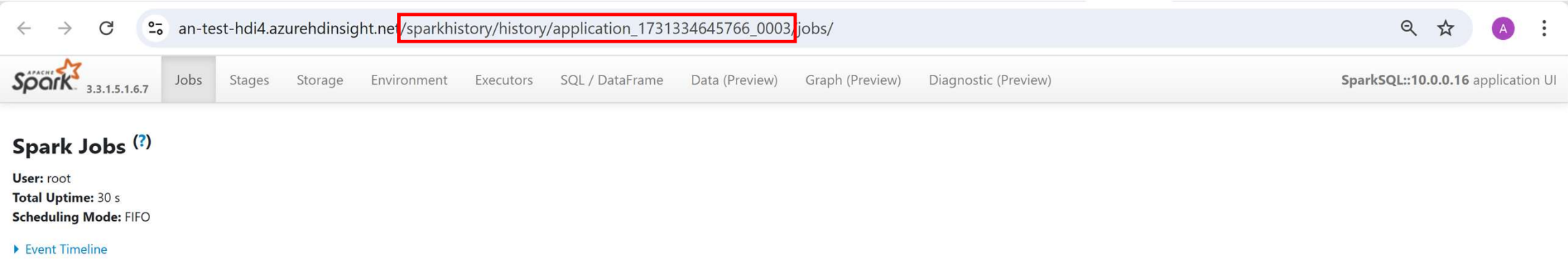
Aplication Attempt (finished)
=> click on "History"



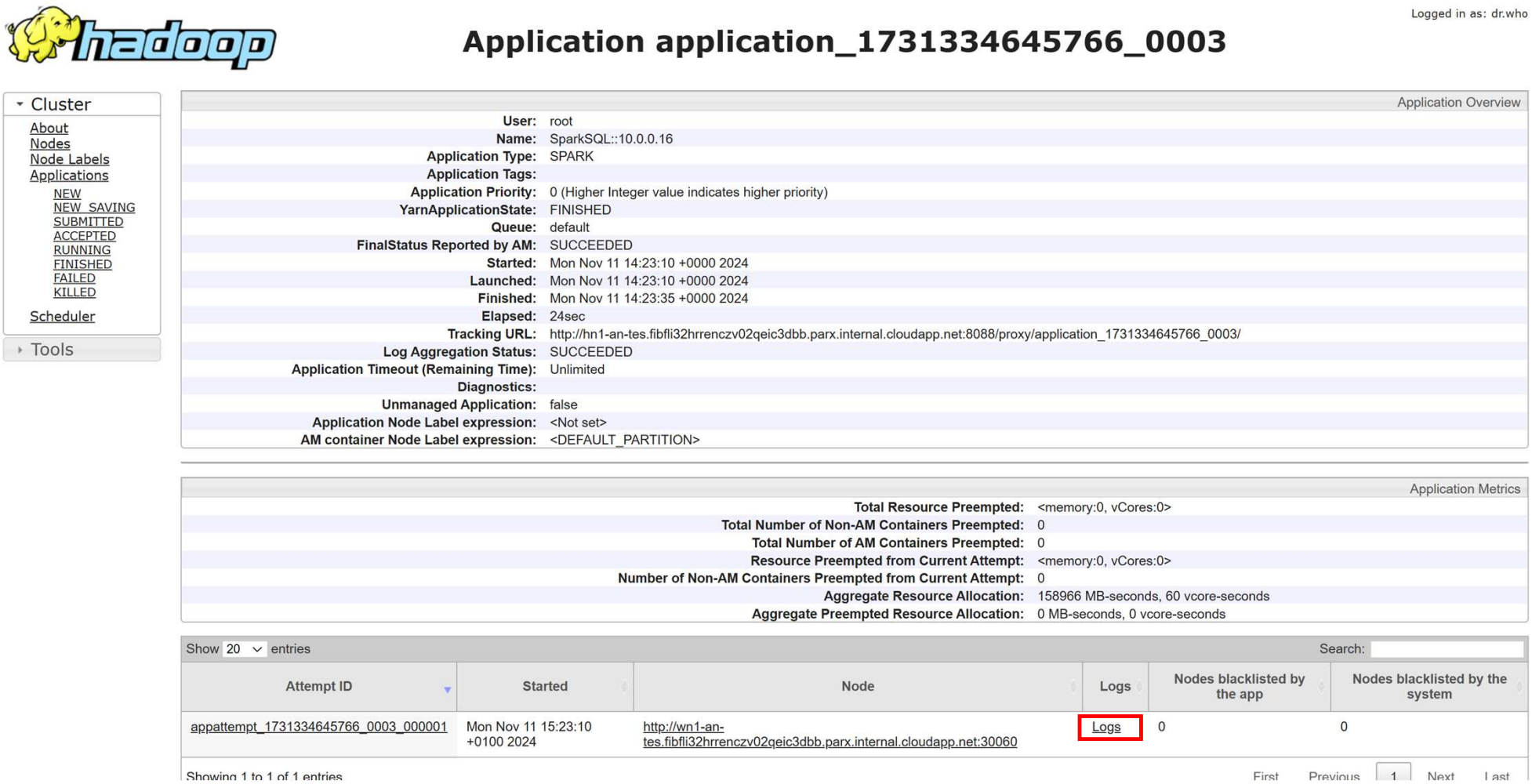
equivalent Direct Link via "Tracking UI" column



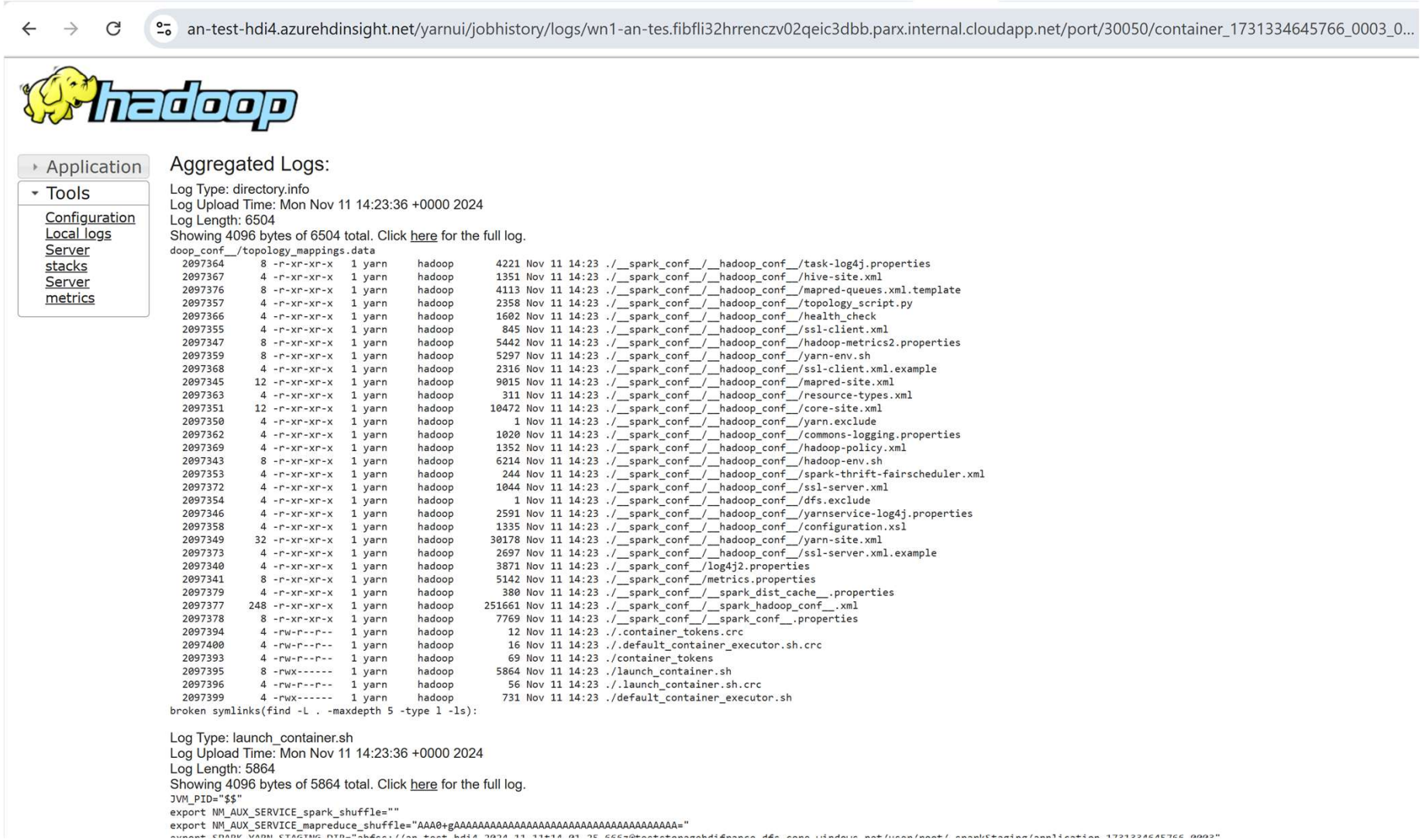
Application History... for type Spark = SparkUI
url: /sparkhistory/application_<<id>>



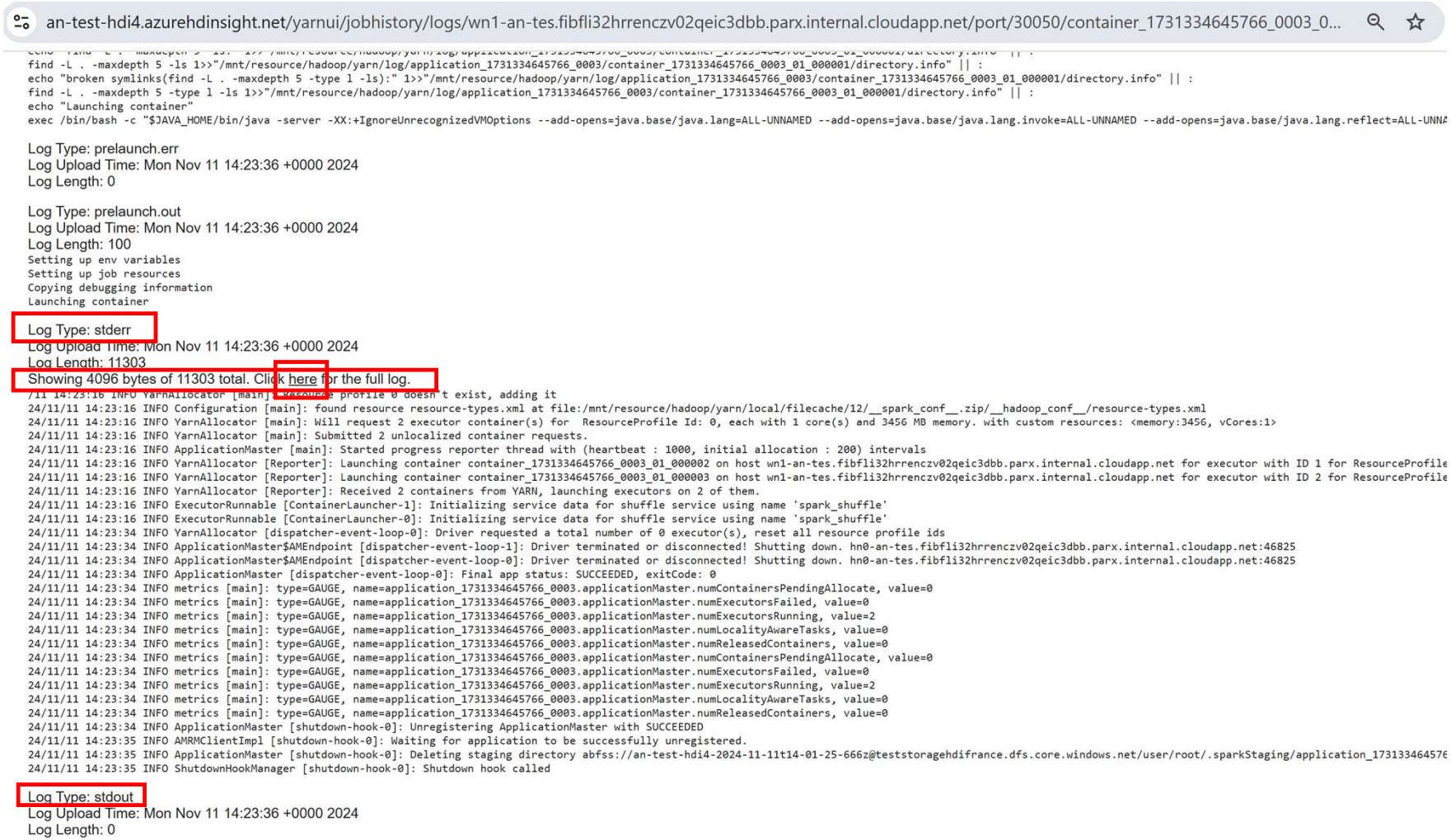
Click on Application > Attempt > "logs"



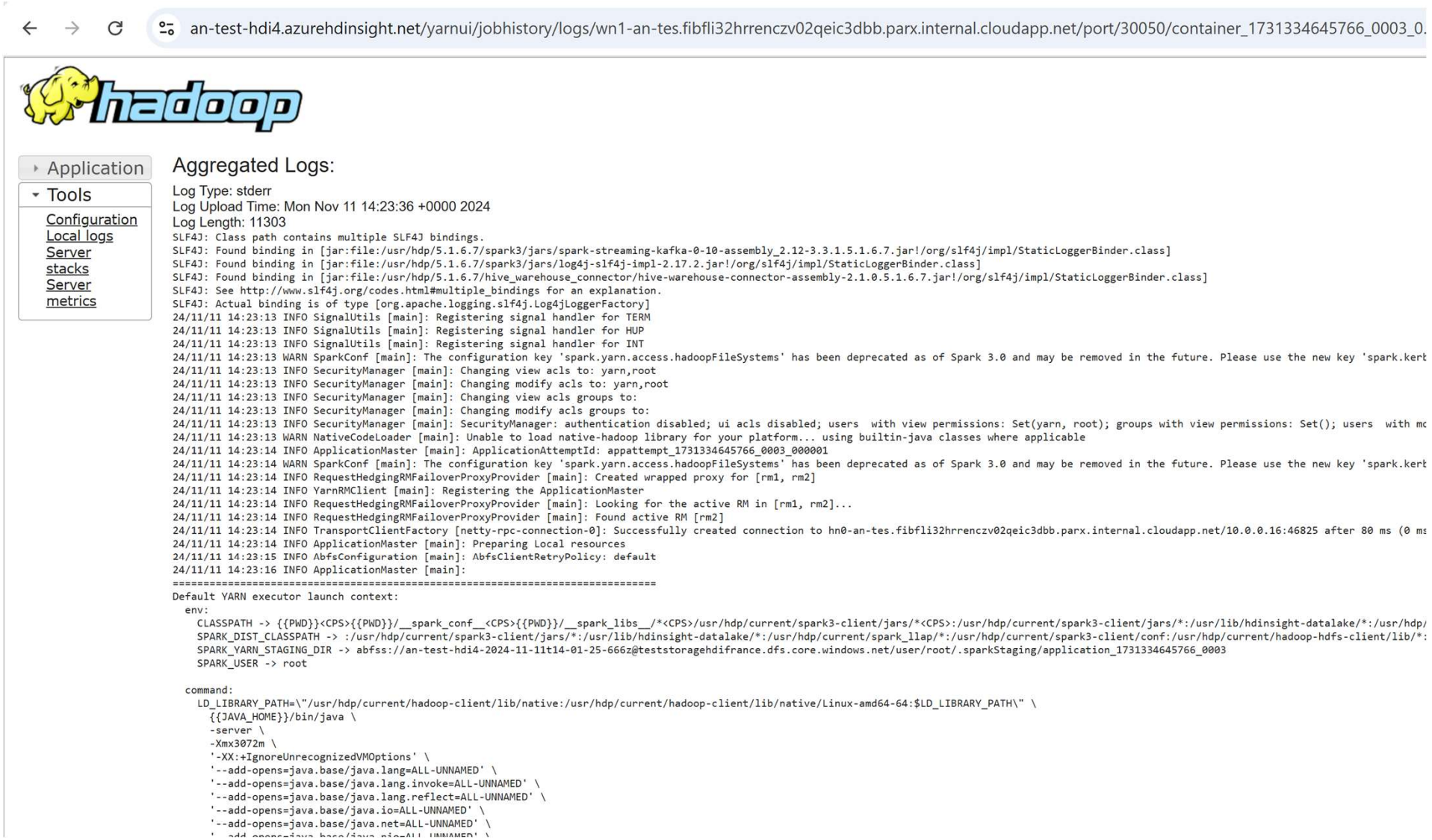
Yarn logs


an-test-hdi4.azurehdinsight.net/yarnui/jobhistory/logs/wn1-an-tes.fibfli32hrrenczv02qeic3dbb.parx.internal.cloudapp.net/port/30050/container_1731334645766_0003_0...
hadoop
Application
Tools
Configuration
Local logs
Server stacks
Server metrics
Aggregated Logs:
Log Type: directory.info
Log Upload Time: Mon Nov 11 14:23:36 +0000 2024
Log Length: 6504
Showing 4096 bytes of 6504 total. Click here for the full log.
doop_conf__/topology_mappings.data
2097364 8 -r-xr-xr-x 1 yarn hadoop 4221 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/task-log4j.properties
2097367 4 -r-xr-xr-x 1 yarn hadoop 1351 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/hive-site.xml
2097376 8 -r-xr-xr-x 1 yarn hadoop 4113 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/mapred-queues.xml.template
2097357 4 -r-xr-xr-x 1 yarn hadoop 2358 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/topology_script.py
2097366 4 -r-xr-xr-x 1 yarn hadoop 1602 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/health_check
2097355 4 -r-xr-xr-x 1 yarn hadoop 845 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/ssl-client.xml
2097347 8 -r-xr-xr-x 1 yarn hadoop 5442 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/hadoop-metrics2.properties
2097359 8 -r-xr-xr-x 1 yarn hadoop 5297 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/yarn-env.sh
2097368 4 -r-xr-xr-x 1 yarn hadoop 2316 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/ssl-client.xml.example
2097345 12 -r-xr-xr-x 1 yarn hadoop 9015 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/mapred-site.xml
2097363 4 -r-xr-xr-x 1 yarn hadoop 311 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/resource-types.xml
2097351 12 -r-xr-xr-x 1 yarn hadoop 10472 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/core-site.xml
2097350 4 -r-xr-xr-x 1 yarn hadoop 1 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/yarn.exclude
2097362 4 -r-xr-xr-x 1 yarn hadoop 1020 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/commons-logging.properties
2097369 4 -r-xr-xr-x 1 yarn hadoop 1352 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/hadoop-policy.xml
2097343 8 -r-xr-xr-x 1 yarn hadoop 6214 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/hadoop-env.sh
2097353 4 -r-xr-xr-x 1 yarn hadoop 244 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/spark-thrift-fairscheduler.xml
2097372 4 -r-xr-xr-x 1 yarn hadoop 1044 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/ssl-server.xml
2097354 4 -r-xr-xr-x 1 yarn hadoop 1 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/dfs.exclude
2097346 4 -r-xr-xr-x 1 yarn hadoop 2591 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/yarnservice-log4j.properties
2097358 4 -r-xr-xr-x 1 yarn hadoop 1335 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/configuration.xsl
2097349 32 -r-xr-xr-x 1 yarn hadoop 30178 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/yarn-site.xml
2097373 4 -r-xr-xr-x 1 yarn hadoop 2697 Nov 11 14:23 ./__spark_conf__/__hadoop_conf__/ssl-server.xml.example
2097340 4 -r-xr-xr-x 1 yarn hadoop 3871 Nov 11 14:23 ./__spark_conf__/log4j2.properties
2097341 8 -r-xr-xr-x 1 yarn hadoop 5142 Nov 11 14:23 ./__spark_conf__/metrics.properties
2097379 4 -r-xr-xr-x 1 yarn hadoop 380 Nov 11 14:23 ./__spark_conf__/__spark_dist_cache__.properties
2097377 248 -r-xr-xr-x 1 yarn hadoop 251661 Nov 11 14:23 ./__spark_conf__/__spark_hadoop_conf__.xml
2097378 8 -r-xr-xr-x 1 yarn hadoop 7769 Nov 11 14:23 ./__spark_conf__/__spark_conf__.properties
2097394 4 -rw-r--r-- 1 yarn hadoop 12 Nov 11 14:23 ./.container_tokens.crc
2097400 4 -rw-r--r-- 1 yarn hadoop 16 Nov 11 14:23 ./.default_container_executor.sh.crc
2097393 4 -rw-r--r-- 1 yarn hadoop 69 Nov 11 14:23 ./container_tokens
2097395 8 -rwx------ 1 yarn hadoop 5864 Nov 11 14:23 ./launch_container.sh
2097396 4 -rw-r--r-- 1 yarn hadoop 56 Nov 11 14:23 ./.launch_container.sh.crc
2097399 4 -rwx------ 1 yarn hadoop 731 Nov 11 14:23 ./default_container_executor.sh
broken symlinks(find -L . -maxdepth 5 -type l -ls):
Log Type: launch_container.sh
Log Upload Time: Mon Nov 11 14:23:36 +0000 2024
Log Length: 5864
Showing 4096 bytes of 5864 total. Click here for the full log.
JVM_PID="$$"
export NM_AUX_SERVICE_spark_shuffle=""
export NM_AUX_SERVICE_mapreduce_shuffle="AAA0+gAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA="

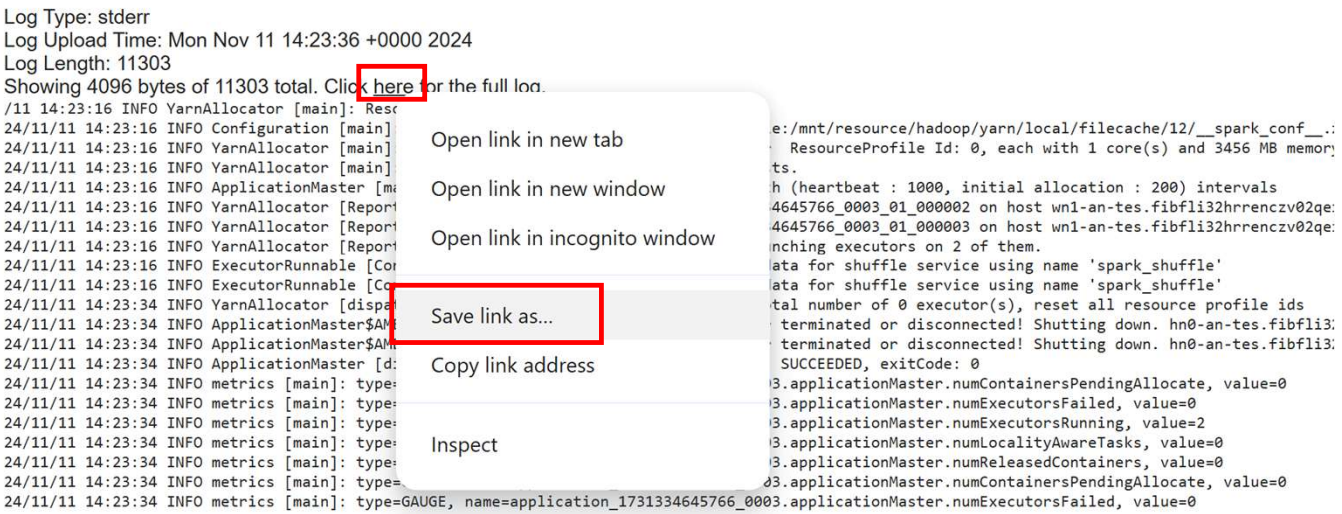
logs ... summary "stdout", "stderr"



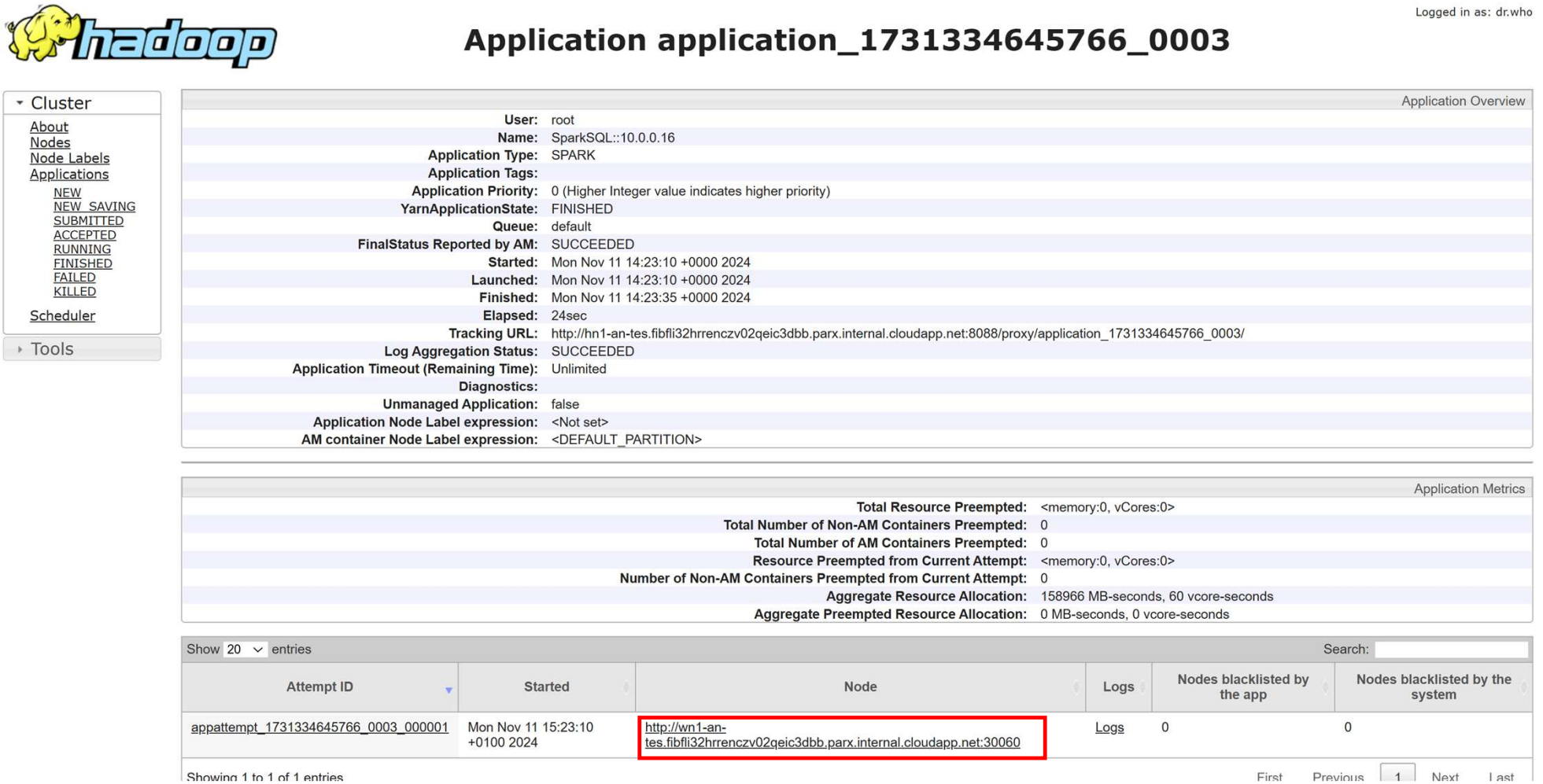
Click on stderr details "here"



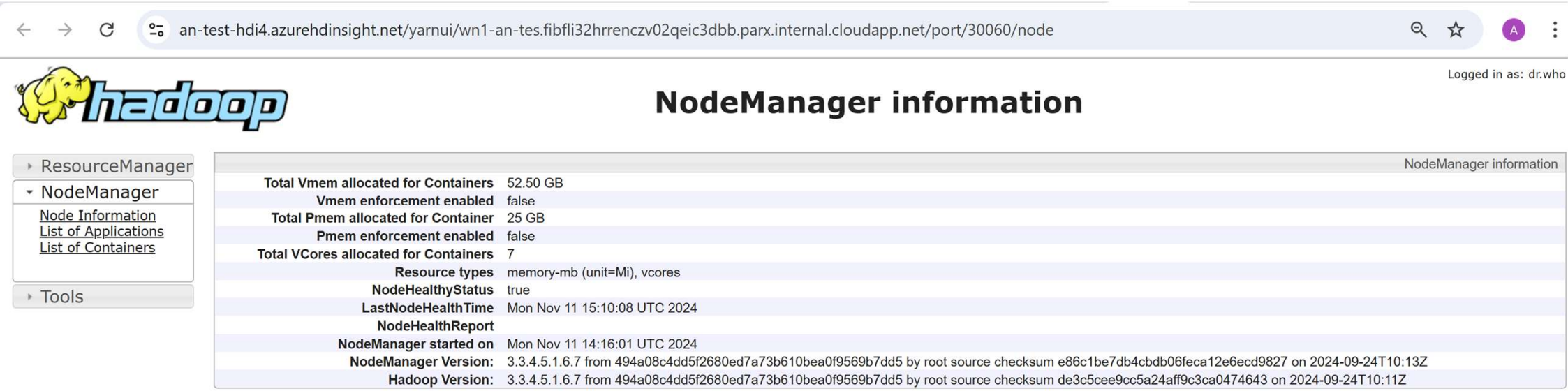
Using "Save As..." instead of click "here" for untruncated (10 Mega?)



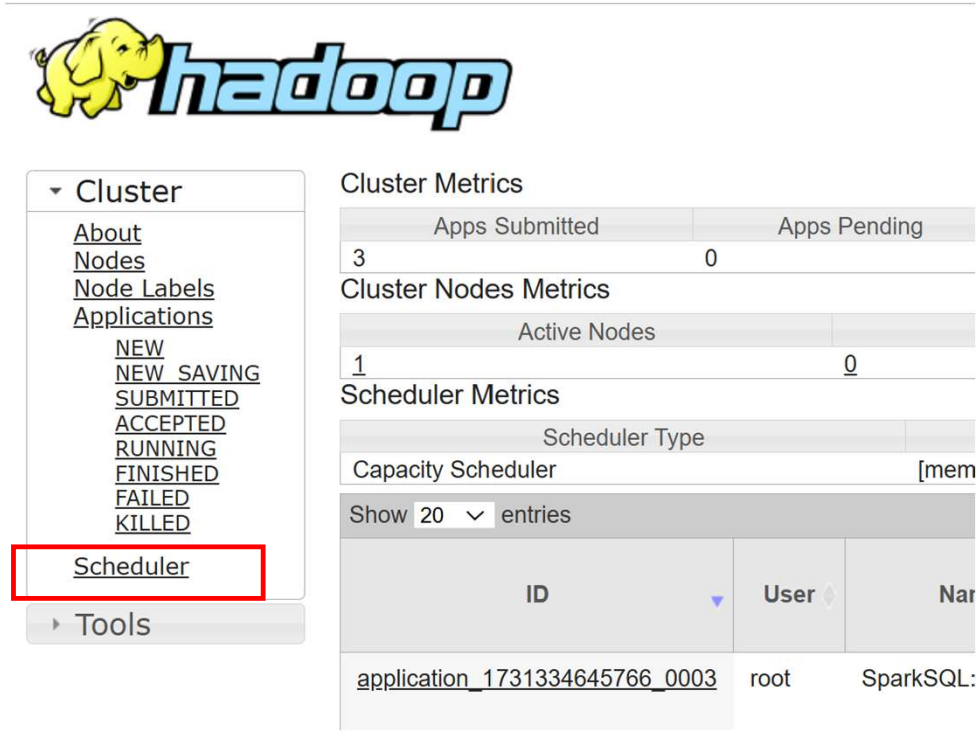
Cilck on Application > Attempt > "node"



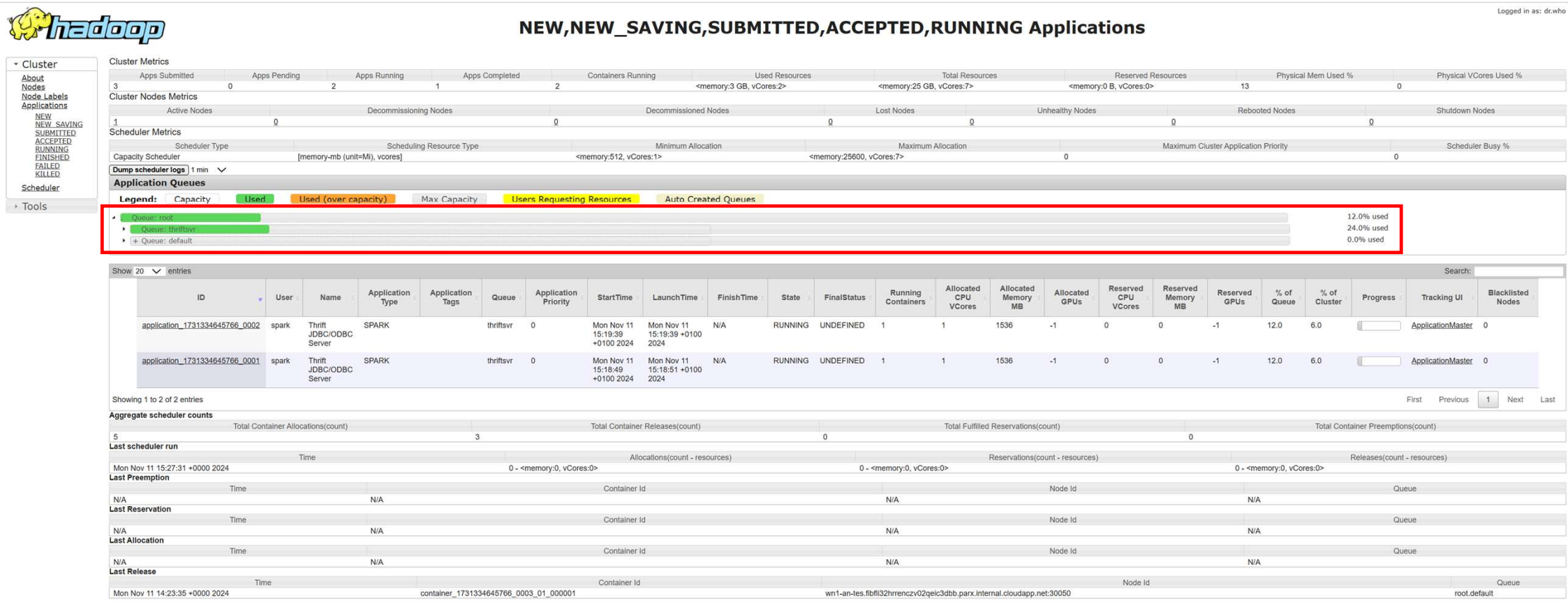
Yarn NodeManager "wn1"



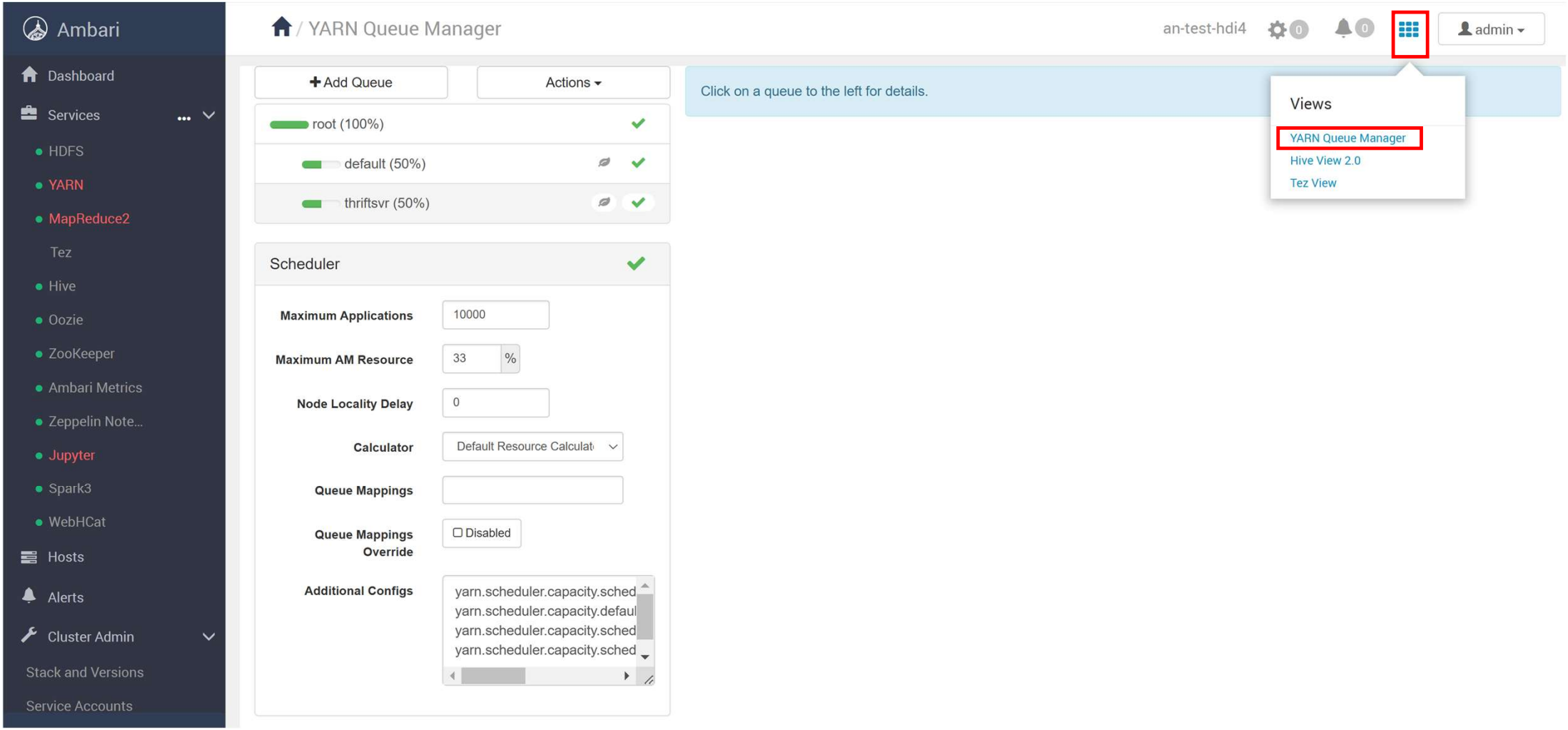
click on "Scheduler"



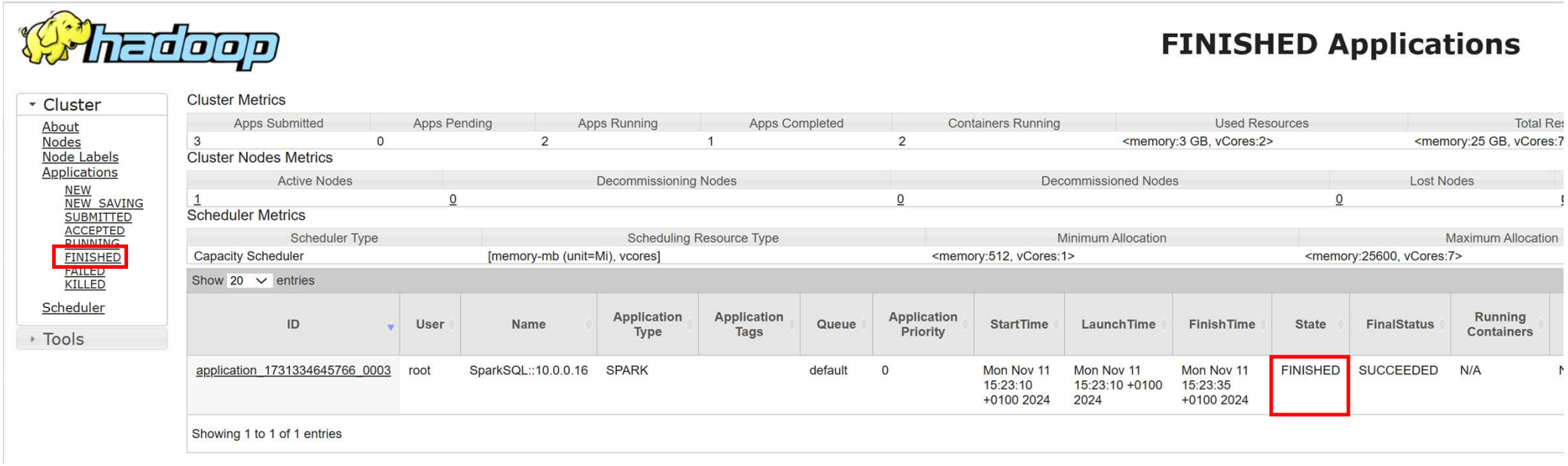
Yarn Scheduler Queue



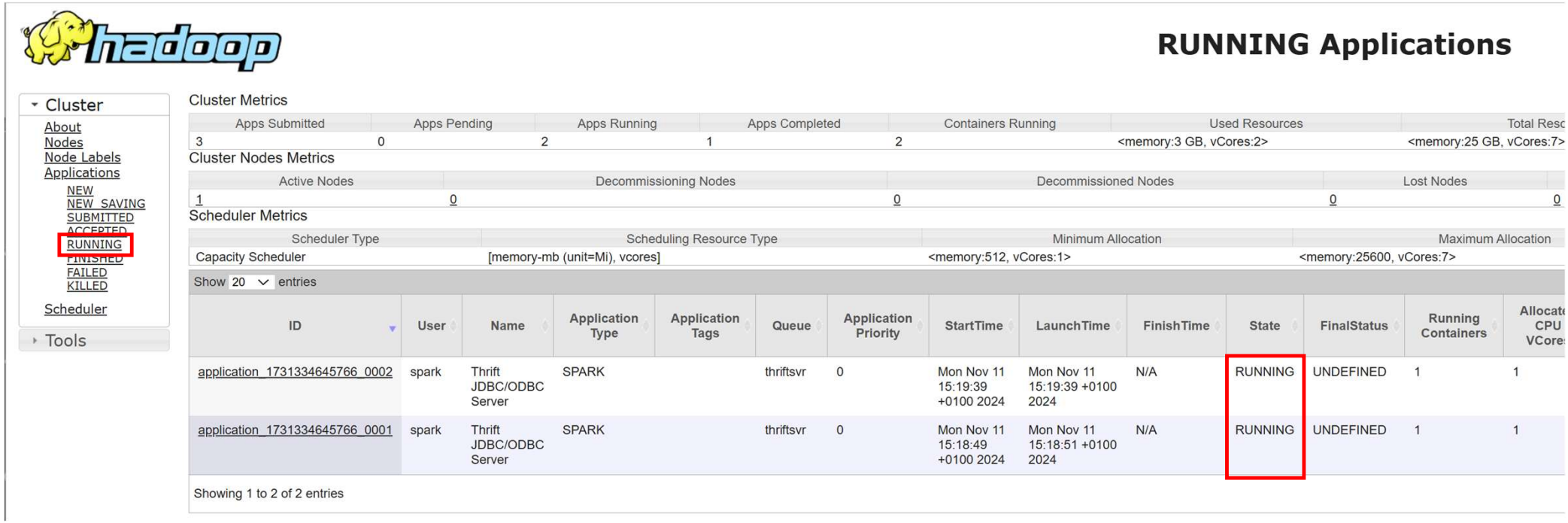
(Admin) Edit Yarn Queue



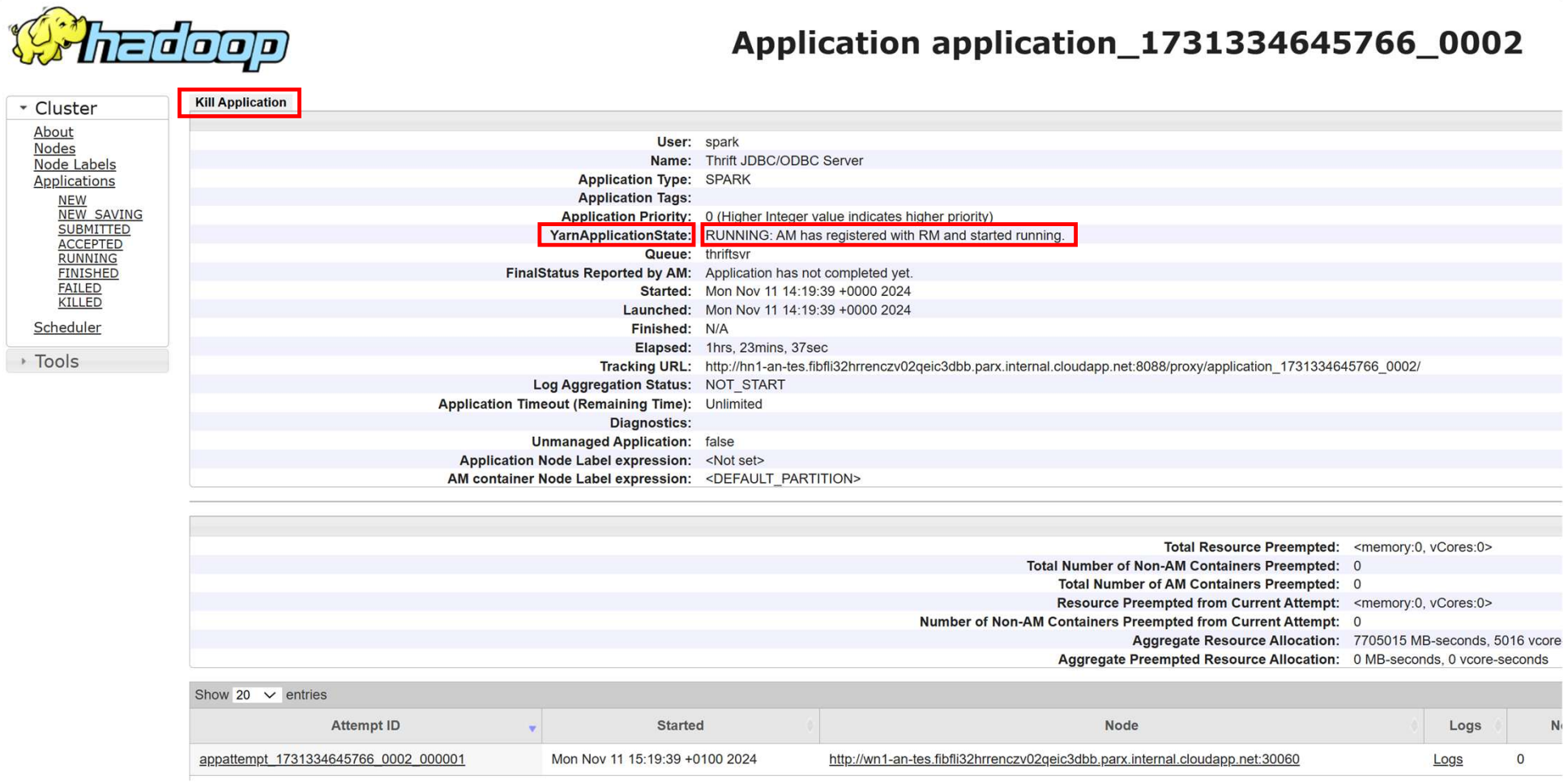
filter "FINISHED" Applications



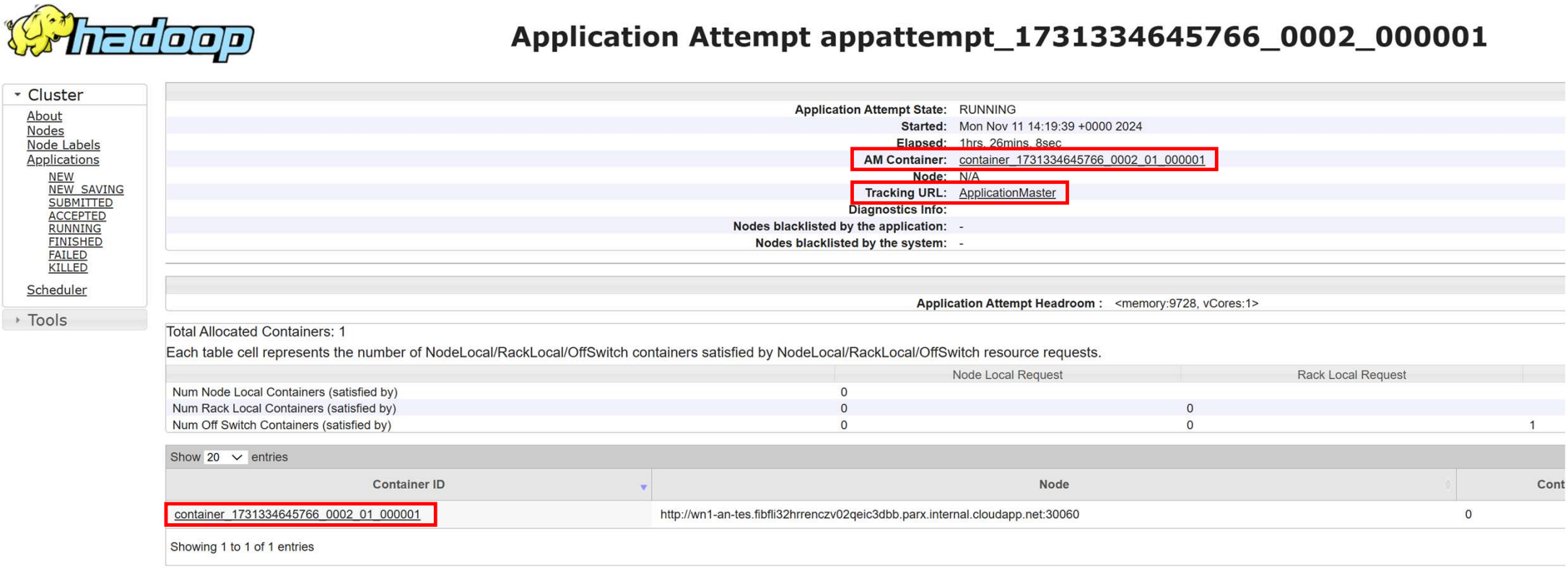
filter "RUNNING" Applications



click a RUNNING Yarn Application

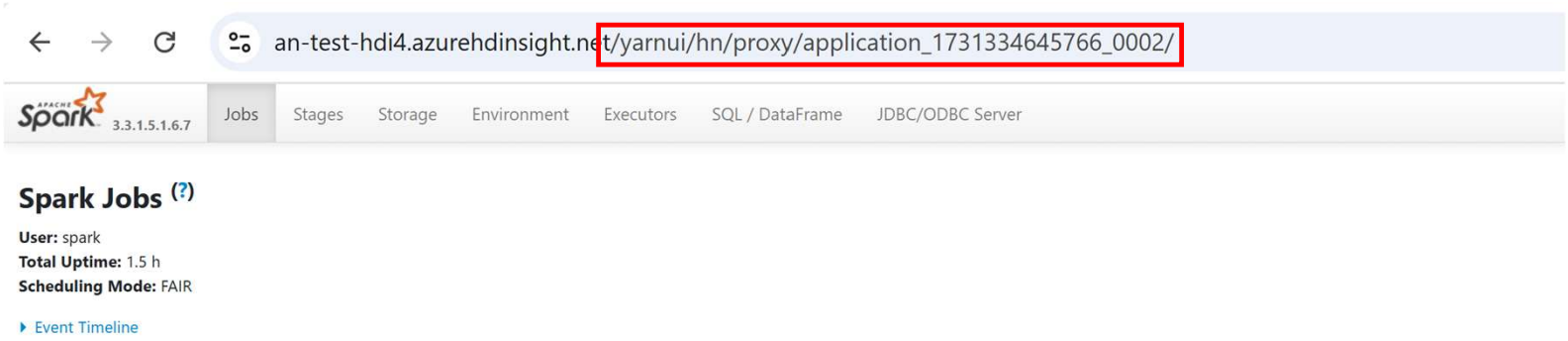


RUNNING Application > Attempt

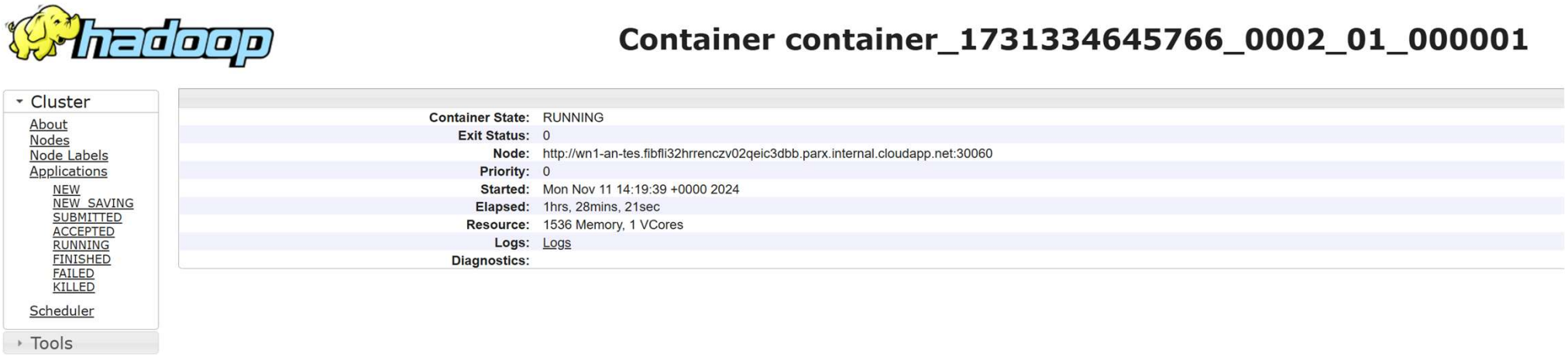


when Applications are still running => "/proxy" redirect to real process instead of "History" server

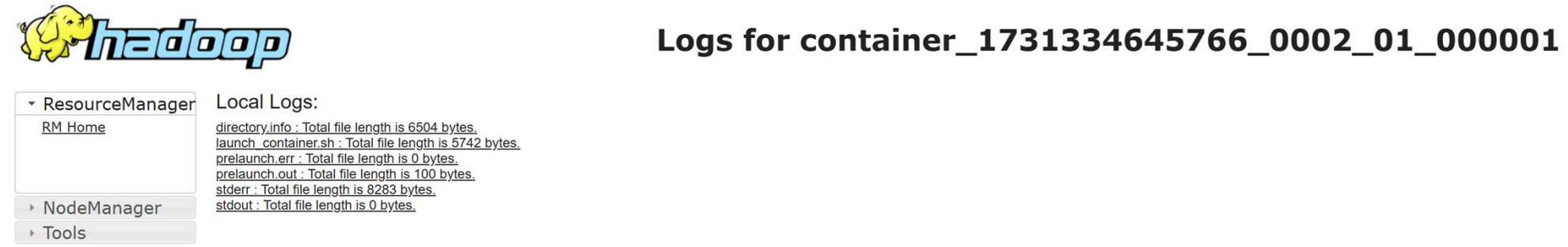
url: /yarnui/hn/proxy/application_<<id>>



RUNNING Application > AM Master > Container



Container > logs



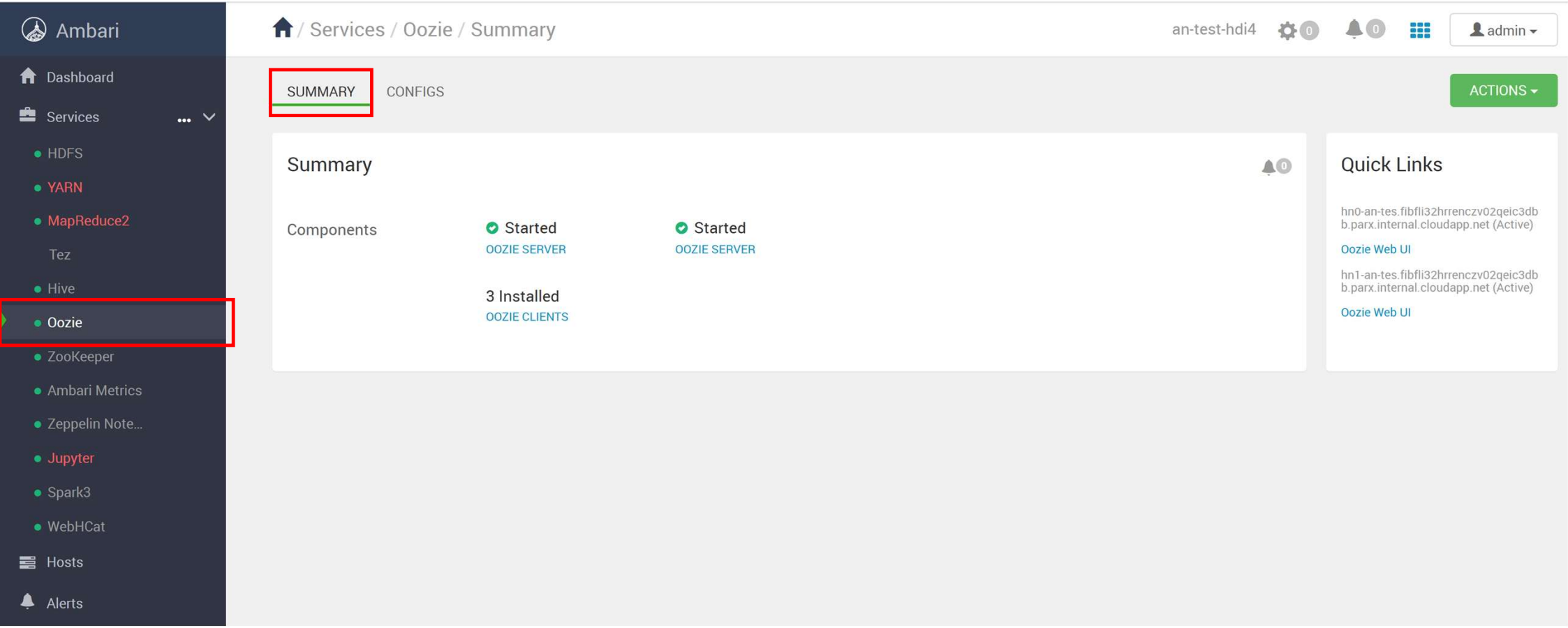
UI Experience for browsing Log in Yarn ...

(and not Search by Text or Regexp,
no selection by Time Range
... nothing)

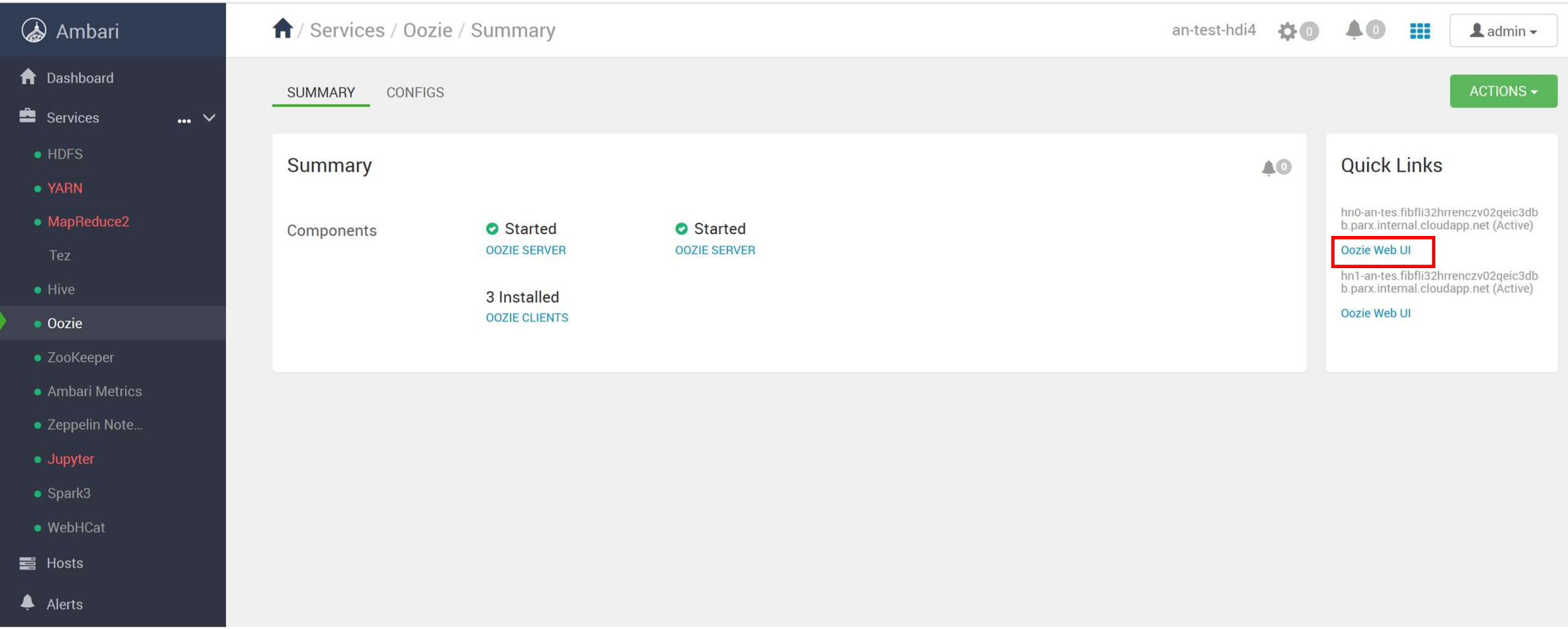


Ambari > Oozie

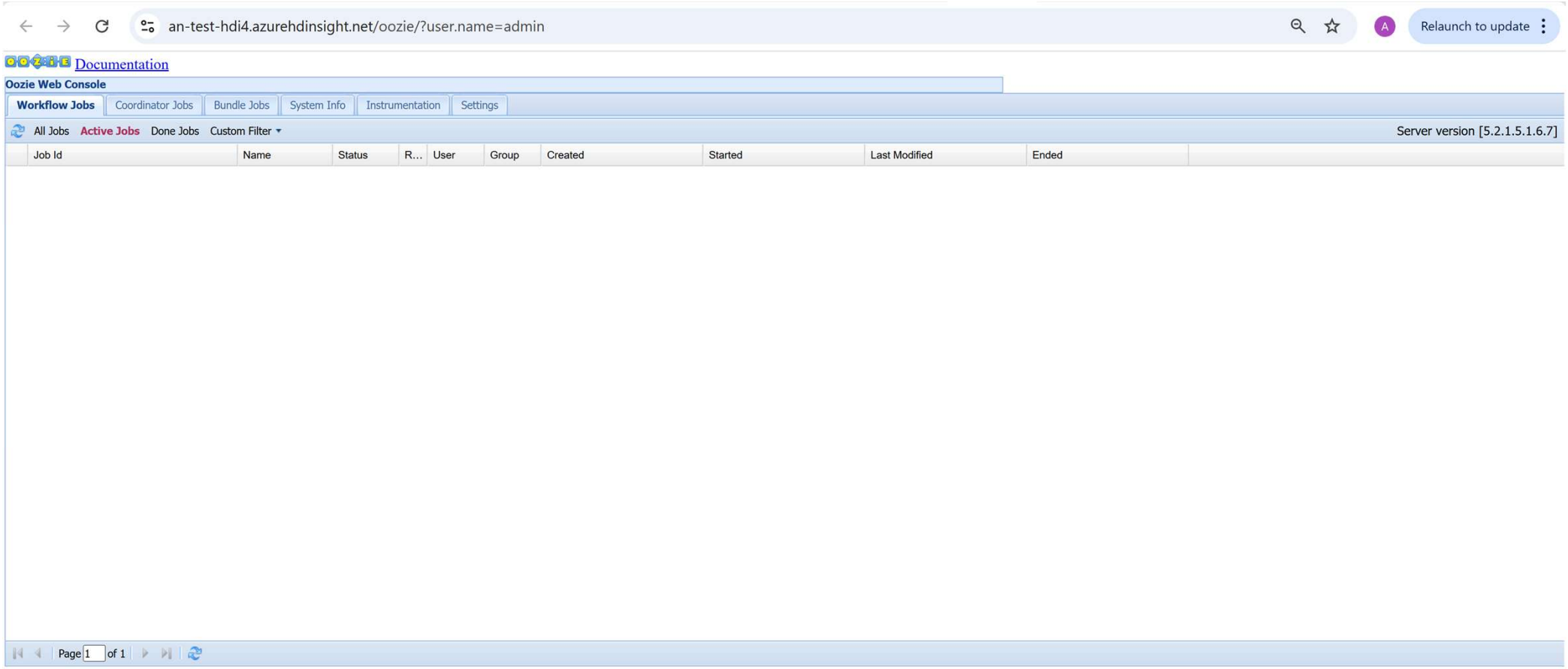
Ambari > Oozie > Summary



Ambari > Oozie > Quick Link "Oozie Web UI"



Oozie UI

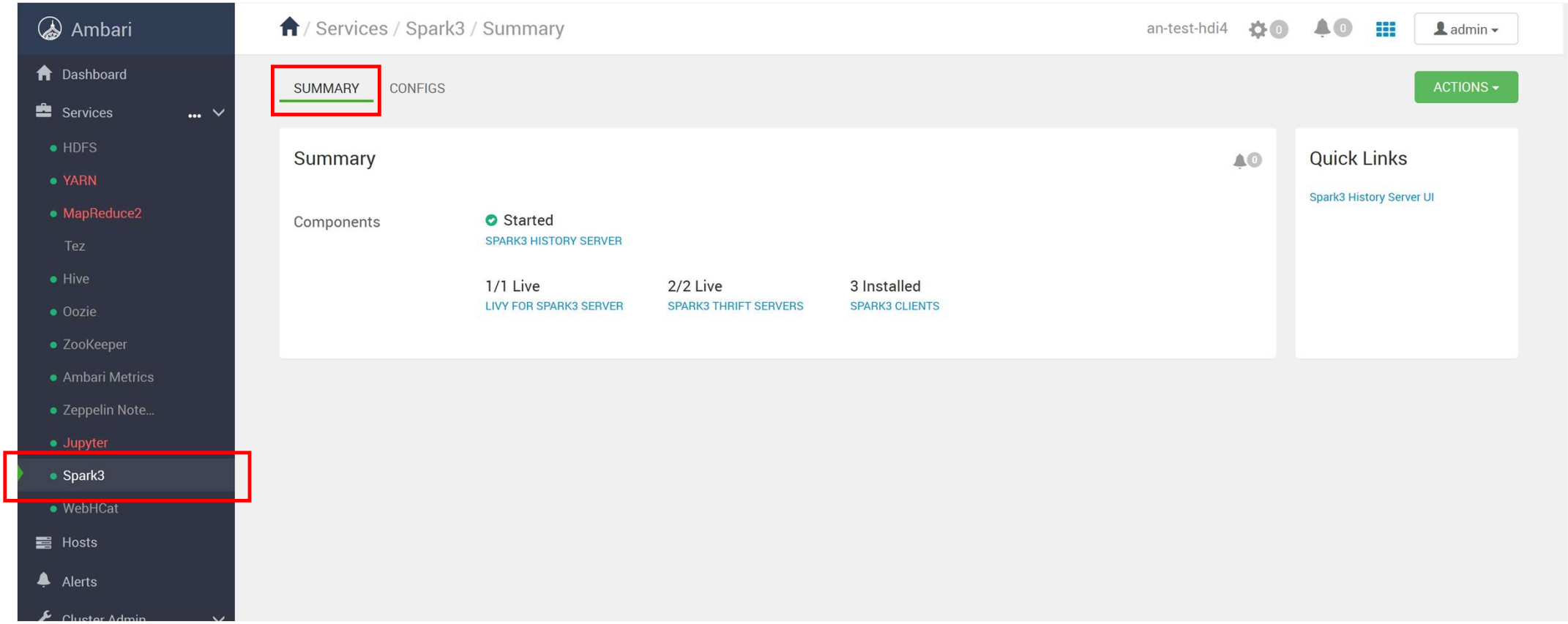

an-test-hdi4.azurehdinsight.net/oozie/?user.name=admin
Relaunch to update
Documentation
Oozie Web Console
Workflow Jobs
Coordinator Jobs
Bundle Jobs
System Info
Instrumentation
Settings
All Jobs
Active Jobs
Done Jobs
Custom Filter
Server version [5.2.1.5.1.6.7]
Job Id
Name
Status
R...
User
Group
Created
Started
Last Modified
Ended
Page 1 of 1

Oozie .. no job launched yet,

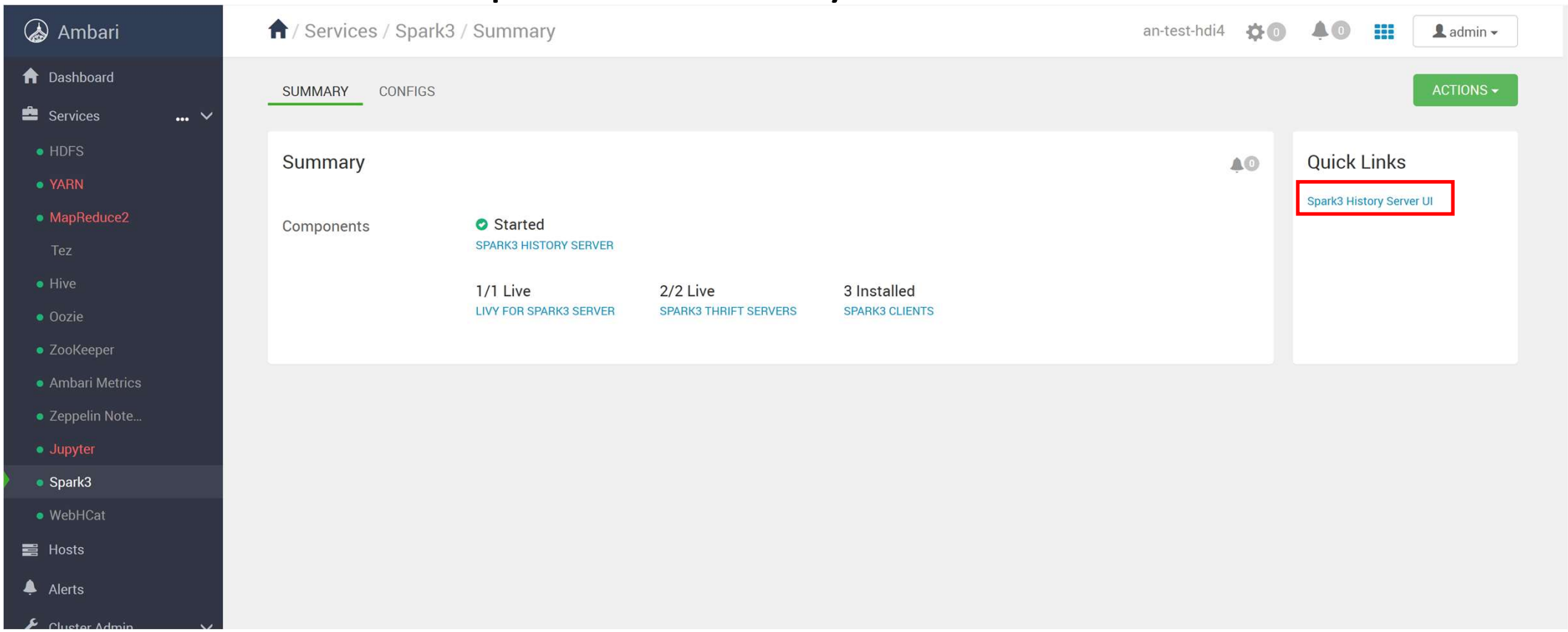
need "workflow.xml" file
+ submit (not from UI !)

Ambari > Spark

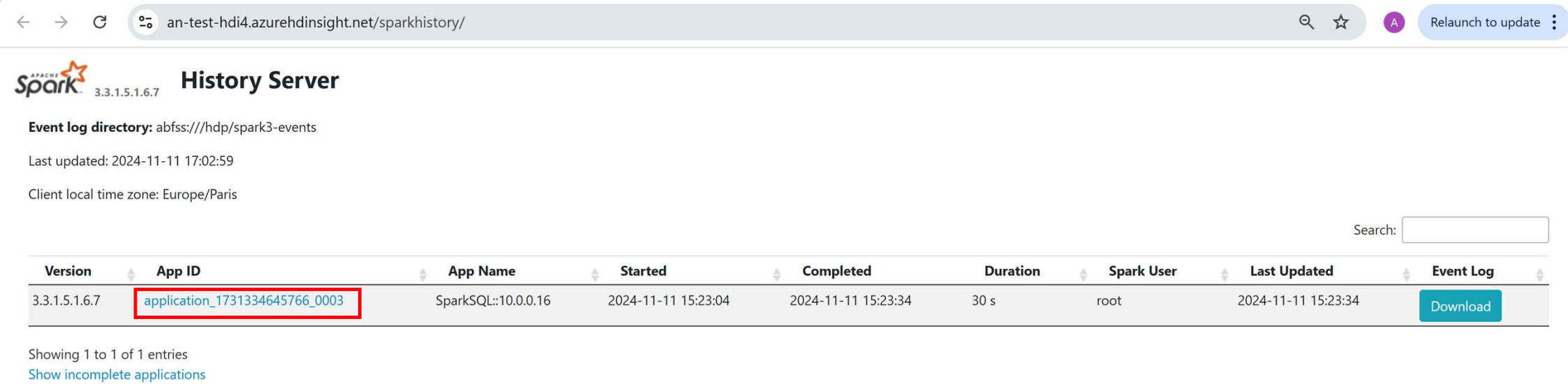
Ambari > Spark > Summary



Ambari > Spark > Quick Link > "Spark History Server UI"



Spark History Server
url : /sparkhistory

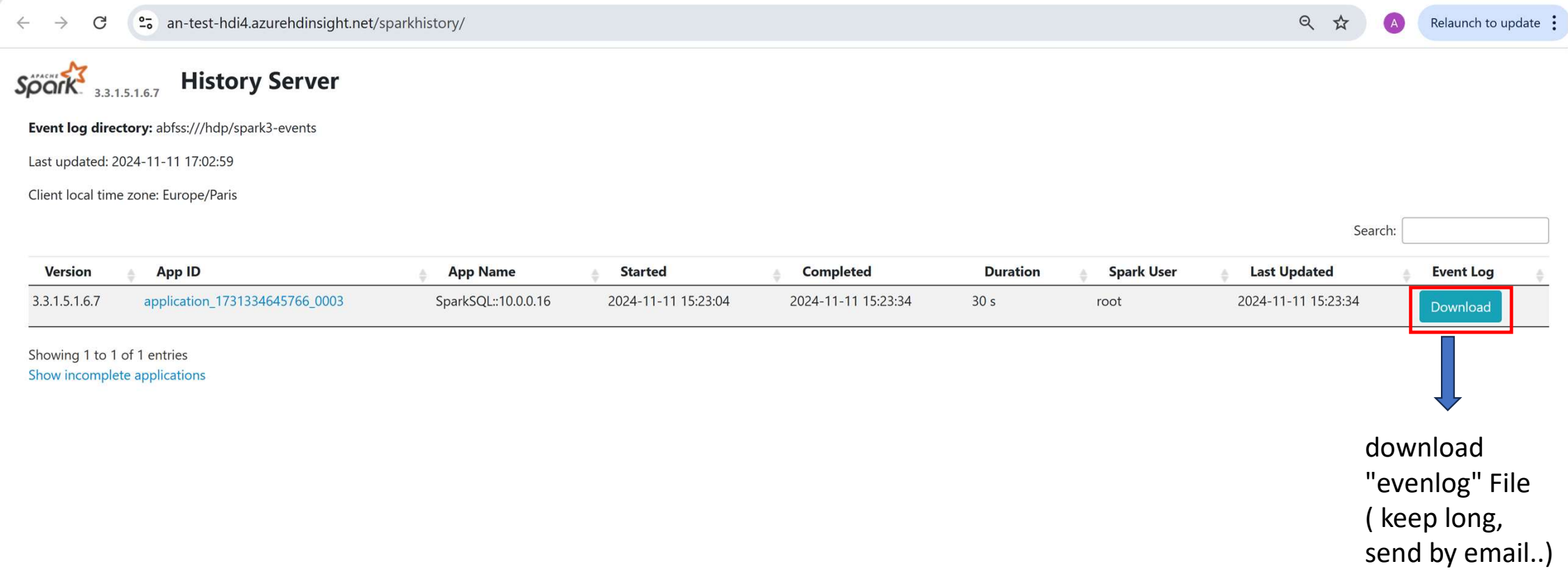


Click on SparkHistory > Application
=> see Spark UI

(same URL as Tracker URL from Yarn)

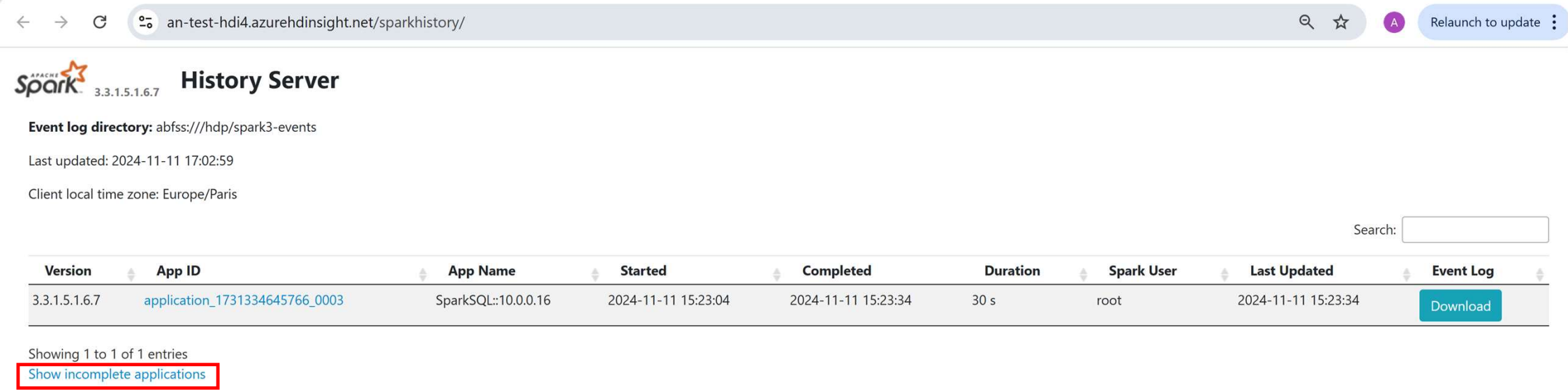


Spark History Server
Download "eventlog" Files

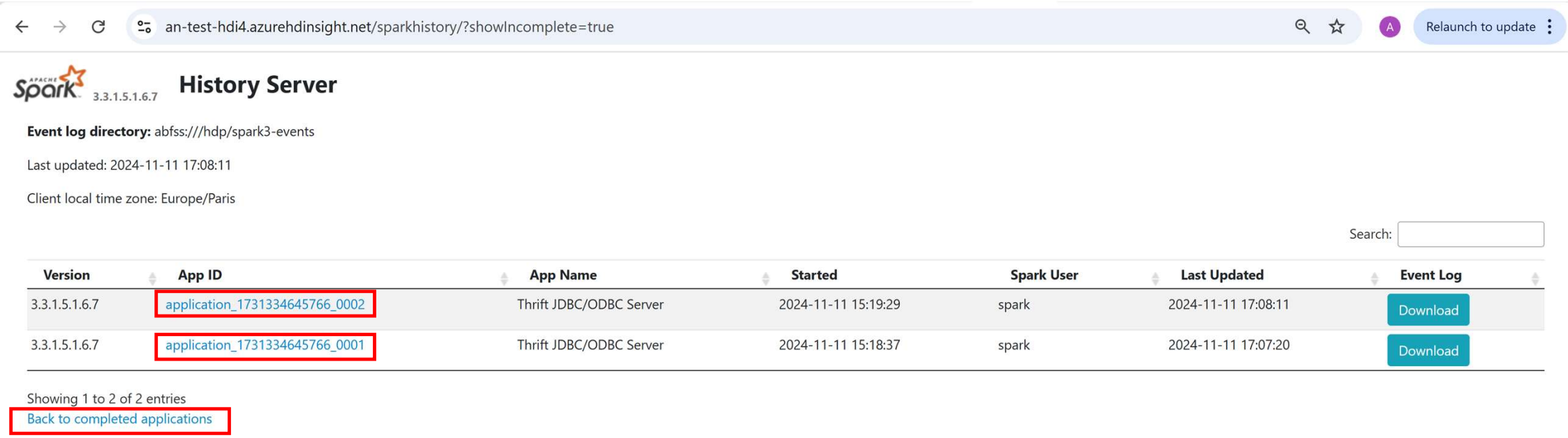


download "evenlog" File (keep long, send by email..)

Spark History Server
click "Show Incomplete Applications"



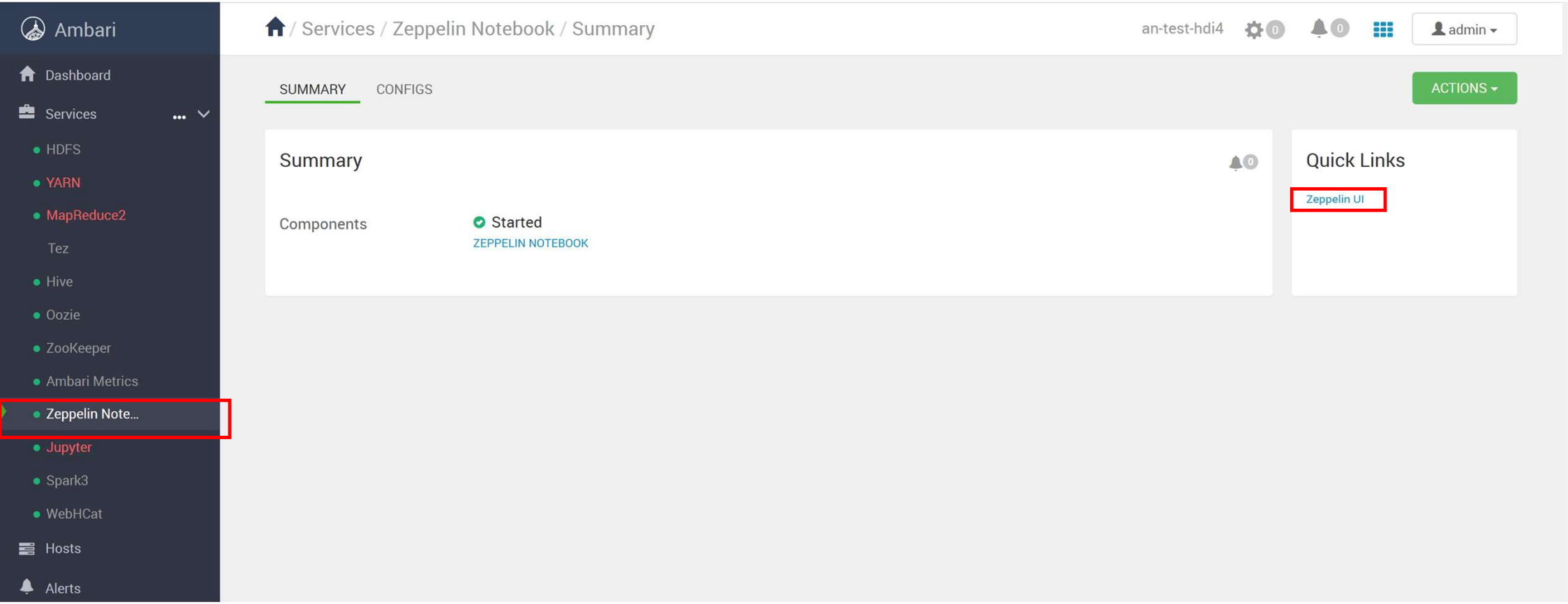
Spark History > Incomplete Applications



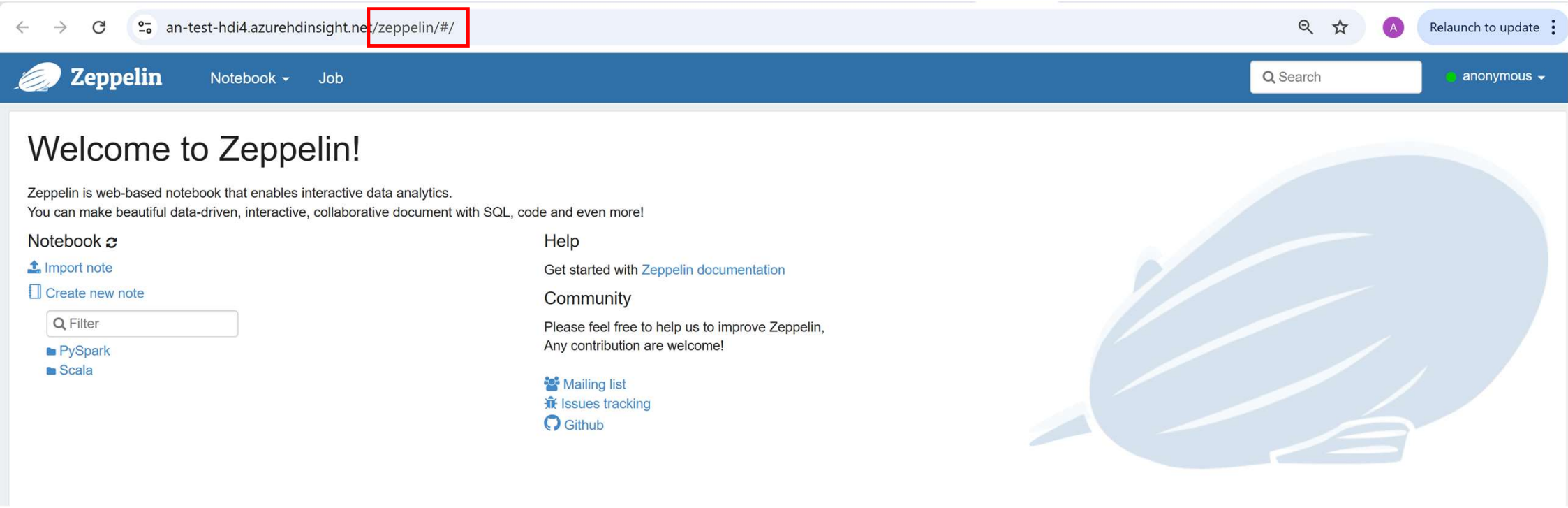
Notice: Spark History may discover "eventlog" Files for RUNNING apps, with a delay of ~5 minutes

Ambari > Zeppelin

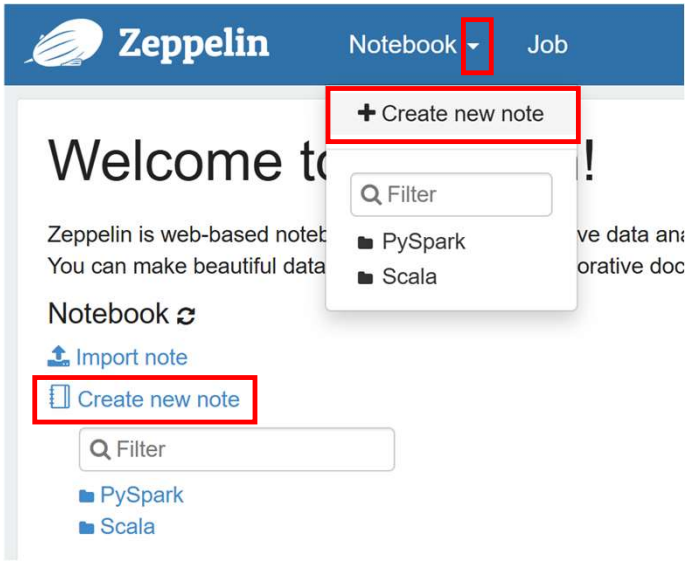
Ambari > Zeppelin > Summary



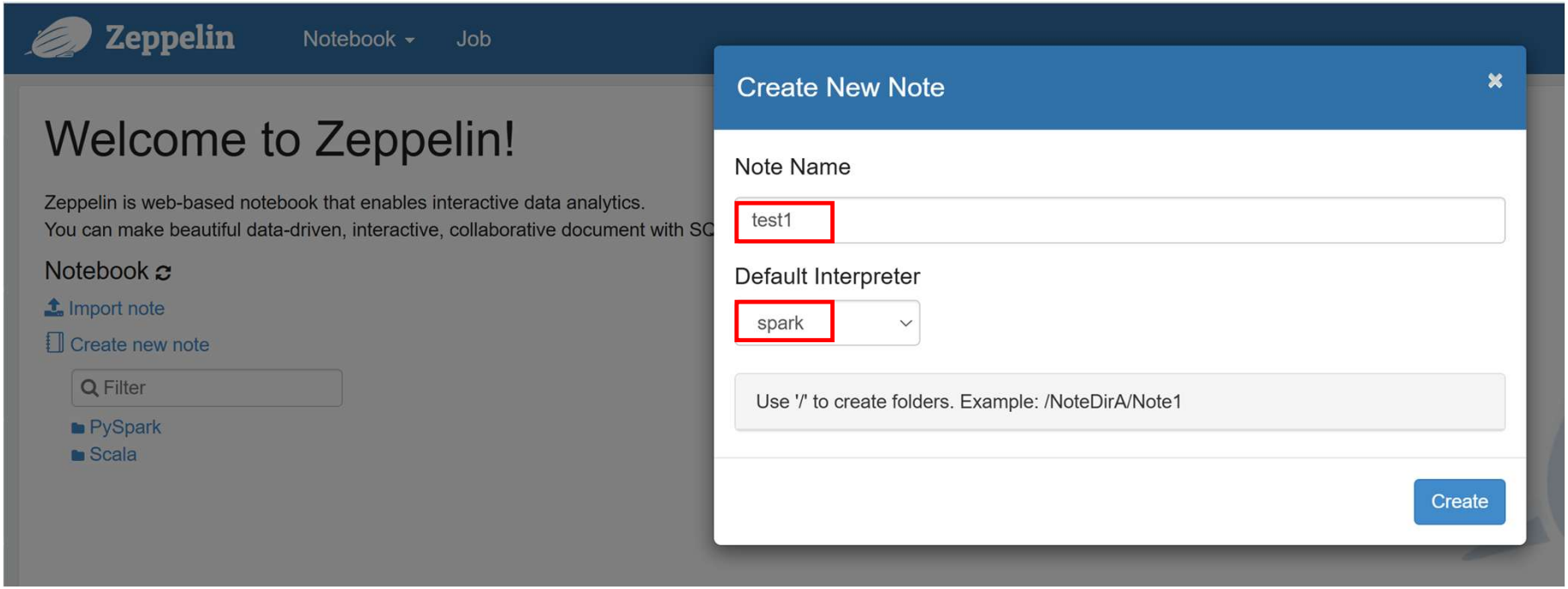
Zeppelin
url: /zeppelin/#



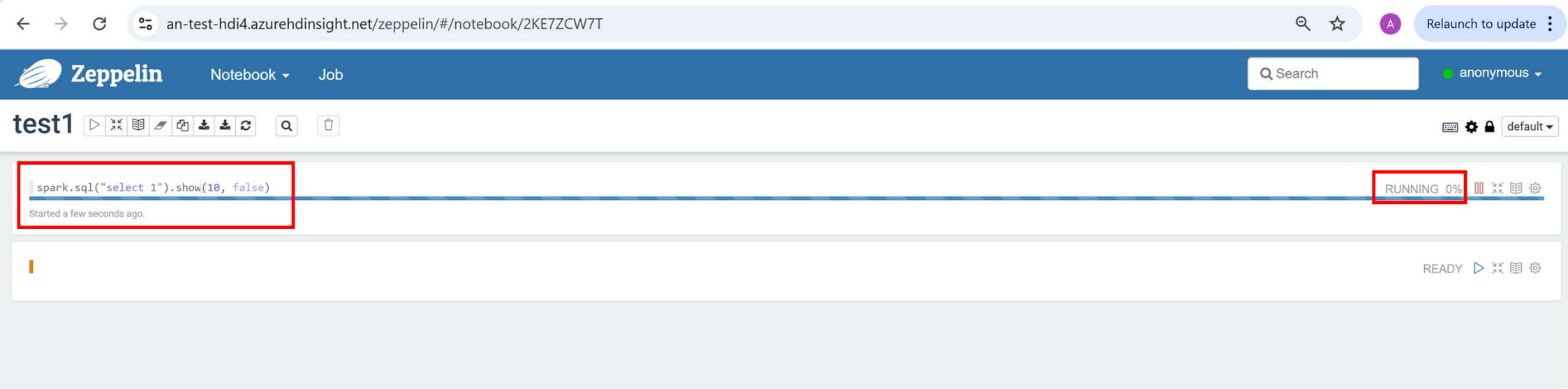
Zeppelin > New Notebook



Create Zeppelin Notebook (Interpreter=Spark)



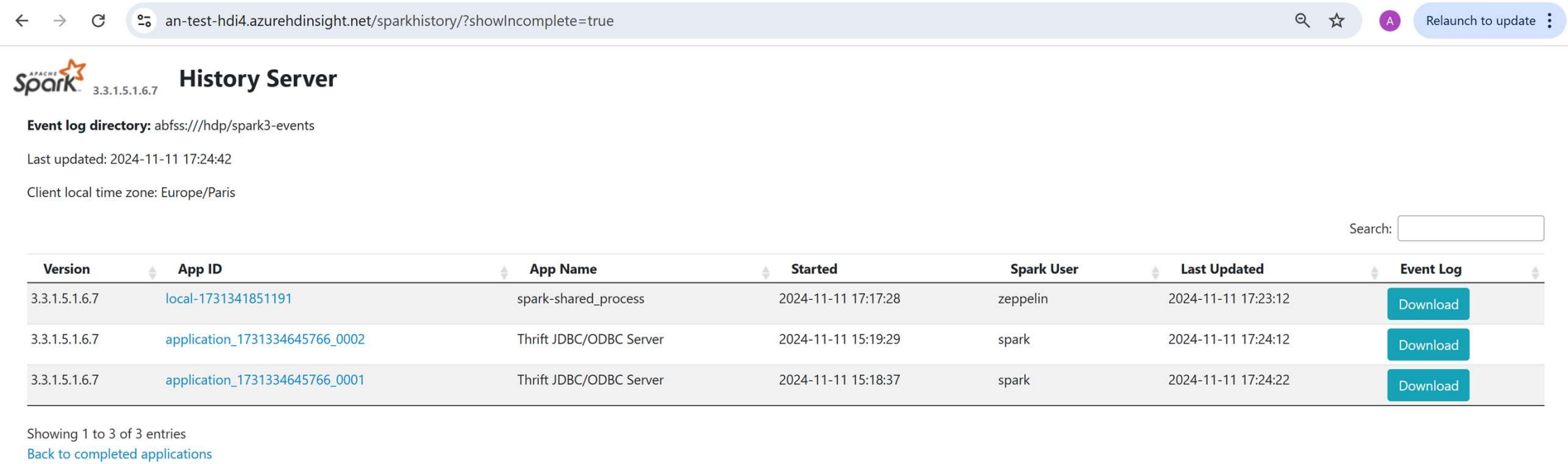
Zeppelin > Cell > "Shift-Enter" to execute



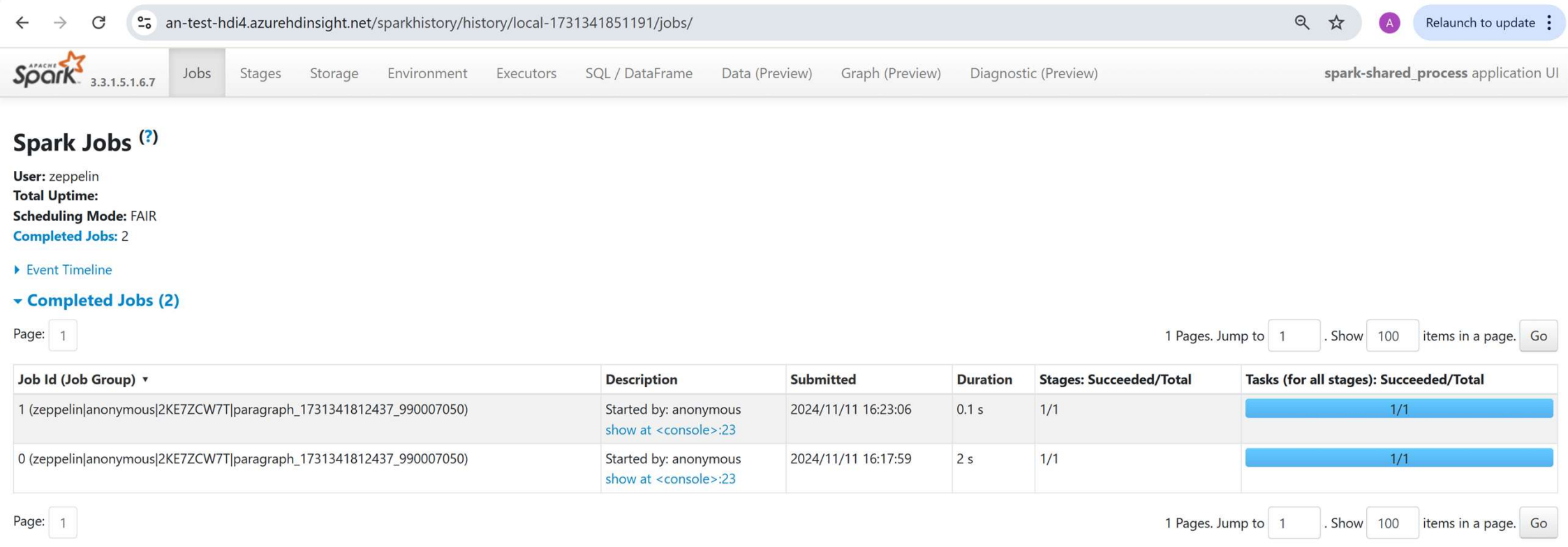
Zeppelin -> Spark Job



Zeppelin - Spark Job launched by "local" (not visible in yarnui) or by livy

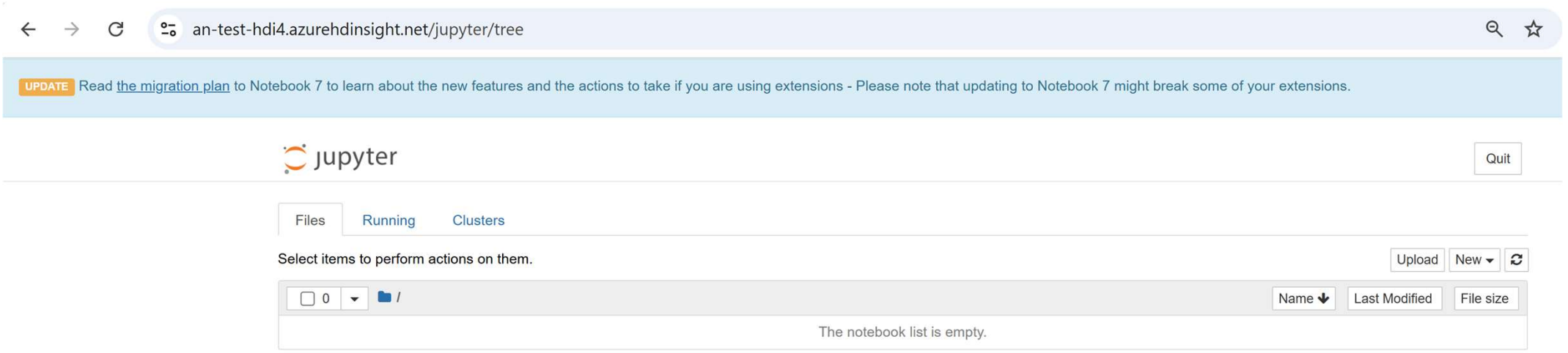


Zeppelin Spark Job in SparkUI



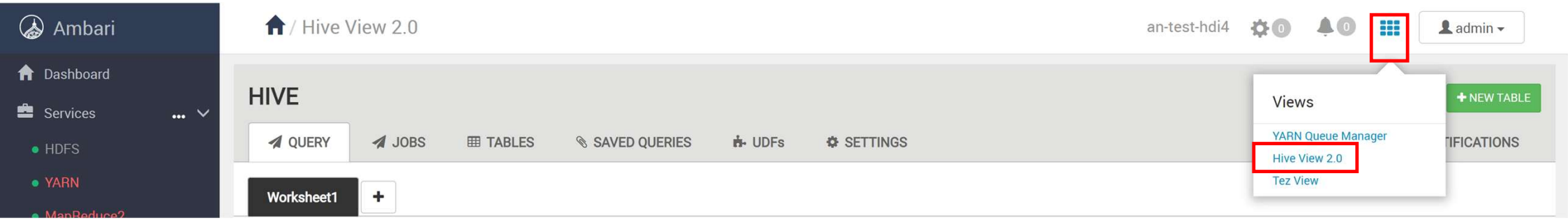
Jupyter

Jupyter
url :/jupyter

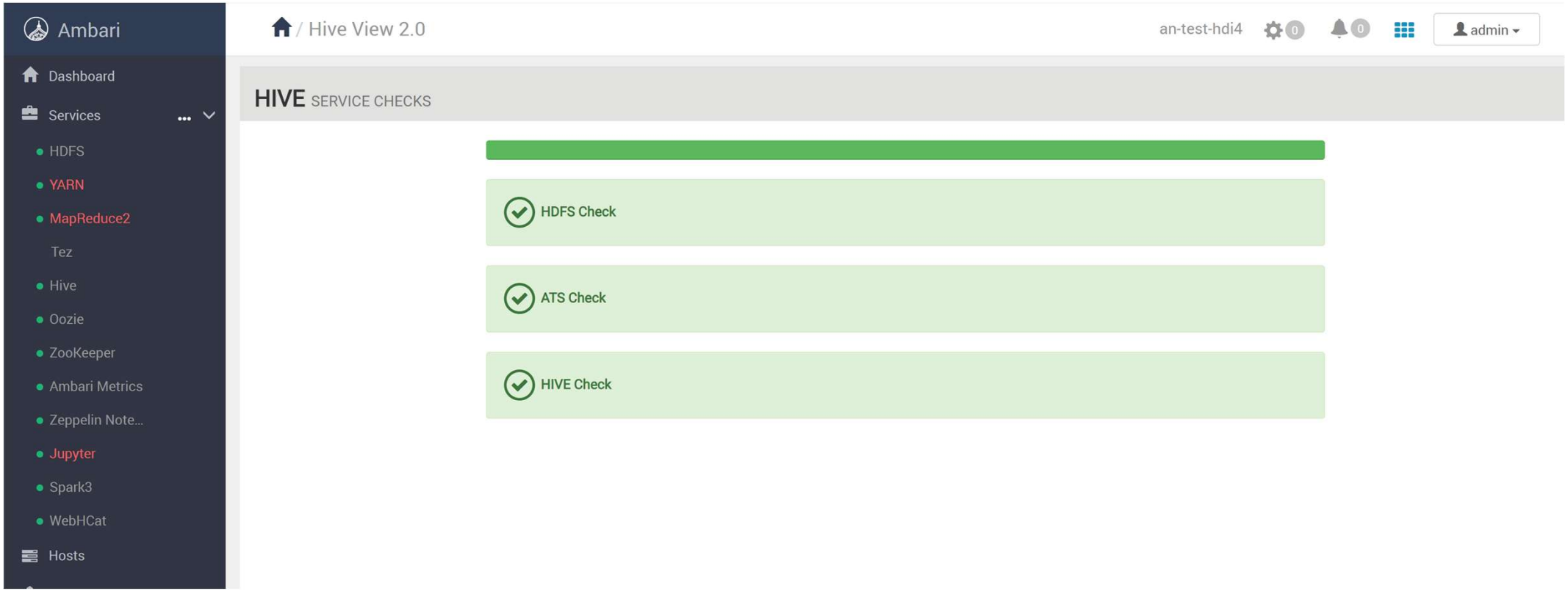


Ambari > Hive View
(NOT RECOMMENDED, prefer Spark)

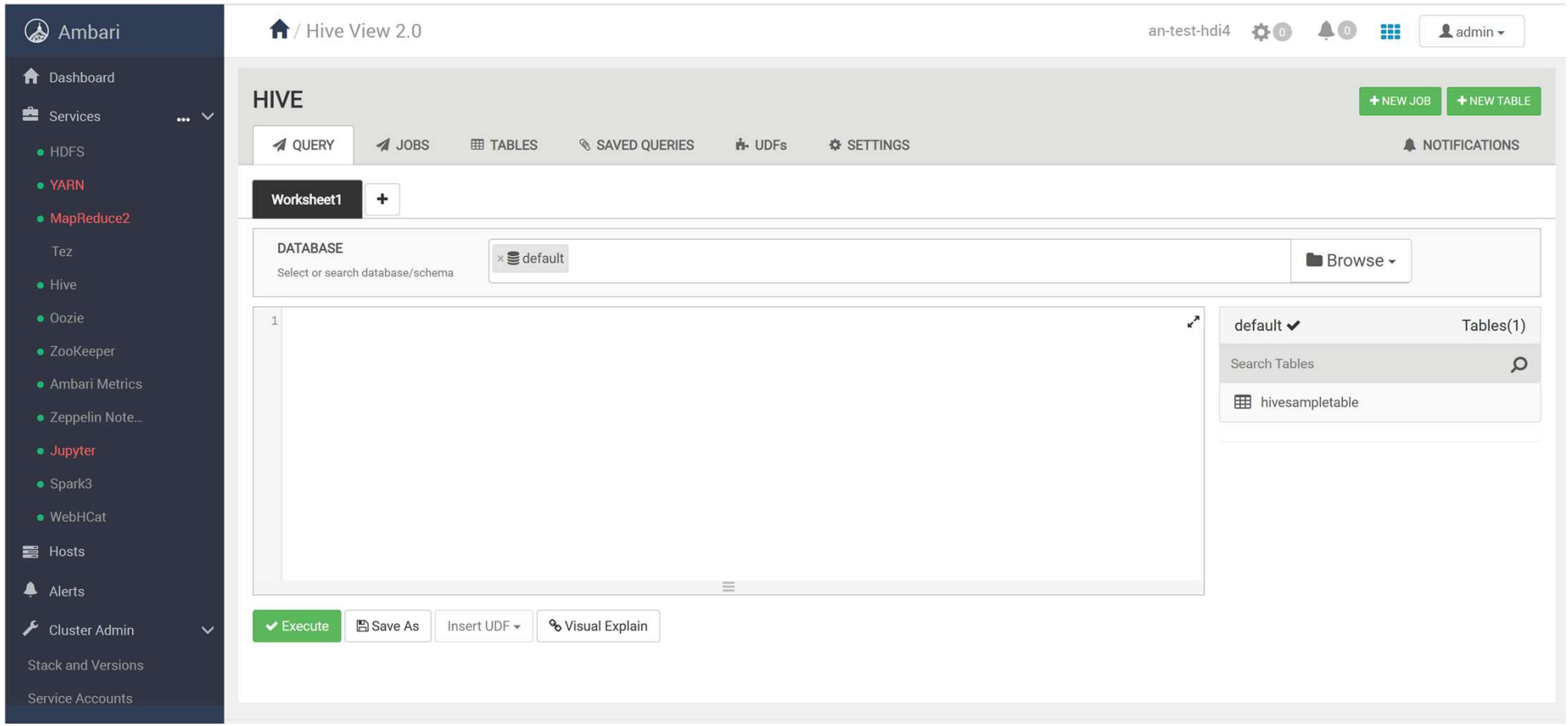
Ambari > Hive View 2



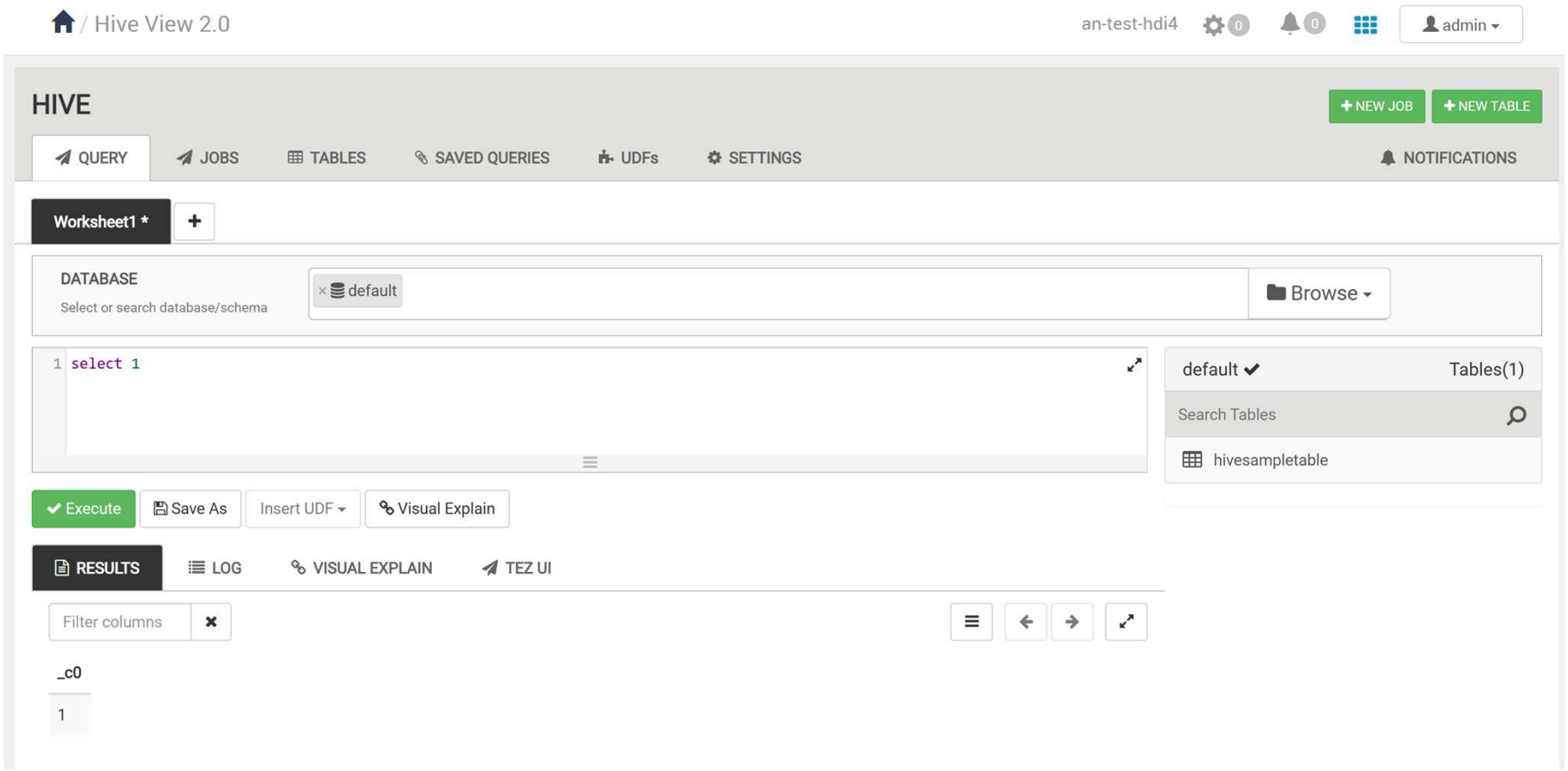
Ambari > Hive View (page loading...)



Ambari > Hive View

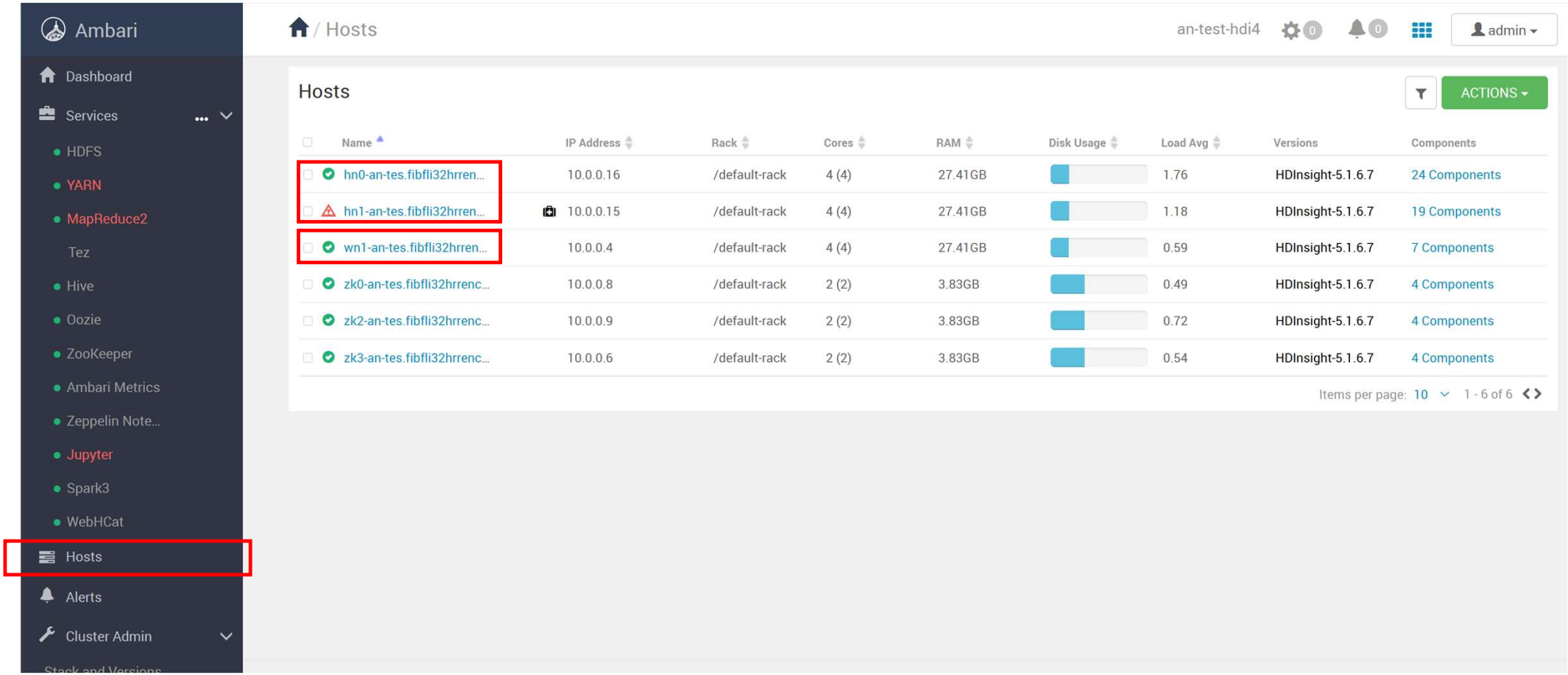


HiveView > Execute SQL Query



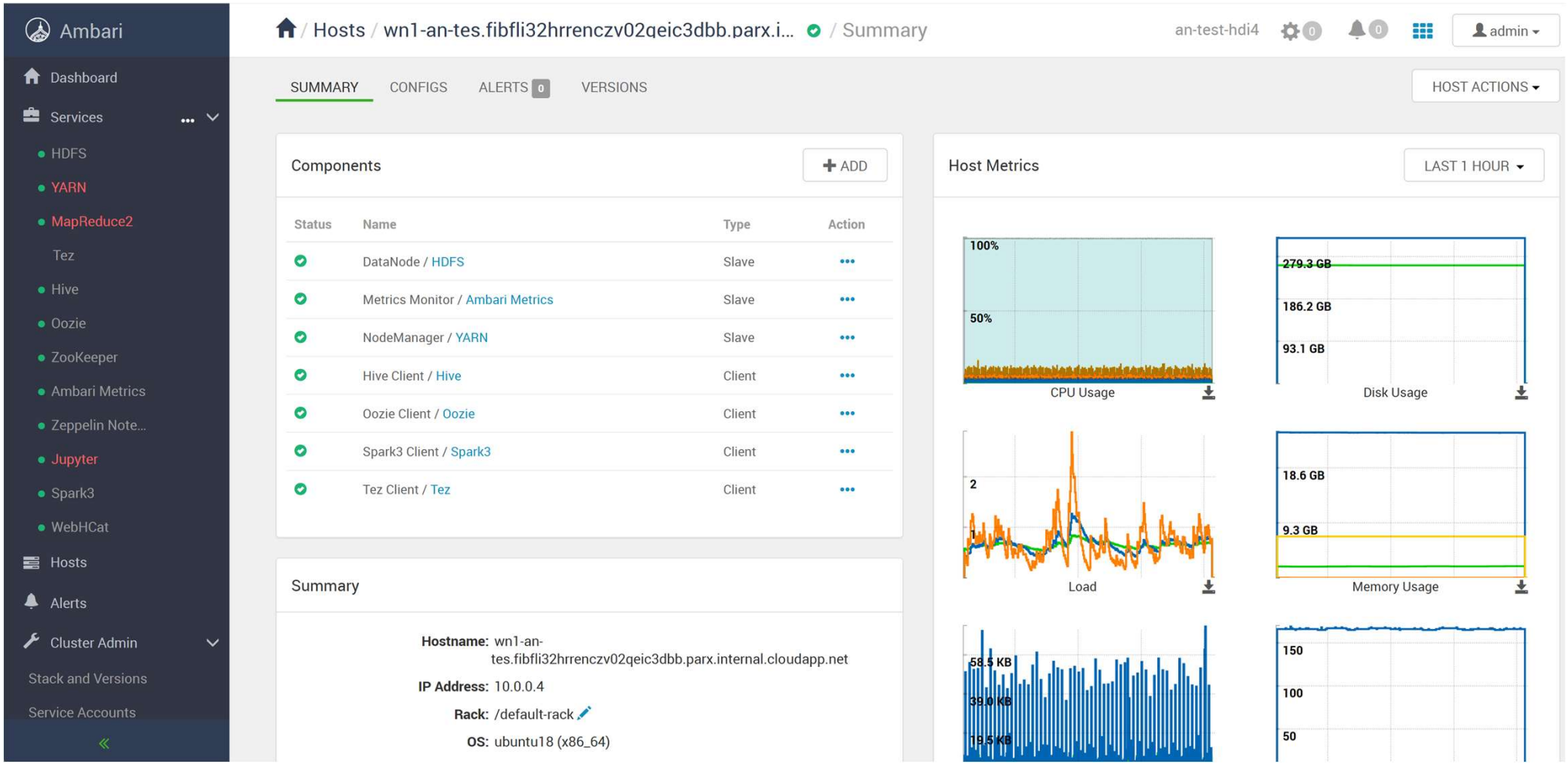
(ADMIN) Ambari > Hosts

Ambari > Hosts (admin)



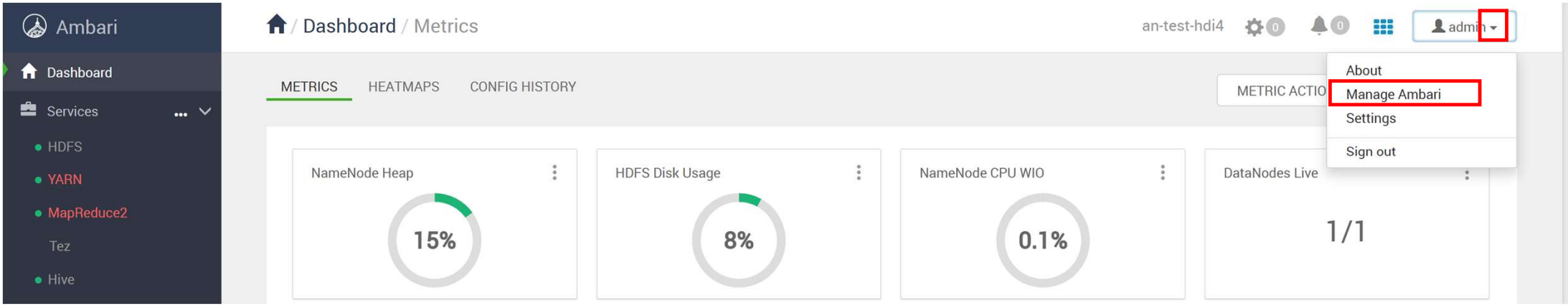
Hosts : 2 Head Nodes (hn0, hn1) + 3 Zookeeper Nodes (zk0, zk1, zk2) + N (>=1) Worker Nodes (wn0, wn1, ...)

Ambari > Hosts > wn1 Summary

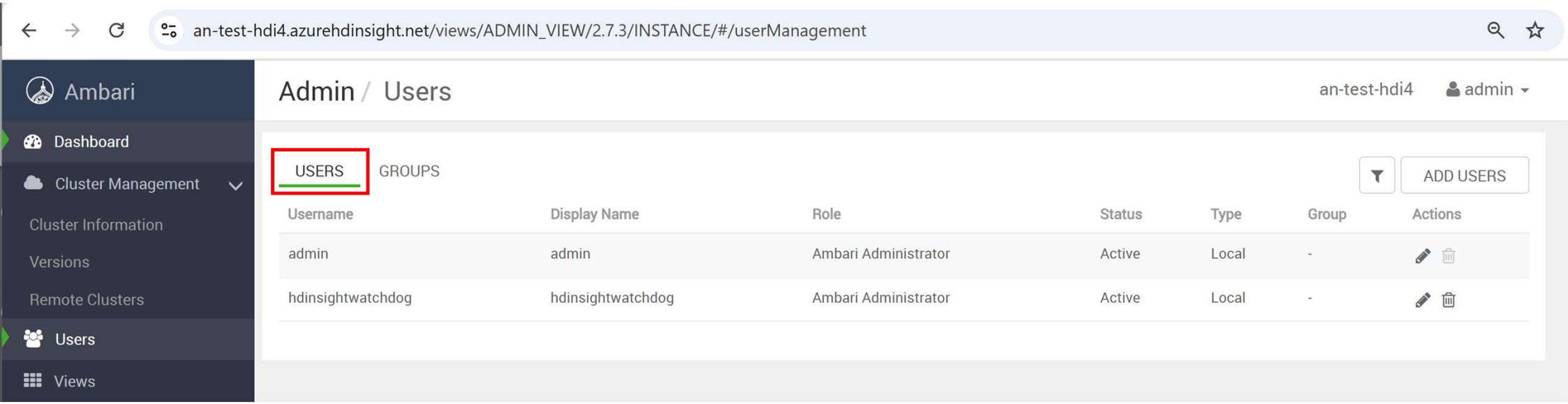


(ADMIN) Ambari > Users & Groups

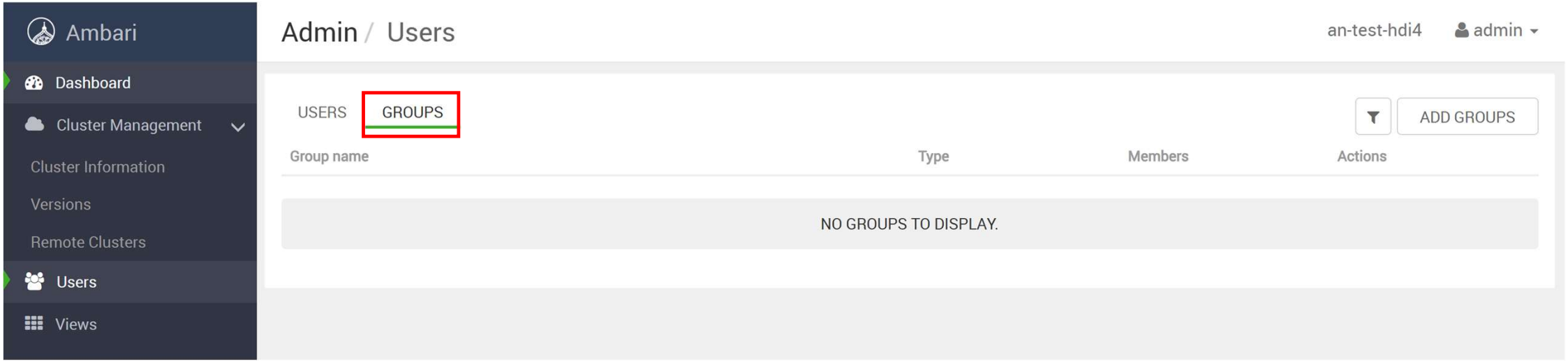
(ADMIN) Ambari > Login > Manage Amari



Ambari > Admin > Users

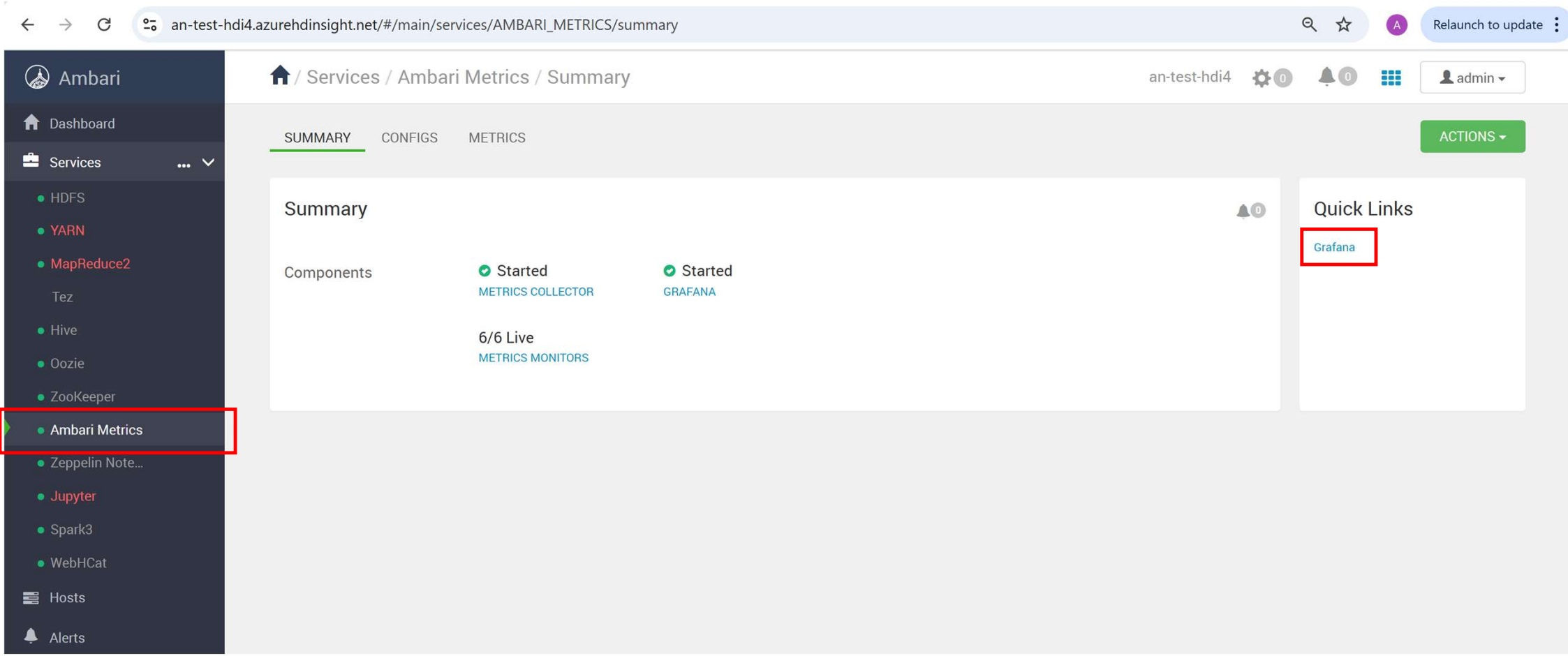


Ambari > Admin > Groups

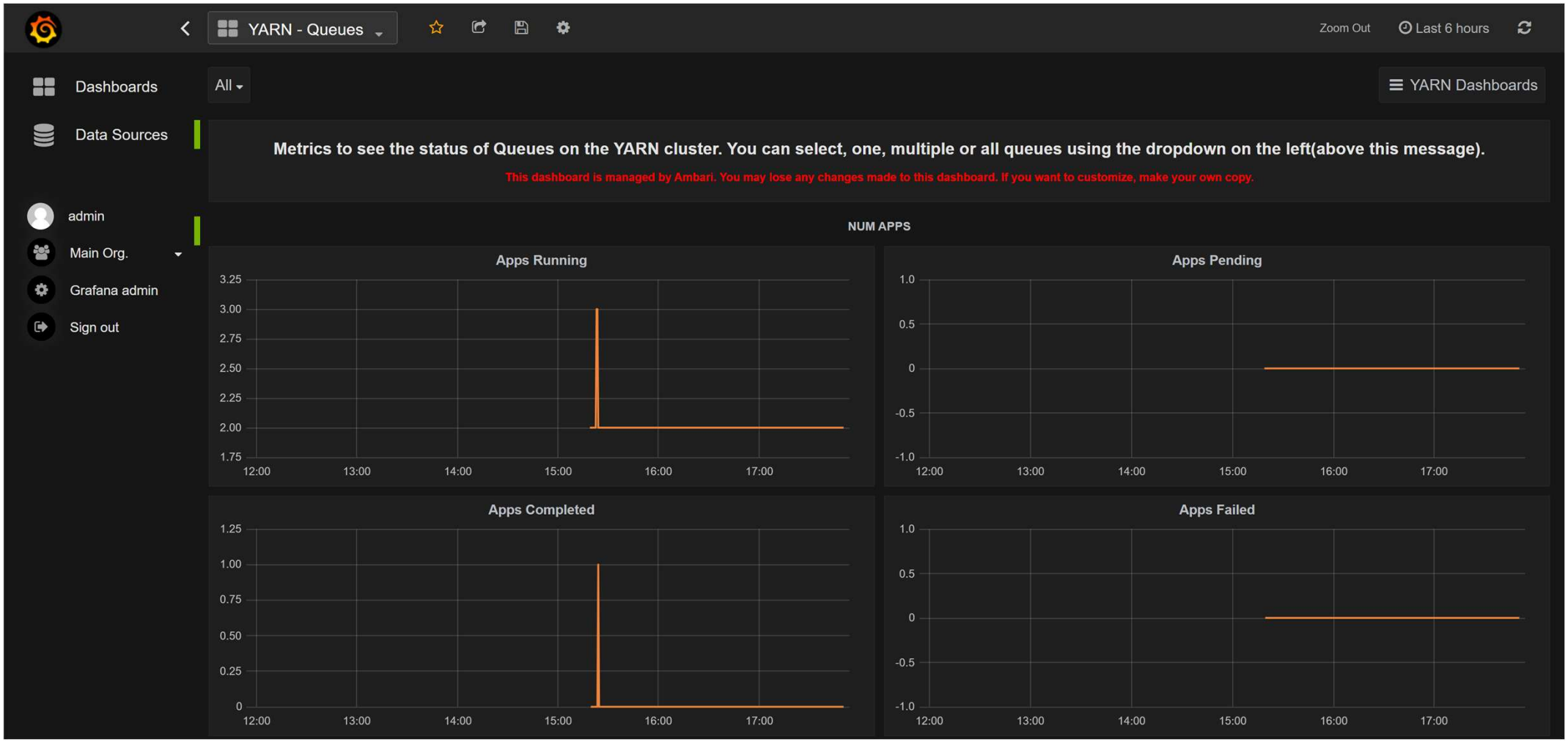


(ADMIN) Ambari > Metrics > Grafana

(ADMIN) Ambari Metrics > Grafana



Grafana


YARN - Queues
Zoom Out
Last 6 hours
Dashboards
Data Sources
admin
Main Org.
Grafana admin
Sign out
All
YARN Dashboards
Metrics to see the status of Queues on the YARN cluster. You can select, one, multiple or all queues using the dropdown on the left(above this message).
This dashboard is managed by Ambari. You may lose any changes made to this dashboard. If you want to customize, make your own copy.
NUM APPS
Apps Running
Apps Pending
Apps Completed
Apps Failed
12:00
13:00
14:00
15:00
16:00
17:00

Conclusion

HD Insight is a ready to use all-in-one Hadoop environnement.
Hadoop would be extremely difficult to setup yourself on your PC (ok on linux, not windows).

Hadoop Yarn is overly complex.

Kubernetes is Better than Hadoop Yarn.
And Unfortunatly, there is NO standard "pre-package" all-in-one environment for Spark on k8s + Jupyter + ...
But, with a custom environment,

Spark on Kubernetes is Better/Cheaper/Easier than Spark on HD-Insight.

Questions ?

arnaud.nauwynck@gmail.com